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Engineering of Additive Manufacturing Features for Data-Driven Solutions

Sources, Techniques, Pipelines, and Applications

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Abbreviations

3D	Three-Dimensional
AE	Autoencoder
AJP	Aerosol Jet Printing
AM	Additive Manufacturing
ANOVA	Analysis of Variance
ASTM	American Society of Testing Materials
ATCS	Automated Computed Tomography Segmenter
BJ	Binder Jetting
BOG	Bag of Words
CAD	Computer-Aided Design
CADD	Computer-Aided Defect Detection
CAM	Computer-Aided Manufacturing
CCD	Charged Couple Device
CFRP	Carbon Fiber Reinforced Polymer
CII	Contrast Improvement Index
CMOS	Complementary Metal Oxide Semiconductor
CNN	Convolutional Neural Network
CoC	Characteristics of Concern
CTFD	Computational Thermal Fluid Dynamics Model
CVAE	Conditional Variational Autoencoder
DA	Domain Adaptation
DBN	Deep Belief Network
DED	Directed Energy Deposition
DfAM	Design for Additive Manufacturing
DIKW	Data-Information-Knowledge-Wisdom
DL	Deep Learning
DM	Digital Material
DPSP	Design–Process–Structure–Property
DT	Digital Twin
DTW	Dynamic Time Wrapping
EEMD	Ensemble Empirical Mode Decomposition

EPMP	Enhanced Phase Measuring Profilometry
FEM	Finite Element Method (or Model)
FEP	Feature Engineering Pipeline
FET	Feature Engineering Technique
FT	Fourier Transform
GAN	Generative Adversarial Network
GP	Gaussian Process Modeling
Grad-CAM	Gradient Weighted Class Activation Mapping
HIZ	Heat Influence Zone
HOG	Histogram of Oriented Gradients
ICA	Independent Component Analysis
ICME	Integrated Computational Materials Engineering
IR	Infrared
ISO	International Organization for Standardization
KB	Knowledge Based
LASSO	Least Absolute Shrinkage and Selection Operator
LENS	Laser Engineered Net Shaping
LOGO-CV	Leave One Group Out Cross Validation
MAM	Metal Additive Manufacturing
MEX	Material Extrusion
MFC	Mel-Frequency Cepstrum
Micro-CT	Micro-Computed X-ray Tomography
MJ	Material Jetting
MPCA	Multi-Linear Principal Component Analysis
NIST	National Institute of Standards and Technology
NLP	Natural Language Processing
NSOM	Near-Field Scanning Optical Microscopy
OHE	One-Hot Encoding
PAM	Plasma Arc Additive Manufacturing
PBF	Powder Bed Fusion
PCA	Principal Component Analysis
PNG	Portable Network Graphics
RAM	Resonant Acoustic Method
RF	Random Forest
RL	Reinforcement Learning
RNN	Recurrent Neural Network
ROI	Region of Interest
ROS	Random Over Sampling
RUS	Random Under Sampling
SAX	Symbolic Aggregate Approximation
SDL	Spare Dictionary Learning
SEM	Scanning Electron Microscopy
SFT	Symbolic Fourier Transform
SIFT	Scale Invariant Feature Transform
SL	Sheet Lamination

SMOTE	Synthetic Minority Oversampling Technique
SP	Signal Processing
SR	Speech Recognition
STC	Short-Time Correlation
STD	Standard Deviation
STFT	Short-Time Fourier Transform
STL	Standard Tessellation Language
SVD	Singular Value Decomposition
TC	Thermocouple
TIFF	Tag Image File Format
TL	Transfer Learning
VAE	Variational Autoencoder
VLAD	Vector of Locally Aggregated Descriptors
VP	Vat Photopolymerization
VPCA	Vectorized Principal Component Analysis
WT	Wavelet Transform
XCT	X-ray Computed Tomography
XRD	X-ray Diffraction
ZCR	Zero Crossing Rate

Chapter 1
Introduction



1.1 Additive Manufacturing: Motivation, Challenges, and Potential Solutions

Contrary to the subtractive approach of material removal, additive manufacturing (AM) or three-dimensional (3D) printing fabricates part in a layer-upon-layer fashion [1]. Its default and added benefits include tool elimination, material savings, design freedom, cost reduction, part consolidation, prototyping ease, mass customization, and production efficiency. These disruptive AM features have a huge economic potential [2]. While recent advancements have resulted in some level of commercialization for the AM technology, it is still far from reaching its true potential. A critical bottleneck toward the realization of AM potential is the existing lack of process reliability which hinders its widespread commercial adoption [3]. For instance, metal AM (MAM) can fabricate metallic parts from 3D designs and has the potential to replace the conventional manufacturing at the industrial scale. It however faces challenges that are caused due to its distinctive nature of simultaneous material-part fabrication where both material and part are designed at the same time [4] and are often not mature enough to meet stringent consumer constraints on quality, strength, and aesthetics. While knowledge on AM and its associated technologies (e.g., welding) has existed for a long time, there still remains a lot of engineering know-how to discover to meet the industrial push. As a result, AM has a rich research landscape with multi-disciplinary efforts in software, hardware, and material's development being made to make it a better competitor to its mature counterparts.

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American Society of Testing Materials (ASTM) defines seven major categories of AM as material extrusion (ME), powder bed fusion (PBF), material jetting (MJ), binder jetting (BJ), vat photopolymerization (VP), sheet lamination (SL), and directed energy deposition (DED) [5]. Interested readers are referred to the introduction by [1] where each of these techniques are described and their merits are compared with each other. These processes have found unprecedented applications in aerospace [6], medical [7], construction [8], energy [9], and tons of consumer products. PBF and DED, the main representative MAM[1] processes, are being used to manufacture parts not possible with conventional subtractive approaches [3]. However, the large-scale production of functional parts has not been realized. Zhu et al. identified macro (e.g., geometric deviations) and micro (e.g., porosity, cracks) mechanical defects as major obstacles to producing high-quality metal parts [10]. ME techniques, fused-filament fabrication (FFF) and fused deposition modeling (FDM), are typical for plastic-based component fabrication [1]. These too involve several challenges impeding their total acceptance. Oleff et al. provided a summary of defects that are commonplace in ME [11]. These defects include bubbles, overfills, scars, underfill, warpage, shrinkage, and incorrect bead deposition position. Smart, reliable, pragmatic, scalable, and economically feasible solutions to these challenges are urgently needed to make AM defect free and an industrial reality.

The potential solutions to existing AM challenges can take different forms. AM knowledge can be developed through extensive experimental studies aimed at refining design, process, and property windows [4]. A trial and error-based random exploration as such is extremely expensive and has limited applicability in commercial setups. The complex physical phenomenon (e.g., heat transfer, mass transfer, phase transformation, and so on) of AM processes can be modeled through representative numerical or analytical techniques [12, 13]. These domain or physics-driven approaches can significantly expedite process development with respect to key characteristics of concern. Although physics-based approaches bring interpretability and are superior to their model-free predecessors, their inherent limitations of computational expenses, oversimplistic assumptions, and longer run times might overshadow the potential gains [14]. In the last decade, an increasing proportion of AM researchers and scientists has taken the alternate route of empirical models for solving complex AM challenges [15]. These include both conventional statistical tools as well as advanced machine learning (ML) and deep learning (DL) approaches. This category of models is referred to as data-driven for the remainder of this text and is extensively discussed in the subsequent portion of this introduction.

[1] MJ and SL are other AM techniques which could be used to manufacture metallic parts.

1.2 Status of Data-Driven Additive Manufacturing

Data-driven models learn from data and make automated predictions on tasks of interest. The "data", the "learning process", and the "tasks of interest" largely determine the nature of these models. It should be noted that the term "data-driven" is being used to categorize a broad range of models. In the field of artificial intelligence (AI), these can be simple machine learning (ML) models trained on relatively small datasets (e.g., Gaussian processes) to make informed decisions [16]. These can also be DL models with sophisticated architectures (CNNs, RNNs, and so on) which learn from big data and perform well on relatively complex tasks [17]. The learning process can be supervised with the provision of targets (regression for numerical outputs and classification for categorical outputs) to which inputs can be mapped by discovering functions or representations in the form of model parameters (or weights). On the other hand, unsupervised learning (mostly clustering in the case of AM) can discover patterns from the data in the absence of targets (or labels). Reinforcement learning (RL) models are also driven by data and learn to take a sequence of decisions which can optimize a custom-designed reward or penalty function [18]. Simple statistical models based on small data (as opposed to big data[2]) are also part of the data-driven paradigm [19]. Simply put, the patterns and information in data can be exploited to develop models for AM-specific tasks. Since these models are based on data of AM which follows the established flow of process–structure–property in materials science, we shall arrange these applications accordingly, by affixing design to it (e.g., design–process–structure–property or DPSP), in the later sections on this text.

The empirical nature of data-driven approaches has numerous advantages over physics-based analytical or numerical techniques. First, they are computationally efficient at the expense of absolute correctness. So much so that sometimes ML is used to speed up simulations while maintaining acceptable accuracy [20]. Second, these methods have a natural synergy and temporal overlap with theme "Industry 4.0". This has led to the development of several cloud-based prototypes on industrial AM applications and to advances on sophisticated concepts such as digital twin (DT)-driven AM [21]. Third, data-driven models are usually domain independent and do not require extensive domain knowledge to start with. This may change with the emerging paradigm of physics-driven ML where architectural components of model design can be domain-inspired [22, 23]. Fourth, their efficient prediction ability makes them an ideal candidate for on-the-fly deployment which is an active research area under the banner of in-situ or in-process smart monitoring [10]. Fifth, data-driven models are capable of knowledge exchanges and updates irrespective of it being implicit (e.g., feature level) or explicit (e.g., decision level). The adaptation aspect is evident from several examples in the parent domain of AI [24]. This enables such models to be adapted to new scenarios and applications (e.g., change of machines, materials, or geometries) with significantly different data distributions. These attributes have made data-driven approaches a popular choice in AM as depicted by the upward trend in Fig. 1.1.

[2] Big data is usually categorized based on its volume, velocity, variability, variety, and value.

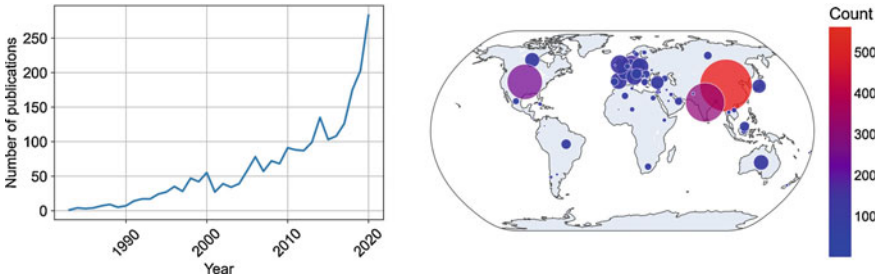


Fig. 1.1 Published literature and corresponding geographical distribution for data-driven manufacturing in the last four decades. Data collection details can be found in [23]. Figure used with permission from Elsevier

The plethora of AM literature inspired by the strengths of data-driven approaches has resulted in several efforts to summarize its status. These can be general or specific with a focus on AM or data-driven aspects. Reviews with broader scope provide an overall status of the field with key research directions. Specific reviews, on the other hand, focus on subtopics in AM, AI, or both. The bulk of the existing surveys fall in the category of being specific. AM subtopics include process [25], material [4], application [26], or system. AI subtopics have focused either on the data [27] or the learner [28]. At the timing of writing this text, we have reviewed over fifty surveys that are directly (e.g., AI in AM) or indirectly (e.g., smart monitoring) related to data-driven AM. These are summarized in Table 1.1. Reviews by Qin et al., Tian et al., and Meng et al. are representative efforts to provide a general summary of the status of AI in AM [15, 29, 30]. AI-driven smart monitoring is the most prevalent category among specific review topics. In this regard, Lin et al. and Zhu et al. have comprehensively focused on reviewing MAM condition monitoring [10, 26]. Another contribution in summarizing data-driven techniques in AM has been provided by Johnson et al. with a specific focus on material's development [4]. An algorithmic equivalent of their work is the recent review on data-driven process, structure, and property modeling by Wang et al. [31]. Some researchers have analyzed the existing literature through the lens of data-driven techniques and provided a summary of the most important aspects. Qi et al. reviewed neural network (NN)-based AM applications [32] while Joshi et al. summarized supervision-based learning in AM [33]. A data-centric review was also conducted where image-based monitoring of PBF processes was thoroughly considered [27].

Table 1.1 Summary of existing reviews on data-driven AM with their respective featurization levels[3]

Review category	Review subtopic	Focus of review	Featurization level	References	Year published
Specific	**AM data**	**"AM data for ML applications"**	**High: data for ML**	[34]	**2022**
General	NA	"Research and application of AI in AM"	None	[15]	2022
General	NA	"Non-destructive evaluation"	None	[35]	2022
Specific	CNN	"Model architecture and applications in AM"	Medium	[28]	2022
Specific	**LAM**	**"ML algorithms for defect detection"**	**High: data fusion**	[36]	**2022**
Specific	MAM	"Physics-informed ML"	Medium	[37]	2022
Specific	**MPBF**	**"In-process monitoring and control"**	**High: feature extraction and feature selection**	[38]	**2022**
General	**NA**	**"Data-driven modeling of process, structure, and property"**	**High: input feature types**	[31]	**2022**
Specific	**MAM**	**"Condition monitoring methods"**	**High: feature extraction methods**	[26]	**2022**
Specific	FDM	"Applications of ML in FDM"	Medium	[25]	2022
General	**NA**	**"Recent trends of ML applications in AM"**	**High: multi-sensor data**	[39]	**2022**
General	NA	"Digital image correlation-based monitoring in AM"	Low	[40]	2021
Specific	**PBF**	**"In-situ monitoring"**	**High: image processing**	[41]	**2021**
Specific	**WAAM**	**"Defect types, detection, and data fusion"**	**High: data fusion**	[42]	**2021**
Specific	**MAM**	**"ML for condition monitoring in MAM"**	**High: feature extraction**	[10]	**2021**
Specific	MAM	"In-situ monitoring"	Low	[43]	2021

(continued)

[3] The featurization levels have been explained in the subsequent text.

Table 1.1 (continued)

Review category	Review subtopic	Focus of review	Featurization level	References	Year published
General	**NA**	**"Image-based monitoring"**	**High: image processing**	[27]	**2021**
Specific	LPBF	"In-situ monitoring"	Medium	[44]	2021
Specific	MAM	"Quality control methods"	Medium	[45]	2021
Specific	LPBF	"Monitoring and control"	Medium	[46]	2021
General	NA	"Data-driven approaches in smarter AM"	Low	[29]	2021
General	NA	"ML in AM"	Low	[47]	2021
General	NA	"ML for predicting mechanical behavior in AM"	Medium	[48]	2021
General	NA	"DT and AI in AM"	Low	[49]	2021
Specific	**MEX**	**"Process monitoring"**	**High: data handling techniques**	[11]	**2021**
Specific	LPBF	"Perspective of machine learning"	Medium	[50]	2021
General	NA	"ML for error compensation in AM"	Low	[51]	2021
General	NA	"AI-based methods and future perspectives"	Low	[52]	2020
General	**NA**	**"In-situ monitoring through acoustic techniques"**	**High: data processing techniques**	[53]	**2020**
General	NA	"ML status and perspective"	Medium	[54]	2020
Specific	FFF and DED	"Online monitoring and closed-loop control"	Low	[55]	2020
Specific	DED	"Sensing signals, process signatures, and monitorable qualities"	Medium	[56]	2020
General	NA	"DT status and perspective"	Low	[57]	2020

(continued)

Table 1.1 (continued)

Review category	Review subtopic	Focus of review	Featurization level	References	Year published
General	NA	"Non-destructive quality control"	Medium	[58]	2020
General	**NA**	**"AI for materials development in AM"**	**High: data-specific featurization**	[4]	**2020**
General	NA	"ML for advanced AM"	Medium	[59]	2020
General	NA	"ML in AM review"	Medium	[30]	2020
Specific	**NN**	**"NN applications in AM"**	**High: featurization challenges**	[32]	**2019**
Specific	DED	"In-situ monitoring and characterization"	Low	[60]	2019
General	**NA**	**"Review of ML applications in AM"**	**High: data oriented**	[61]	**2019**
Specific	DED	"Online monitoring techniques"	Low	[62]	2019
Specific	SL	"Supervised learning in AM"	Low	[33]	2019
Specific	PBF	"ML applications"	None	[63]	2018
General	NA	"Big data-based ML applications in AM"	Low	[64]	2018
General	NA	"AI in AM overview"	Low	[65]	2017
Specific	MAM	"Process monitoring and inspection"	Low	[66]	2017
Specific	DED	"Online monitoring and control"	Medium	[67]	2017
Specific	MPBF	"Measurement science needs for real-time control"	Low	[68]	2017
Specific	MAM	"In-situ monitoring and metrology"	Low	[69]	2016
Specific	PBF	"In-process sensing"	Medium	[70]	2016
Specific	DED	"Sensing and control systems"	Low	[71]	2015
Specific	MAM	"Applications of thermal monitoring and control"	Low	[72]	2014

The highlighted reviews have been identified as having high level of featurization discussions while no review has focused on data preparation in AI-driven AM

1.3 Why Feature Engineering?

A major part of developing data-driven models is related to data handling. The data handling operations can range from initial data collection or acquisition to the point where data (or its transformed form) is finally used to make predictions. The gray area between these steps contains numerous conventional as well as novel and AM-specific techniques which play a significant role in the success of data-driven methods. These techniques are greatly dependent on the nature of the data (e.g., domain, scale, computer representation) and the task for which data is being prepared (e.g., shallow vs deep learning, supervised vs unsupervised). In the field of AI and especially ML, the term "feature[4]" is used to describe an individual measurable property or characteristic of a phenomenon [16]. Some AM researchers also defined the feature as "a measurable property that can be quantified and recorded [73]". We will use this term to refer to any processed form of raw data contributing to the learning process in the digital pipeline of AM informatics. These feature forms can be implicit or explicit, learned or designed, extracted or selected, low-level or high-level, real-world or synthetic, and temporal, spatial, or both depending on data sources and processing methods used to "engineer" them. Ling et al., while developing data-driven model from microstructural images, defined featurization as "the process of transforming the raw input data into more meaningful or information-rich features that can be used for machine learning [74]". In ML, great attention is given to engineering features that can help improve the quality of learning tasks in a direct or indirect manner. Feature engineering is an umbrella term used for all processing techniques in support of data-driven AM solutions. Figure 1.2 highlights the scope of feature engineering in a typical data-driven pipeline. A refined categorization of these methods will be presented in Chap. 2 as some AM-specific (e.g., data registration) and advanced learning methods (e.g., representation learning) are also considered within the scope of this text.

The featurization environment in AM is becoming increasingly rich. The upper level of their complex taxonomy can be best captured by following the well-known DPSP model. Some novel and representative featurization efforts are highlighted here. For example, Gu et al. encoded voxels of 3D printed materials (e.g., digital material (DM)) as input features for the CNN model trained to predict the toughness of a composite DM [76]. In structural design, Wang et al. optimized topology for composite microstructures by learning a physics-dominant latent space of geometric features with inputs directly mapped to responses of interest [77]. Lin et al. proposed a melt pool motion feature for smart monitoring in laser PBF processes [78]. Zhao et al. learned sparse representations of key melt pool features in DED for automated and unsupervised anomaly detection [79]. At the structure phase, an interesting effort to classify SEM images by mean featurization of CNN layers was reported by Ling et al. [74]. Recently, in an effort to segment carbide particles from SEM images of an additively deposited composite, feature freezing was used to fix transferred

[4] Synonymous of "variable" or "attribute".

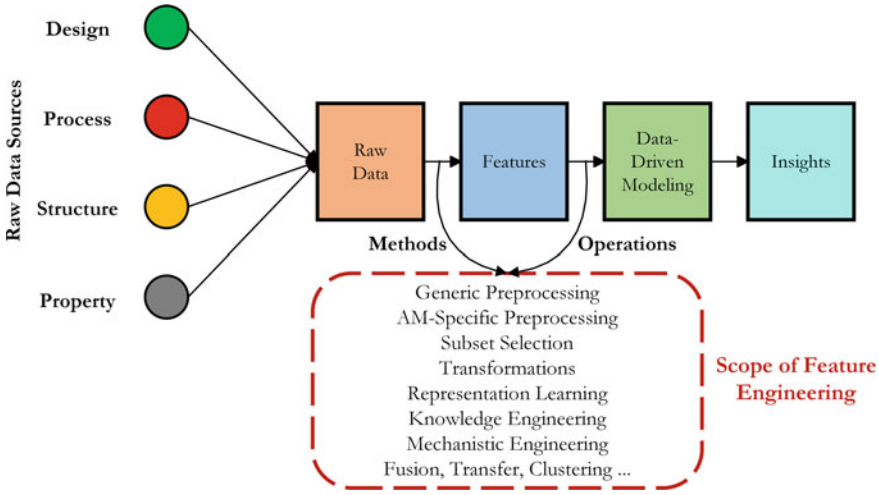


Fig. 1.2 Scope of feature engineering in data-driven AM. The identified feature engineering techniques for AM raw data are explained in a subsequent chapter. The figure is repurposed for AM from Zheng and Casari's book on feature engineering [75]

ImageNet features in the encoder part of a U-Net model [80]. Li et al. extracted time–frequency features from monitoring data and later used RF-based feature selection to accurately predict surface roughness of parts in extrusion-based AM [81]. Chan et al. predicted the cost for additive manufacturing by extracting geometric and non-geometric features directly from G-Code [82].

In addition to known techniques of feature engineering, AM domain expertise play a significant role in processing raw data. To the best of the author's knowledge, there exist no reference text encompassing these techniques in detail. Roughly, these are grouped under AM-specific preprocessing (mapping, alignment, fusion, registrations, and so on) and AM-proficiency-driven (e.g., knowledge engineering, mechanistic engineering) feature engineering in this text. Overall, preparation of AM data for ML applications is the backbone of data-driven AM. These techniques fundamentally transform raw data into new spaces or representations making the learning process much more efficient and effective. Each application of data-driven AM involves feature engineering in one form or another. Since AM is a multi-disciplinary engineering field lying at the intersection of several domains and paradigms, a comprehensive overview of featurization is needed to identify useful techniques (w.r.t. lifecycle phases, and/or applications) and existing research issues. In this regard, we cover data preparation techniques used in the published AM literature by systematically arranging them according to AM lifecycle phases and identifying major sources of AM features.

1.4 Review Specifics

To begin the survey of feature engineering in AM, state-of-the-art reviews of Table 1.1 were evaluated and compared for their featurization content. A four-level featurization spectrum was used for this purpose. The reviews with no featurization text were put in the "none" category. Reviews with useful discussions on data-driven AM but no direct mention of feature engineering were ranked "low". Reviews with the presence of feature engineering relevant text were ranked "medium". While no review had feature engineering as its focus, some works have presented specific aspects of FEPs for the scope considered and these reviews were ranked "high" on the featurization spectrum. Out of fifty or more selected AM reviews, fourteen were categorized in the "high" category. The common themes among these aspects were multi-sensor data and its fusion, data handling, feature engineering techniques (FETs), and feature types. Johnson et al. highlighted data-specific featurization in their comprehensive review on material's development [4]. Qi et al. presented featurization challenges while surveying NN applications in AM [32]. The remaining reviews of the high-level category were either related to general AM data [34] or its specific types [27]. The significance of FETs in data-driven AM is evident from their growing popularity. It was also interesting to analyze the perspective of these AM reviews where the majority have highlighted feature engineering-oriented challenges in one way or another (e.g., ML enabled data processing). However, these perspectives are not reviewed in this text.

In the absence of an AM-focused feature engineering text and the past reviews on AM informatics [83] being insufficient in the data-driven paradigm, a comprehensive effort is made to provide a systematic summary of FE. Specifically, this text contributes to:

1. Summarize existing state-of-the-art reviews on data-driven AM for researchers, small and medium enterprise (SME) engineers, and AM learners.
2. Introduce FE ecosystem in data-driven AM.
3. Present a comprehensive review of existing FE techniques in data-driven AM following the well-known design–process–structure–property paradigm.
4. Identify major sources, engineering techniques, engineering pipelines, and corresponding applications of features at each AM lifecycle phase.
5. Provide useful perspectives on FE for future research as well as its role in the adoption of data-driven AM.

Scopus being the world's largest abstract and citation database for scientific literature was used to collect peer-reviewed articles to summarize feature engineering techniques in AM [84]. The review methodology is detailed in Fig. 1.3. Key aspects of data-driven AM research were first identified. These aspects were then transformed into representative keywords for Scopus search. "Additive manufacturing" was taken as the root keyword, and following subkeys were used with it: "data-driven", "artificial intelligence", "machine learning", "deep learning", "monitoring", "data fusion", "feature engineering", "computer vision", "modeling", "metal", "supervised learning", "unsupervised learning", "clustering", "data processing", "advanced", and

"knowledge". A two-level filter search was done where "additive manufacturing" was kept as the root keyword and remaining keywords were applied iteratively. Each search resulted in one group of articles. The resulting groups were scanned for inclusion and exclusion criteria after being examined for repetitions. These criteria were taken and modified from the recent review of Qin et al. where research and applications of ML in AM were summarized [15]. Specifically, the exclusion criteria are modified to consider papers where machine learning may be absent, but AM data is processed in its support (e.g., AM data preprocessing, knowledge engineering). The inclusion criteria were left unchanged since these consider techniques that can improve the performance of existing ML technologies and feature engineering falls under this definition. The resulting literature is finally checked to identify any article not fitting with the chosen criteria (e.g., monitoring setup instead of monitoring algorithms). A total of 264 articles were found to either have direct application of ML in AM or processed AM data in support of ML.

The rest of the text is divided into five chapters: Chap. 2 introduces the extent of FE in data-driven AM and provides a reference chart for researchers and practitioners alike by systematically structuring the existing FE techniques across the AM digital thread. Chapter 3 summarizes state-of-the-art featurization efforts in data-driven AM. There works are arranged in pre (design), in (process), and post (structure, property, and performance) process categories. Representative applications are highlighted in each subsection. Chapter 4 analyzes trends for insights into design, process, and post-process feature spaces. Chapter 5 provides future perspectives on FE in data-driven AM by discussing challenges and potential opportunities. Chapter 6 concludes the review.

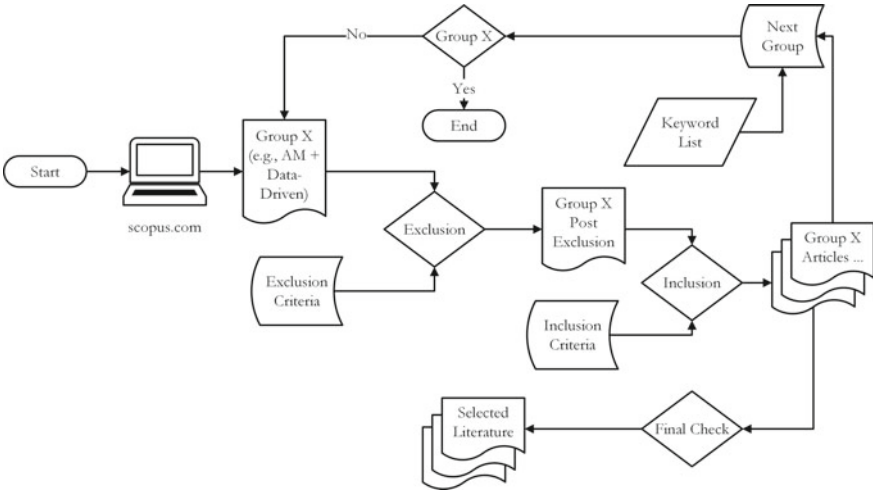


Fig. 1.3 Methodology used to collect literature for the review

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Chapter 2
Feature Engineering in Additive Manufacturing



2.1 Domains and Paradigms

It is challenging to present a unified view of feature engineering in data-driven AM as it lies at the intersection of multiple domains (signal processing (SP), computer vision, 3D or geometric ML, informatics, statistics, AI/ML/DL) and paradigms (e.g., integrated computational materials engineering (ICME), data-information-knowledge-wisdom (DIKW), DPSP). The text is certainly not an effort to provide a self-contained and comprehensive introduction to feature engineering. Rather, it systematically identifies specific feature engineering techniques that are pertinent for AM data types and tasks. This is done by organizing feature engineering techniques according to AM digital thread (e.g., information components across product lifecycle phases) and associated data types. The life cycle of an AM process follows the DPSP where data is exchanged between subsequent phases [1]. In addition to these exchanges, new data is generated at each step which could be added to the existing digital thread. This information-rich thread governs data-driven solutions. Researchers at the National Institute of Standards and Technology (NIST) have made initial efforts to organize the flow [2] and semantics [3] of AM digital thread [4]. Kim et al. proposed a system for streamlining information flow throughout the AM product realization process [2]. Their end-to-end digital thread contains all phases and highlights key information transactions among all phases (e.g., design-process, process–product). These phases can be analyzed for data types to base FETs upon. Lu et al. derived a four-tier self-improving AM knowledge management framework from the well-known data-information-knowledge-wisdom (DIKW) model [3]. The

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framework semantically organized AM knowledge components. Roughly, data represents raw data (e.g., signals, images), information corresponds to processed or organized data (e.g., signal or image features), knowledge is based on useful information (e.g., defect-prone feature), and application is driven by knowledge-specific use cases (e.g., fault detection). Accordingly, feature engineering in AM concerns information layer but is driven by data and relates to knowledge, and respective applications. We will structure feature engineering ecosystem in AM according to this hierarchy.

Figure 2.1 represents the overview of feature engineering ecosystem in AM where vertical direction has semantic levels and horizontal direction follows AM lifecycle phases. The rest of the activities and techniques are arranged within. It should be noted that Fig. 2.1 does not provide a comprehensive list of all feature engineering techniques possible. In fact, feature engineering is often left out of the most ML texts and books as pointed out in [5] which is a compositional text on the topic. Interested readers are referred to this short book which compiles different feature engineering techniques for ML. A similar effort was made to organize data-specific feature engineering techniques [6] and a relatively early text on FETs exists as well [7]. The lower level in Fig. 2.1 highlights data sources corresponding to each lifecycle phase that are arranged according to DPSP linkages. The data coming from these sources can be in different formats. Zhang et al. identified four major AM data types for ML applications as tabular, graphic, 3D, and spectrum, with spectrum data mainly comprising electromagnetic and ultrasonic spectra [8]. There have been other efforts to summarize AM data formats but are not directly concerned with ML applications [9]. We will use sequence instead of spectrum to be generic (e.g., timeseries, sequences, or text) and place collected raw data at the next level. Collected data is not always in a ready-to-engineer format and usually requires preparation that can be generic or AM-specific. This has been identified as the next step in feature engineering pipelines. These data types can be engineered in several ways to help data-driven tasks. Feature engineering level recognizes transformations, selection, learning, integrated, and knowledge-based techniques as major categories of FETs in AM. The specific feature engineering technique used could be from either of these categories and depends on the data being processed as well as the task at hand. The resulting features can be classified as design, process, structure, or property features depending on the life cycle phase of AM data. There is an increasing trend of using data from sources across lifecycle phases where resulting features cannot be classified into one pure category. The next level highlights operations possible with learned, selected, or transformed features. This level corresponds to the knowledge layer of Lu et al. which drives AM applications [3]. The last level corresponds to applications of engineering features.

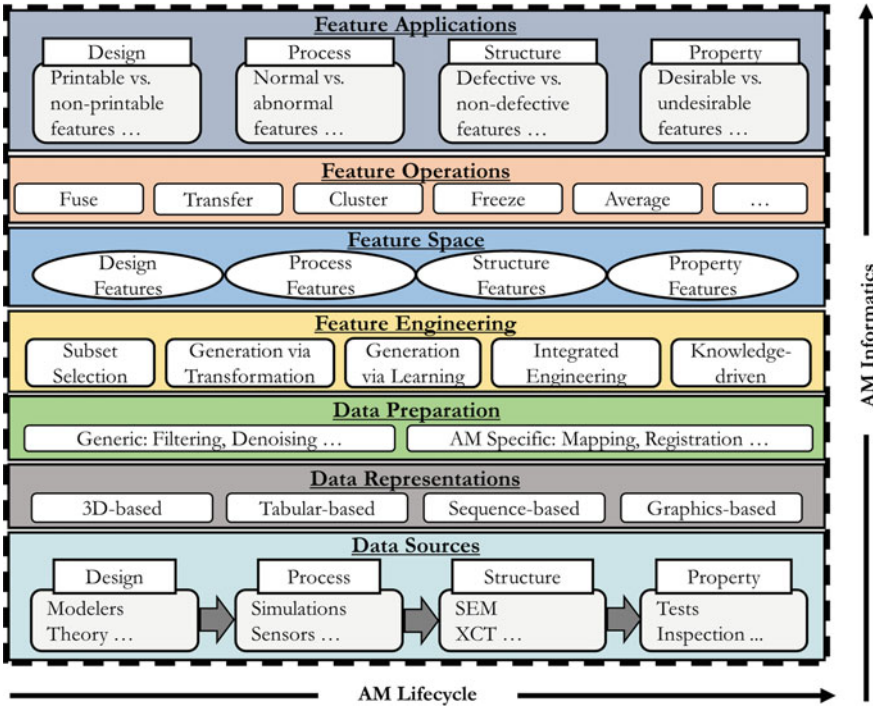


Fig. 2.1 Scope of feature engineering in additive manufacturing with respect to DIKW model. The horizontal axis follows lifecycle of a typical AM process while vertical axis depicts semantic-aware operations and levels

Figure 2.2 highlights major domains associated with feature engineering in AM. This is primarily determined by the representations of raw AM data. In this regard, computer vision, signal processing (SP), geometric ML, tabular transformations, knowledge engineering, and data mining-based preprocessing cover majority of the techniques. Signal processing is concerned with the featurization of timeseries measurements from a physical phenomenon [10]. Computer vision overlaps with AM graphic data transformed through conventional, customized, or advanced algorithms [11]. Geometric or 3D ML is deep learning for 3D data. In AM, it is mostly concerned with the processing of as-designed models, structures, and materials [12]. Tabular transformation (linear or nonlinear) groups most frequent engineering tools for features in tabular representations [13]. Knowledge engineering in AM has several forms. It has been used to organize raw AM data [14] or develop special features from it. Finally, data mining represents generic data preparation [15].

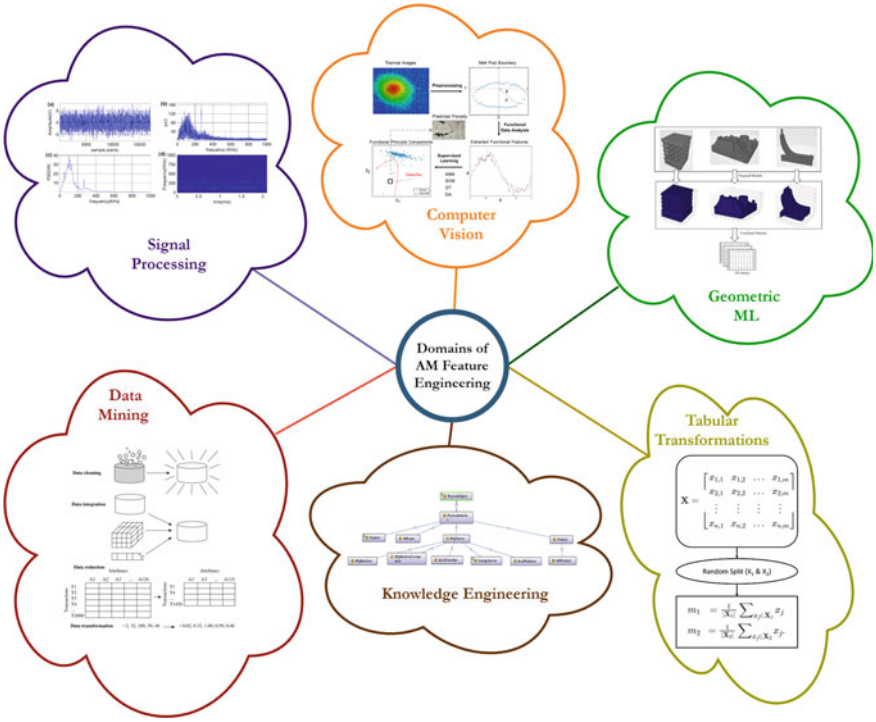


Fig. 2.2 Domains associated with major AM data representations and FETs. Signal processing (used with permission from Elsevier [10]), computer vision (used with permission from Elsevier [11]), geometric ML (used with permission from Elsevier [16]), tabular transformations (used with permission from Elsevier [13]), knowledge engineering (used with permission from Elsevier [17]), and data mining-based preprocessing (used with permission from Elsevier [15])

2.2 Feature Sources

Scientific modeling and measurement science are two main sources of information in data-driven AM. In the recent years, AM has benefited from novel contributions to these fields. Additionally, some improvements (increased sensor and simulation fidelities) were purely inspired by measurement science and modeling needs of AM. In modeling, high fidelity and accurate simulations are needed to generate high-quality data. At the same time, reliable simulations can help validate the results of data-driven solutions. Scientific modeling results in timeseries data (thermal histories) which is usually represented in tabular forms. Sometimes, graphic (e.g., artificial melt pool image) and 3D (e.g., microstructure simulations) data can be obtained

from these surrogate models. Measurement science presents an enriched toolset at our disposal enabling massive datafication of AM. As a result, there are abundant sources of information which results in diverse data types with different representations. The most common sequence data in AM is signals captured in-situ. These timeseries signals describe different physical phenomena and can be obtained from electromagnetic or ultrasonic waves such as their respective spectrums [18]. 1D in-situ signals usually measure temperature or vibration during AM processes. The most exhaustive efforts made to collect in-situ signals have been in MAM where melt pool or layer is the key object to monitor [10]. Text sequences are relatively less common in AM. Figure 2.3 provides an overview of both sources and properties of resulting features.

Graphic data with pixel topology is probably the most common AM data overall. These images can contain information on design (e.g., multi-angle view of CAD models), process (e.g., build platform, powder layers or streams, aerial melt pool, sideways melt pool, under-build part geometry), structure (e.g., SEM, fabricated part)

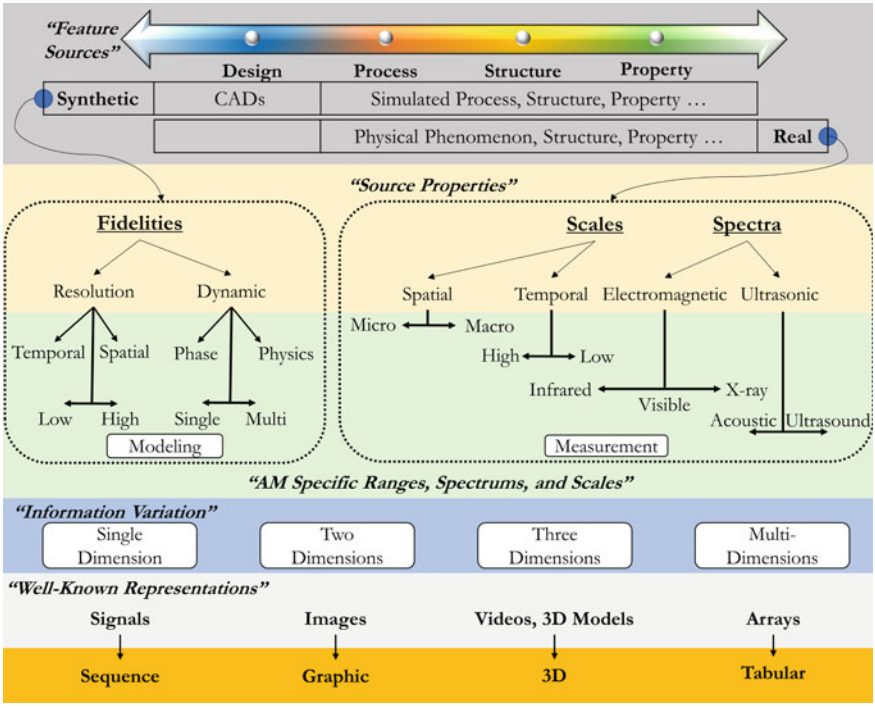


Fig. 2.3 Scientific modeling and measurement science as main sources of AM features

and property (e.g., stress–strain curves). In-situ graphic data mostly captures infrared (IR) or visible (Vis) radiations by cameras based on charged-couple device (CCD) or complementary metal oxide semiconductor (CMOS). Pyrometer-based (e.g., diode arrays) graphic data is commonly produced from thermal monitoring systems as well [19]. In addition, X-ray, simulation, and 3D-based images (e.g., voxel-based) are also available for use in data-driven solutions. 3D data (sometimes called spatial) mainly corresponds to as-designed models from available modeling tools. Further, printed parts can be scanned in-situ or ex-situ resulting in 3D data (e.g., X-ray computed tomography (XCT) or 3D scanners). Tabular data with column-row arrangements can come from numerous sources (raw data, operator feedback, analytical or numerical solutions, product properties) at all phases of AM lifecycle. Reviews on data types [8], data-specific applications [20], condition monitoring [21], and non-destructive evaluations [22] can provide a detailed overview of these sources to interested readers.

2.3 Feature Engineering Techniques

Feature engineering techniques have terminology overlap which could be confusing. Liu and Motaba identified subset selection and transformation as two generic techniques of feature engineering. Their definition of transformation contains both extraction and construction of new features [7]. In a rather recent text on feature engineering, generic feature engineering techniques identified were based on selection, transformation, learning, and pattern [5], with pattern-based techniques being specific to the data type. Zhang and Casari organized feature engineering techniques for ML according to data types [6]. The three techniques are not mutually exclusive. Furthermore, feature engineering has been generally taken as a domain-specific problem with explicit techniques developed for SP, computer vision, speech recognition (SR), natural language processing (NLP), and 3D ML. Data-driven AM has applications of these fields and data types. As such, a generic taxonomy is needed to organize feature engineering in AM literature. This review establishes the existing feature engineering techniques of AM into five categories, namely *subset selection, generation through transformation, generation through learning, knowledge-driven feature engineering, and integrated feature engineering*. Figure 2.4 provides a taxonomy of feature engineering techniques in AM. These techniques are discussed in the next sections, and most popular approaches for each category are introduced.

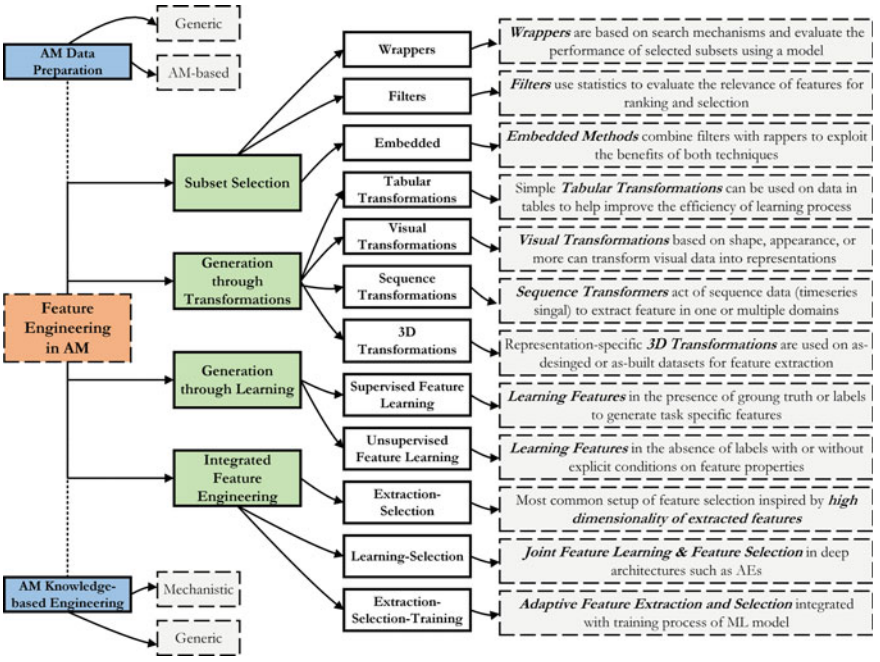


Fig. 2.4 Rough taxonomy of feature engineering techniques in data-driven additive manufacturing

2.4 Generic Data Preparation

The purpose of feature engineering is to improve the quality of the data-driven solution or the efficiency of the learning process by improving available features. In this regard, there are common techniques that are sometimes referred to as feature engineering, data processing, data cleaning, or data handling. These techniques are discussed under the umbrella of preprocessing before the introduction of generic feature engineering techniques. For instance, the way around for missing values is usually imputation where categorical values are imputed by the most common entity and numerical values are imputed by the mean [23]. Existing feature values are sometimes better processed by encoding them into specialized formats. One-hot encoding, where each value is encoded as zeroes (absence) and ones (presence), is the most common approach in this regard [24]. A feature can be split into meaningful components or merged with another feature to improve the accuracy of the learning task. Data instances with extremes or outliers can be either removed or updated with techniques such as imputation. A skewed feature can be transformed to its less skewed form with different transformations such as log transform, power transform, square root transform, and Box Cox transform [25]. Significant variations of magnitude across feature values can be handled by scaling. Normalization (Min–Max) and standardization (Z-Score) are two important techniques in this regard. Continuous

data values are sometimes discretized to improve ML performance. Discretization is popular for many AM data types such as binning of tabular data, temporal subsequencing in signals, segmentation (pixel grouping) of images, and voxelization of 3D data. Some preprocessing techniques are specific to images. These may include cropping, resizing, rotating, binarizing, denoising, smoothing, blurring/unblurring, simple segmentation, or flipping of images. Han et al., in their book on data mining, provide a chapter (third) on data preprocessing which covers most of AM-based data preparation techniques [15]. They categorized data preprocessing into four types (cleaning, integration, reduction, and transformation) with a note of caution that these are not mutually exclusive. Another useful resource on preprocessing is a recent review by Alasadi et al. [26].

2.5 AM-Specific Data Preparation

The regular data preprocessing techniques, discussed in the previous section, assume that representative and accurate data is available for the learning task. However, this is not the case, especially in AM. In this section, we summarize AM-specific data processing activities that cannot be treated as typical feature engineering approaches. These techniques process data and elevate its semantic level, making it more suitable for downstream data-driven tasks. More importantly, these techniques highlight critical research issues which need to be pursued for the adoption of data-driven AM. Among these techniques, registration (sometimes mapping or correspondence) of AM data is common and can be done in an inter (structure–design, structure–process) or intra (process–process) phase fashion which is often guided by the application at hand. According to International Organization for Standardization (ISO), registration is an alignment process where data is transformed to a neutral or standard reference [27]. This helps to build correlations among data captured or generated across process phases. Such correlations can help arrange data into ML-ready format (e.g., accurate input–output pairs) or provide a basis for comparison and subsequent characterization leading to ground truth generation [28].

There is a growing trend of registering AM data in support of ML applications. Figure 2.5 highlights the layer-wise image registration against CAD slices [29, 30]. Francis and Bian, while predicting distortions in laser-based AM, registered thermal images with coordinates of distortions (in point cloud representations) [31]. This helped in identifying inputs (nearby stack of images) for the deep learning models trained to predict point-wise distortions. Feng et al. proposed meta-data for the construction of a data repository that could help register multi-sensor data and provide correct correlations for data analytics [32]. A five-step solution to registering

melt pool image data with laser spot position was proposed for spatial correlations between as-designed and as-built models [33]. In an example where data-driven models of AM were developed to characterize design defects, as-built XCT data was registered with slices of as-designed models to bring it into the same coordinate system [34]. In their most recent work on data registration, researchers at NIST published a data registration process to support downstream data-driven tasks [35]. Examples of microstructural data registration are also common such as registration of micro-XCT data with pyrometry data [36] and registration of pores with CAD models [37]. In almost all instances, registration supports subsequent learner by improving the accuracy of registered data. Another aspect of raw AM data is its huge size (e.g., Okaro et al. used downsampling to avoid collecting 400 GB of data per built [38]), particularly at the process phase, as noted by several researchers. Hardware and/or software-induced constraints limit the use of AM big data. As a result, AM researchers and practitioners face the difficult choice of casually discarding data which could be useful for the learning task. Salloum and colleagues at Sandia National Laboratories developed AM data compression method using adaptive Alpert tree-wavelets [39]. Based on the features that should be preserved, they reported one to three orders of magnitude compression applied to simulation and experimental datasets of a laser engineered net shaping or LENS process.

Several approaches, similar to the registration process, have been proposed for AM data preparation. Roh et al. introduced a sensor ontology which related feature sources, feature representations and their associated characteristics of concern (CoCs) through resulting physical phenomenon [14]. The purpose of the study was to find a way where sensor selection can be optimized for a given application. Their insights on statistically important sensor data are useful for arranging the sources of features throughout the DPSP linkages. AM data improvement methods specific to

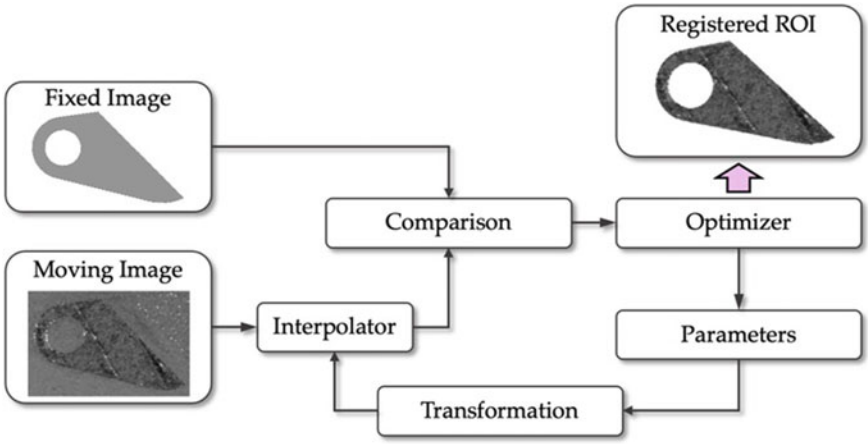


Fig. 2.5 Example of data registration prior to analytics tasks. Image registration for mapping layer-wise images to CAD slices (used with permission from Elsevier [30])

sensors for in-site and ex-situ measurements have been proposed as well. Scime et al. proposed coordinate transformation to improve the utility of melt pool data captured from a high-speed camera with a fixed field of view [40]. Inspired by the variety of data collected during an AM process, Feng and colleagues proposed a generic data alignment process as a precursor to process monitoring and control using data analysis [41]. They identified related data representations (melt pool image to scan path being one such pair), grouped all pairs for coordinate transform, sequenced all pairs, performed transformation, and suggested the use of a persistent identifier to reference resulting chains in the next processing steps. Their work also reviews research on AM sensor data alignment which some readers may find useful. These methods are expected to improve the quality of captured raw data which can make it ready for feature engineering and subsequent ML applications.

In addition to systematic data registration with spatiotemporal considerations, data acquisition, fusion, storage, and management have been considered in support of data-driven models. Yang et al. proposed a multi-scale (point, track, layer, and part-wise) data fusion framework to support the usage of multi-modal in-process data for monitoring and control applications [42]. Data registration was defined as a prerequisite to the multi-scale fusion process resulting in ready-to-fuse representations coming from different sensors. Mies et al. outlined technical requirements for a big data storage architecture while discussing AM information components arranged according to its digital thread (e.g., information across life cycle phases) [43]. Liu et al. implemented a digital twin-based collaborative data management system [44]. The data management system was populated with data model of a MAM process. A deep learner was later integrated into the system for predicting layer defects which validated the usefulness of their data management system. Overall, their system is comprehensive and considers data sources across AM life cycle. Based on the literature review, there are relatively few projects and reported works that have focused on AM-specific data preparation. As data-driven AM matures and R&D activities yield robust ways to ensure high-volume AM production, these solutions are expected to be adopted by SMEs. To facilitate a smooth transition from lab to market, AM data processing methods increasing its readiness for feature engineering and ML applications will come handy.

2.6 Feature Subset Selection

Feature engineering through selection is concerned with the discovery of a subset of existing features that can either improve the performance of the learning task or reduce the computational expense or accomplish both at the same time. As opposed

to generation-based approaches, no new feature space is created. Interested readers are referred to the comprehensive and structured review on feature selection by Fernández et al. [45]. Overall, feature selection techniques fall into three categories: filter, wrapper, or hybrid (embedded). Filter methods (univariate or multi-variate) use statistics to evaluate the relevance of features which results in their ranking and consequent selection of the subset. Popular filter methods in AM include chi-squared test for categorical or nominal data [46], maximal information coefficient (MIC) and Pearson correlation coefficient (PCC) [47], and analysis of variance (ANOVA) [48, 49]. Wrapper methods work directly with the learner and use its performance metric to find the optimal subset through a search mechanism. The search mechanism can take different forms such as random, complete, and sequential search. A common example of wrappers in AM includes genetic algorithms (e.g., NSGA-II) for parameter selection [50]. Hybrid methods, as is implied by their name, combine merits of the previous two approaches and fall between them in terms of efficiency. Hybrid methods use the performance of a learner (as in wrapper methods) but also select the most appropriate features (as in filter methods) based on some penalty. Examples in AM include use of Least Absolute Shrinkage and Selection Operator (LASSO) and tree-based (random forest or RF) [51, 52] feature subset selection (surface roughness prediction in MEX [53]).

2.7 Feature Generation Through Transformation

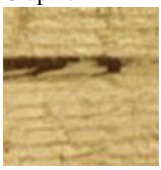
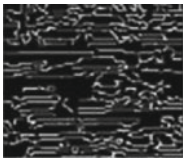
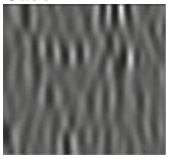
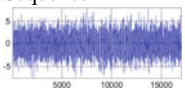
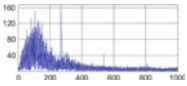
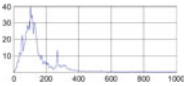
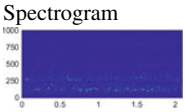
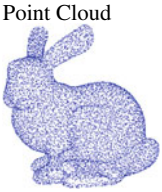
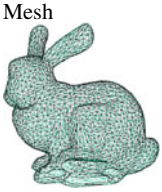
Transformation-based feature engineering involves creation of new features through mathematical or domain-inspired functions (or mappings). Feature extraction, feature construction, and feature design are commonly used to refer to this category of feature engineering [7]. An important distinction between feature transformation and feature learning is that the former does not need to learn transformations and the mappings are usually data specific. Unlike feature selection methods which select a subset of existing features, transformation-based techniques generate new features from raw data and are therefore dependent on its format. As highlighted earlier, 3D, graphic, sequence, and tabular data types are most common in AM [8]. Feature extraction according to each AM data type constitutes a vast field of its own and would not be summarized under the scope of this review. Some of these fields, SP and computer vision, are quite active with research being done for efficient feature extraction in ML applications. Interested readers are encouraged to consult relevant surveys or texts in this regard (feature extraction for computer vision [54] and SP [5]). In AM, the most popular method for feature extraction is principal component analysis (PCA) [55]. The PCA works by reducing the dimension while creating features (or principal components) which capture the maximum variance in the original feature space. Variants of PCA such as multi-linear, functional, or vectorized PCA are also common in AM. PCA is frequently used with tabular and graphic representations but can be applied to timeseries and 3D-based AM representations as well.

In their recent survey on computer vision-based defect detection in conventional manufacturing, Gao et al. identified the following four categories of standard image feature extractors: statistical, structural, filter-based, and model-based [56]. Lin et al. while surveying condition monitoring in MAM found graphic feature extractors to be based on pattern, color, or shape in images [10]. These categories are based on different references as methods to extract visual features can vary. Graphic feature extractors could work in different contexts such as local (point), global (area), appearance (color or texture) or shape (geometry). In addition to spatial considerations, extractors based on frequency domain can be used as well (e.g., Fourier transform). We categorize image feature extractors of AM based on the applications of resulting features and in this regard identify three distinct categories of conventional, advanced, and customized graphic feature extractors. Conventional techniques are based on well-known extractors related to shape, appearance, or pattern. The features extracted from these are not necessarily optimized for the task at hand and may result in a course of dimensionality. Common examples in AM include Scale Invariant Feature Transform (SIFT) [57], Bag of Words (BoW) [58], Gaussian, Gabor [59], Canny, and Sobel [60] image feature extractors. Advanced image feature extractors can generate optimal features directly from the image with prime example being PCA [61, 62]. Customized feature extractors directly capture features of interest to drive empirical solutions. Melt pool motion feature introduced by Lin et al. to monitor and control PDF process is one example of customized image feature extraction in AM [63].

Contrary to the static nature of tabular and graphic datatypes, sequence data can bring information on process dynamics, especially for the in-situ physical aspects. Timeseries signals, the most common sequence data in AM, can be transformed into features using a multitude of featurization techniques. So much so that this richness of techniques inspired a comprehensive effort to unify them based on 9000 different features (with each feature corresponding to one method) extracted from 30,000 scientific timeseries by researchers. These global transformations as summarized in [64] and can be used for timeseries featurization. However, the usefulness of features generated by these methods or algorithms is completely dependent on the problem at hand. This may require subsequent feature selection from the resulting feature vector. Generally, timeseries featurization can be done with respect to time domain, frequency domain, or both. It can be global (whole signal) or local (interval or subsequence) in its scope as well. In temporal domain, simple statistical features (e.g., mean, STD, minimum, maximum) can be locally extracted to compare the signals and be used in tasks such as classification.

Table 2.1 presents a summary of generic transformations commonly used for AM representations. More AM-specific aspects of the transformations will be discussed in the next section.

Table 2.1 Major data representations in AM and associated well-known transformations for feature generation

Major AM data representations	Well known transformations		
Tabular ($X =$) $\begin{bmatrix} 12 & 00 & 25 & 1 & 9 \\ 22 & 13 & -40 & 0 & 85 \\ 90 & 03 & 0 & 11 & -2 \\ 00 & -10 & 2 & 45 & 4 \\ -15 & 25 & 8 & -18 & 0 \end{bmatrix}$	PCA $X = U\Sigma W^T$ or singular value decomposition	Z-score $X' = \frac{x-\mu(\text{mean})}{\sigma(\text{std dev})}$	Min–Max $X' = \frac{x-\min(x)}{\max(x)-\min(x)}$
Graphic 	Canny 	Sobel 	Gabor 
Sequence 	FFT 	PDS 	Spectrogram 
3D 	Voxel 	Point Cloud 	Mesh 

Graphic and sequence data images from [10, 56] are used with permission from Elsevier respectively. Images of 3D Stanford bunny model are taken from [65] with permission from Elsevier

Temporal importance of simple subsequence features may be insufficient to capture the underlying process phenomenon. Therefore, timeseries signals are commonly transformed to frequency domain for analysis and feature extraction. This can be done with the well-known Fourier transform (FT) which generates constituent frequency components of a signal [66]. One common feature of this transformation is power spectral density (PSD) which is obtained by normalizing FT's output. Highly complex and non-stationary AM process results in aperiodic signals which could be better analyzed in a composite fashion (time & frequency domains). This is usually done by first analyzing signal's subsequences individually and then jointly plotting the results in a time–frequency spectrogram. Time–frequency analysis can be done with fixed (with respect to time, frequency or both) and variable transformations. To this end, one common example of fixed transformation is short time Fourier

transform (STFT) [67]. A drawback caused by the fixed nature of STFT is that a randomly chosen window or step size may result in features that are not representative of the underlying frequency distribution. As a result, wavelet transforms (continuous or discrete) or WT, can be used with varying window and step sizes for local time–frequency analysis [68, 69]. Search of the optimal wavelet is an issue that AM researcher may need to deal with. Videos of AM processes (e.g., melt pool, powder bed) represent timeseries signals and can be transformed to generate both spatial as well as temporal features [70].

Engineers (with design or manufacturing backgrounds) may be most familiar with 3D data. Long before featurization of modeled (as-designed) or reverse-engineered (as-built) 3D data started for ML applications, a myriad of research on data structures, formats, exchange standards, evaluations, and schemas was already existing. The list is certainly not exhaustive. Similarly, AI is no stranger to 3D data as it has been used in fields such as medical imaging where 3D data is directly processed in deep models for tasks such as segmentation [71]. Today, there are extensive ML-based applications of 3D data which are often grouped under the sphere of 3D or geometric ML. Eman et al. surveyed deep learning advances on 3D data, providing a summary of both 3D representations and their applications in DL [72]. Unlike graphic data where particular pixel-based[1] format (TIFF, JPEG, PNG) does not influence feature extraction, different 3D representations greatly impact prospects of featurization. It is because images exist in a pixelic grid structure with equidistant sampling, and a transformer (e.g., a convolutional operator) can easily exploit this property to generate image features. It is however not possible in the case of 3D data owing to its unstructured representations. Common 3D representations used in ML are volumetric, point cloud, multi-view images, RGB-D images, and polygonal mesh.

Most prevalent 3D representations in AM include volumetric (source: CAD modelers and XCT), point cloud (source: 3D scanners), and polygonal mesh (source: Standard Tessellation Language (STL) files). Zhang et al. highlighted the dependence of feature recognition on 3D data types while developing an ML-driven manufacturability analyzer of AM designs [73]. 3D volumetric models of AM can be discretized into voxels or octrees for efficient processing by ML models. The discretized models are often made sparse for the same reason. Slicing of volumetric models is another featurization technique commonly found in practice. It is possible that instead of volumetric models, their image(s) are available for use in data-driven models. In cases as such, image-based feature engineering techniques can be used. Point cloud is another common 3D representation in AM which is usually generated from in-situ or ex-situ scanning. Working with point cloud data can be challenging due to its globally unordered structure resulting in ambiguity and making the transformations variant. As a result, there are research efforts focused on noise reduction from point clouds. One notable example of engineering features from point clouds is SO-Net [74]. SO-Net uses self-organizing maps (SOM) and a subsequent feature extraction technique to generate single feature vector for entire point cloud. Finally, polygonal

[1] Vector-based graphic representations may differ from pixel-based representations during featurization.

meshes are often available in AM where the surface of a 3D model is defined in terms of polygons consisting of faces, edges, and vertices. Like point clouds, polygonal meshes also suffer from the lack of invariance characteristic. Their local patches are usually defined to be processed by feature extractors [75]. A simple feature extraction technique for these models is to use normal vectors (magnitude and direction) of each polygon in a single feature vector. It is also possible to featurize 3D volumetric models w.r.t. their constituent features which is common in primitive or part-based geometric DL [76]. Finally, there are sophisticated featurization pipelines that exist for 3D AM data and will be discussed in the respective sections of feature engineering applications in AM.

2.8 Feature Generation Through Learning

Feature engineering practices discussed so far may be limited, in one way or another, in their ability to extract expressive features for data-driven tasks. For example, a transformation defined for a given 3D representation (polygonal mesh) will be inherently limited in its ability to capture discriminative features from another 3D representation (point clouds). Also, in the case of customized graphic transformations, significant effort goes into the design and development of filters. For this reason, feature selection and feature transformations are sometimes referred to as hand-engineered (or hand crafted) techniques. In contrast, feature engineering through learning can automatically learn the transformations which generate representative features (or representations) to aid ML tasks. Goodfellow et al. defined a good representation as one "which can make the subsequent learning task easier" [77]. This automatic learning of features can be done in a supervised or unsupervised manner which forms the basis for their classification in this review. Other classification criteria also exist as described by [78]. The strength of deep learners in gradually learning discriminative features has made this category of feature engineering extremely popular in fields such as computer vision. Bengio et al. summarized the overall status of feature learning with a focus on DL-based techniques [78]. Recently, some of these feature learning techniques have found their application in AM. It should be made clear that we will only discuss those techniques that are used to engineer features and skip regular deep architectures (e.g., CNN) which have been widely reviewed by AM researchers as identified in Chap. 1. In this regard, distinct scenarios are identified where DL is deployed for feature learning in AM.

Feedforward and supervised networks that are commonly found and have been used in several AM applications learn representations to perform the task (classification such as defect detection or regression such as dilution prediction) at hand. This could be thought of as learning representations toward the end of the network in a way that helps with the prediction task (separable representations for classification). In this way, the features of the last layer(s) could be used in another simpler model such as K-nearest neighbor (KNN) if the task requires classification [79]. Doing so would increase the interpretability of features being learned, features that are usually hidden

under the black box nature of deep learners. There are not many examples of this approach in AM. In a related case, features learned from the images of melt pool were extracted from a CNN and fused with deep features from an encoder for segmenting pool from plasma arc [80]. The leading efforts of feature learning in AM have been in the absence of ground truth or labels and hence belong to the category of unsupervised feature learning. There are several types of unsupervised feature learning techniques but autoencoders (AEs), restricted Boltzmann machines (RBMs), independent component analysis (ICA), and transfer learning[2] (TL) methods are most common [77]. Figure 2.6 shows AE-based feature learning at the design, process, and structure phases of AM processes. In all three examples, feature learning also helps in reducing the high-dimensional input space.

Regular architectures have been modified to learn specific features which represents an intersection of feature learning and AM knowledge-based (KB) engineering. Zhang et al. engineered a hybrid deep setup for utilizing spatiotemporal features of melt pool video stream [81]. A CNN was first used to learn spatial features of each image frame. The learned features were then sequentially arranged to account for temporal features. The resulting 2D array was employed in a second CNN for prediction task. They used a sliding window to group melt pool frames and justified its size and sliding step based on AM knowledge (solidification time) and hardware setup (capture rate). Similarity, a multi-scale CNN was developed to learn features of powder bed at different resolutions [82]. Feature engineering through learning has several notable benefits. Learning representations with deep networks enable imposing explicit conditions that may result in features with desirable properties. One type of AE, conditional variational AE (CVAE), learns features conditional to the provided constraints [83]. Representation learning also enables engineering of features from large unlabeled sets of data first and then using these representations with labeled portion of the data. Such setting is occasionally referred to as semi-supervised learning (SSL) [77]. In a related example, Fathizaden et al. used deep representation learning of melt pool images to annotate unlabeled data using learned features which could be a useful application of feature learning in AM [84].

2.9 Knowledge-Driven Feature Engineering

In the parent domain of AI, knowledge engineering has been an old and fierce competitor to machine learning. At one point in the 1980s, when machine learning suffered at the hands of low compute power, knowledge engineering appeared to be on the verge of taking over the world. However, big data combined with efficient processors caused the tide to turn in the favor of ML and its unstoppable rise ensued. Programming of knowledge by human experts to drive specific applications is prevalent in AM as well. The KB operations discussed here are solely under the

[2] Some texts make distinction between different types of transfer learning (such as domain adaptation).

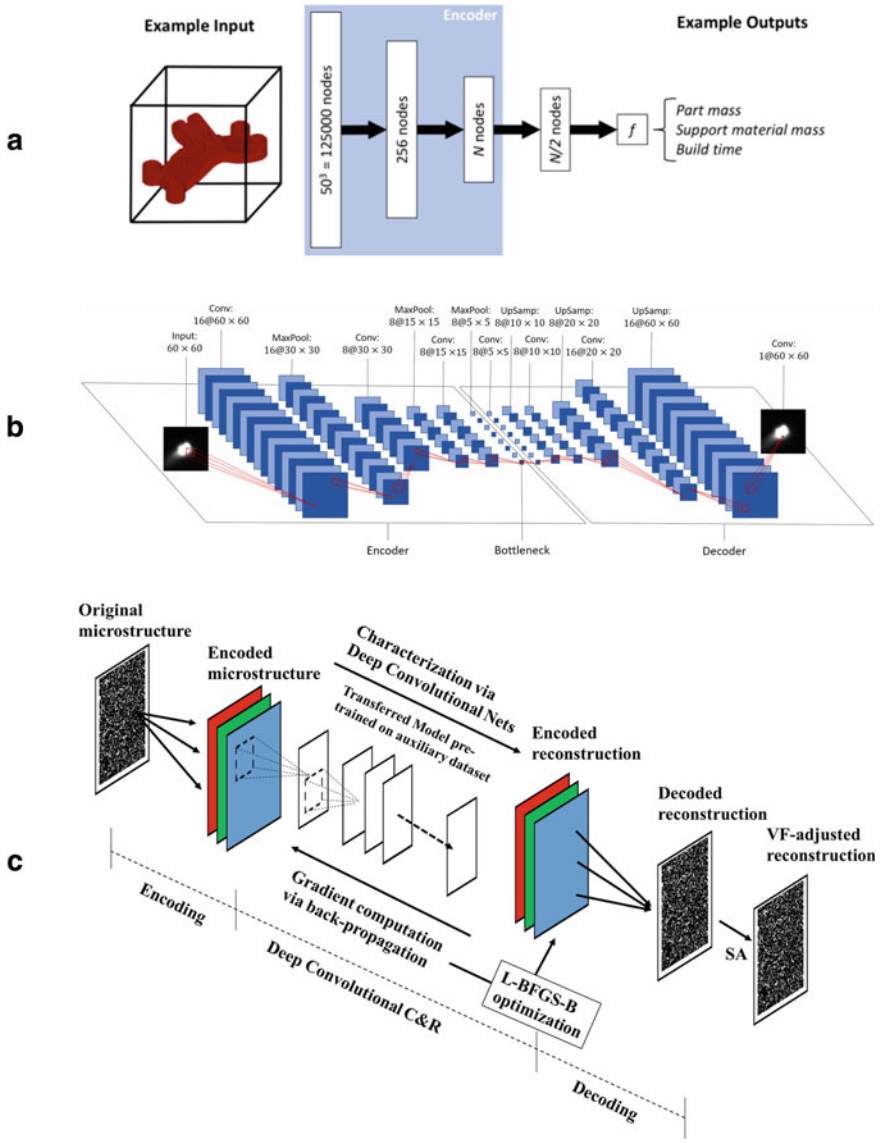


Fig. 2.6 Examples of feature learning from high-dimensional design (**a**) [85], process (**b**) [84], and structure (**c**) [86] spaces of AM. The respective figures are used with appropriate permission from SFFF, Elsevier, and Nature

scope of this review (e.g., improving raw data for data-driven applications). Outside of this scope, there exist operations that are focused on the representation (e.g., design [87], process [88], or planning [89]), management [3], composition [90], and transfer [91] of AM knowledge. In another application, knowledge has been used to identify data-driven opportunities in AM [92]. For some applications such as knowledge mining, AM knowledge serves as input to data-driven tools opening the possibility of its featurization [46]. Authors expect similar research trends to follow given the frequency in which knowledge (implicit or explicit) is being generated from data-driven solutions.

Authors identified two broad categories of knowledge-driven feature engineering in AM. The first category is based on the knowledge of physical mechanisms in AM which could be used for feature engineering. In their survey on mechanistic-AI in advanced manufacturing, Mozaffar et al. pointed out that such knowledge of physics can be embedded in the raw data to convey the process insights [93]. They also introduced physics-informed model development which is outside the scope of this review. Mechanistic data processing was divided into mechanistic feature selection and importance analysis and utilization of data invariants. The first technique is based on the applications of AM knowledge in identifying features which could be useful for a given task. Once such features are identified, their sensitivity for the given task can be evaluated using wrappers or filters leading to the selection of the most critical features. Utilization of data invariants involves transforming raw data in a way, where known unimportant (or irrelevant) correlations are ignored. This can be done by either augmenting the dataset or completely transforming it into a new representation. A possible candidate for KB invariant transformation was identified in using dimensionless groups or numbers where physical dimensions are irrelevant [94]. While this category of feature engineering is becoming common with the advent of physics-informed ML, there are still few examples of these techniques in AM.

Not all AM knowledge is based on process physics. AM knowledge can be specific to a given lifecycle phase (e.g., design knowledge) or a specific data representation (e.g., information in a melt pool image). Feng et al. proposed knowledge engineering context to manage knowledge for smart manufacturing techniques [95]. Since the knowledge comes in different forms and is always encoded for the task at hand, it is nearly impossible to bring a method to the madness introduced by knowledge-driven engineering of AM features. At the data level, knowledge can be used for organization, making the data more suitable for analytics activities [96]. Knowledge can also be used to extract feature directly from raw data (geometric feature extraction from 3D design, defect feature extraction from process images, and so on) [97]. Similarly, it can be used to select features generated from regular FETs. Knowledge has been employed for designing the architecture of feature learners (e.g., CVAE, condition being based on expert knowledge). The specific cases of KB engineering

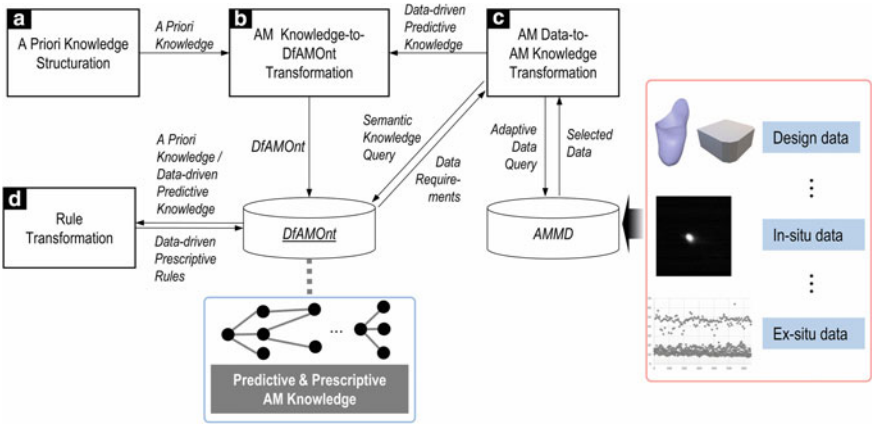


Fig. 2.7 ML-aided knowledge graph-based rule construction proposed by Ko et al. Used with permission from Elsevier [99]

in AM are covered for design, process, and structure-based sources of AM features in the next sections. In more sophisticated settings, knowledge has been used to design or construct features based on AM data representations. For instance, a 3D deviation feature of as-built part to capture the deviation from design information or a melt pool motion feature to capture its dynamic behavior [63, 98]. These and similar features (e.g., microstructural descriptors) can drive data-based applications and are discussed in the respective sections. Figure 2.7 highlights the framework of design rule construction using machine learning and knowledge graphs.

2.10 Integrated Feature Engineering

Engineering AM features is not always about choosing one technique over the other and can be done in a hybrid style. The pipeline is also not sequential and an iterative approach to feature engineering can be adopted until ML model has achieved acceptable performance. This category is usually referred to as integrated feature engineering (IFE) as depicted in Fig. 2.8 which is repurposed from [7]. The concept of integrating different feature engineering techniques is not new. For instance, it is normal to apply feature selection once features have been extracted by appropriate transformations of raw data. Apart from such simpler integrations, adaptive transformations can be developed with variants capable of extracting different features each time. Each individual feature set can be evaluated in the ML model until an optimum

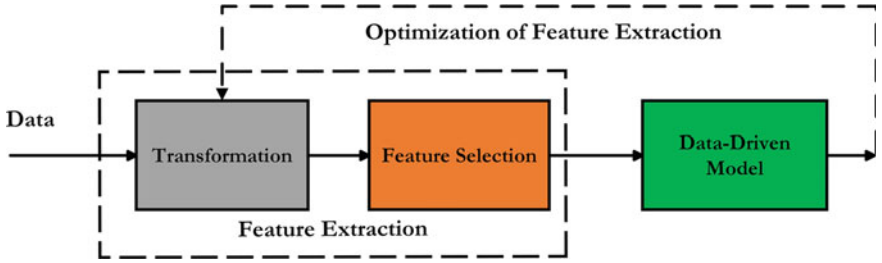


Fig. 2.8 Integrated feature engineering (repurposed from [7])

one is generated by adaptive transformer. An old example of adaptive feature extraction is adaptive wavelets applied to spectrum data [100]. Some advanced and deep feature learners encode this fusion in a rather implicit manner. AEs being the prime example learn new features (learning) but also keep their dimension significantly lower (selection) than the input data so as to learn only those features which are representative of the underlying patterns. Furthermore, DL-based representation learning is also capable of generating features according to multiple criteria which is sometimes referred to as shared representation learning (generative adversarial neural network or GANN being one example). The opportunities are enormous. There are several examples of IFE in AM, and a summary of each pipeline is provided for literature surveyed in the next sections. It is quite common in AM to have integrated feature engineering. These pipelines are explained in the respective applications sections of Chap. 3.

2.11 Feature Operations and Libraries

Feature operation refers to all work with selected, generated, learned, or knowledge engineered features. These works can include clustering, fusion, transfer, visualization, freezing, averaging and other miscellaneous actions on resulting features. Interestingly, the visualization of learned features (or feature maps in computer vision) is common and helps in validating the accuracy of these features for the task at hand. Yang et al. visualized the features from the hidden layer of an AE trained to predict power consumption in AM [101]. They verified that feature values increase monotonically with power consumption. Feature maps learned from melt pool surface images were visualized to prove that CNN models rely on sophisticated and rich features in contrast to simple geometric descriptors [102]. A useful technique for feature visualization is gradient-weighted class activation mapping or Grad-CAM and relates to the visualization of learned features in DL [103]. Several AM researchers have used Grad-CAM for visualizing important features of deep architectures. For instance, Lee et al. used Grad-CAM to visualize discriminative regions in the learned feature

maps for normal and abnormal melt pool images [104]. Figure 2.9 shows GRAD-CAM-based feature maps and visualizes the regions that are important for CNN to identify defective melt pools.

Overall, fusion is probably the most common feature operation and is widely employed to improve the learning process. There can be several ways of fusing features. Similarly, the feature being fused could come from different sources at different levels and lifecycle phases. Figure 2.10 is a mind map of potential fusion scenarios of process features. The authors encountered at least one example of feature fusion from each category. These fusion techniques are identified when summarizing FETs for AM in Chap. 3.

Important feature libraries for AM data can be classified into those of computer vision, SP, multi-dimensional arrays, and geometric ML. Majority of these libraries

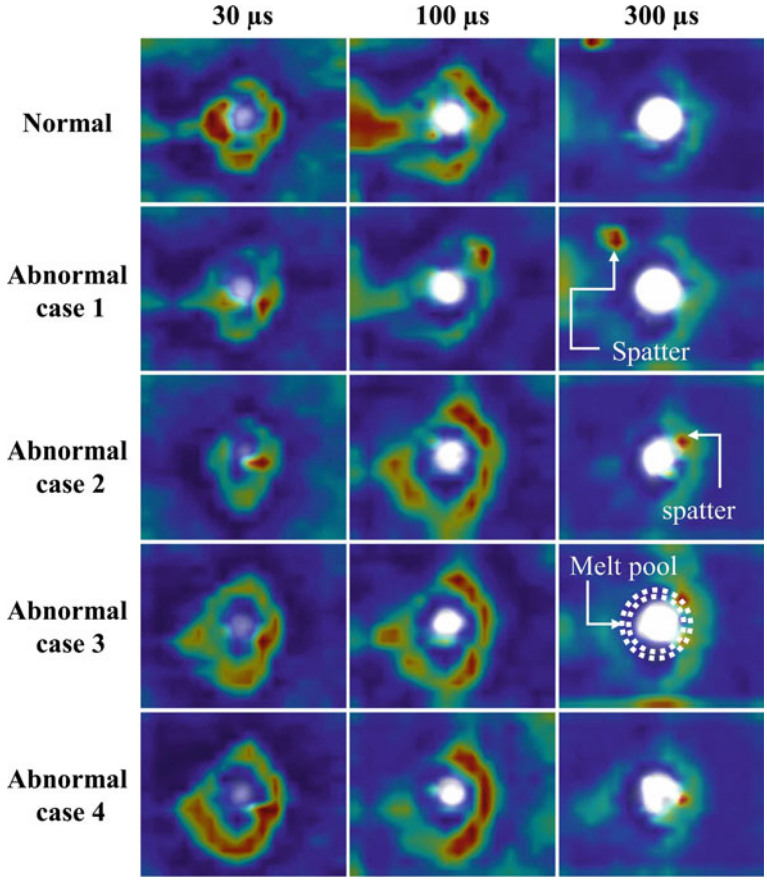


Fig. 2.9 Grad-CAM-based heatmap of CNN features highlighting discriminative regions where more attention is given to make a prediction from melt pool images. Used with permission from Elsevier [104]

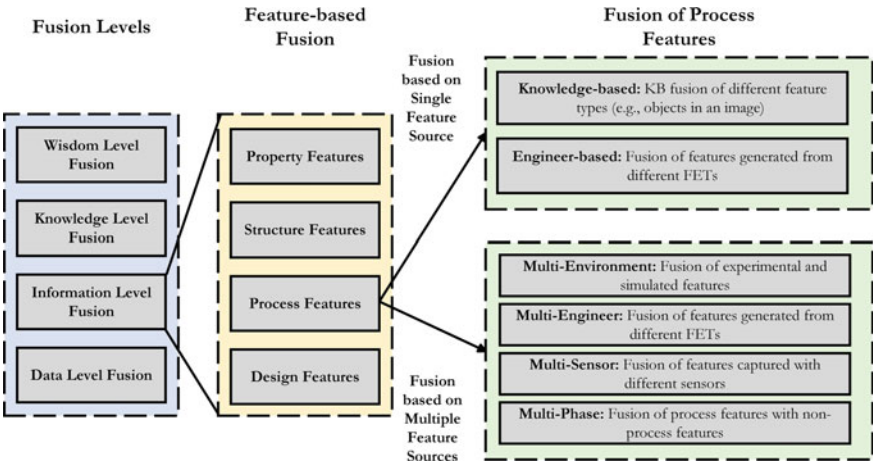


Fig. 2.10 Scenarios of feature fusion at the process phase in data-driven AM

are well integrated with Python programming language environments supporting their usability in implementation environments of data-driven models. Most important libraries of these categories are listed in Table 2.2.

Table 2.2 Generic and well-known libraries for engineering features of AM for a given data type

Domain	Library name and link
Computer vision	OpenCV (https://opencv.org/) Scikit-image (https://scikit-image.org/) Pillow/PIL (https://python-pillow.org/) Mahotas (https://mahotas.readthedocs.io/en/latest/) Pgmagick (https://pypi.org/project/pgmagick/) SimpleCV (http://simplecv.org/)
Multi-dimensional arrays	NumPy (https://numpy.org/) Matplotlib (https://matplotlib.org/) Scipy (https://scipy.org/) Pandas (https://pandas.pydata.org/)
Signal processing	Scipy Signal (https://docs.scipy.org/doc/scipy/reference/signal.html) ThinkDSP (https://github.com/AllenDowney/ThinkDSP) Librosa (https://librosa.org/) PyAudioAnalysis (https://pypi.org/project/pyAudioAnalysis/0.1.3/)
Geometric ML	Open3D (http://www.open3d.org/) Binvox (https://pypi.org/project/binvox/) Point Cloud Library (PCL) (https://pointclouds.org/) Pyntcloud (https://pyntcloud.readthedocs.io/en/latest/) PyMeshLab (https://pypi.org/project/pymeshlab/) Panda3D (https://www.panda3d.org/) PyTorch3D (https://pytorch3d.org/)

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Chapter 3
Applications in Data-Driven Additive Manufacturing



3.1 Engineering of Design Features

Feature engineering at the design phase mainly concerns data for ML applications in support of material or structural design [1]. The major sources of design features can be found in Table 3.1 alongside the details on feature engineering techniques, feature engineering pipelines, resulting features, and their applications. We should highlight the fact that sometimes word "feature" is used at the design stage to refer to a shape construct with an engineering meaning following the feature-based modeling paradigm of CAD/CAM [2]. We will refer to these constructs as "primitives" in the text to avoid any confusion with features of ML tasks. In Table 3.1, engineering of AM design features is presented in two parts by separating feature engineering techniques (Column 2) from feature engineering pipelines (Column 3). The former lists one or more of the generic feature engineering techniques highlighted in the last chapter while the latter provides details on steps involved in the conversion of raw data into ML-ready features. Since most of the featurization efforts in AM are task- and/or data-oriented, a separation as such seems appropriate to handle bulky and task-specific details of feature engineering pipelines. We will follow the same structure for the process and the post-process features as well.

Table 3.1 highlights that 3D data constitutes the majority among all sources of design features. In the process of featurization, 3D data is generally decomposed (a process called voxelization) in a continuous or discrete fashion for usage in ML models. An example is shown in Fig. 3.1. This is largely done in the form of binarized voxels which can be converted into 3D matrices to serve as inputs to ML [28]. Increasing the resolution of voxels can incur high computational cost during the training phase. Zhang et al. made these voxel matrices sparse and showed that both

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M. Safdar et al., *Engineering of Additive Manufacturing Features for Data-Driven Solutions*, SpringerBriefs in Applied Sciences and Technology,
https://doi.org/10.1007/978-3-031-32154-2_3

Table 3.1 Feature engineering techniques, pipelines, resulting features, and their applications at the design phase of AM for major design data sources

Feature source (rep.)	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Design features	Application of design features	References
Functionality-centric AM design knowledge	Multi-categorical encodings	NA	Encoded loadings, objectives, and properties	Design primitive recommendation	[3]
As-designed hierarchical composites (graphic)	Numeric encoding	NA	Encoded assemblage of unit cells	Design of bioinspired hierarchical composites	[4]
As-designed models (3D CAD)	KB feature extraction, tabular transformation	KB geometric features → Sensitivity check → Extraction (PCA)	Model geometry	Energy consumption estimation	[5]
As-designed models (3D STL)	3D transformation	STL → Featureless analysis → Voxelization	Process and material parameters and with voxels	AM printability	[6]
Multi-source: AM user, as-designed model (3D CAD), and resource planning	AM KB engineering	Automated feature extraction from 3D CAD file and user inputs	AM feasibility and potential metrics	AM candidate detection	[7]

(continued)

Table 3.1 (continued)

Feature source (rep.)	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Design features	Application of design features	References
As-designed models (3D STL)	3D transformation	Discrete decomposition of CAD models	Normal vectors of polygons	Build orientation determination	[8]
As-designed model repository (3D CAD)	3D transformation	CAD from parametric templates → STL → Voxelization	Boolean voxels	Design repository effectiveness for ML	[9]
Multi-source: as-designed models (3D CAD) and layer-wise images	Graphic and 3D transformation	Image to CAD masking → Region of interest (ROI) → Resize → Normalization → Characterization	Images from characterization	Quality control	[10]
Metamaterial (structure and material)	AM KB engineering	Extraction of hybrid feature set	Geometry and material	Metamaterial design	[11]
Design space of parametric structure	AM KB engineering	Extraction of geometric features	Geometry	Automation of design space exploration	[12]
As-designed models (3D CAD)	3D transformation, feature learning	Voxelization of CAD models → Feature learning	Design voxels and learned representations	Automation of design for AM	[13]

(continued)

Table 3.1 (continued)

Feature source (rep.)	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Design features	Application of design features	References
Multi-source: as-designed models (3D CAD) and process parameters (tabular)	Graphic transformation	Binarization of layer geometry	Layer geometry	Deviation prediction	[14]
As-designed models (3D CAD)	3D transformation	Parameterized CAD → STL → Voxelization	Voxels	Prediction of parts metrics	[15]
Analytical model of bead geometry (AM knowledge)	AM KB feature engineering	Synthetic and hybrid feature generation from analytical model	Process parameters and polynomial coefficients	Bead geometry prediction	[16]
As-designed models (3D STL)	3D transformation	STL → Voxelization	Primitives and voxels	Build orientation prediction	[17]
As-designed models (3D CAD)	AM KB engineering	Geometry extraction	Geometric features	ML-aided design and optimization	[18]
As-designed models (3D CAD)	AM KB engineering	AM feasibility-based feature extraction → Selection → Transformation	Geometric and economic features	AM feasibility and design update recommendation	[19]
As-designed models (3D CAD)	Feature subset selection, transformation, learning	Geometry → KB selection of geometry vector → Normalization → PCA	Geometry and PCs and VAE representations	Power consumption estimation	[20]

(continued)

Table 3.1 (continued)

Feature source (rep.)	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Design features	Application of design features	References
As-designed models (3D CAD), AM users, ERP	AM KB engineering	Automated features from CAD, inputs from user and ERP	Design, social, economic	AM candidate selection	[21]
As-designed models (3D CAD/STL)	3D transformation	CAD/STL → Binarization → Frequency domain → Max filter → Coordinates with max values	Frequency domain values	CAD model search	[22]
Meta material structure	Graphic transformation, feature extraction	Design modification in finite element model (FEM)	Selected synthetic features	Design of auxetic materials	[23]
Microlattice architecture (graphic)	Graph encoding, feature learning	Adjacency matrix-based graph encoding, feature learning in AE	Graph connection and AE representations	Design and analysis of microlattice	[24]
Meta material architecture (graphic)	AM KB engineering	2D architecture (nodes and edges) to graph representation	Graphs	Meta material design	[25]
AM knowledge (ontology)	AM KB engineering	AM knowledge representation → AM printability knowledge extraction → Design rule construction	AM printability knowledge	Design rule construction	[26]
As-designed models (CAD)	AM KB engineering	Input features from build or geometry	Explicit geometric and build features	Design defect characterization	[27]

(continued)

Table 3.1 (continued)

Feature source (rep.)	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Design features	Application of design features	References
As-designed models (3D CAD)	3D transformation, feature learning	CAD → Binary voxelization → Feature learning	Voxels and AE code	Manufacturability assessment	[28]
Unit cell image (graphic)	Graphic transformation	3 by 3 tiling of the unit cells	Explicit image features	ML for optimal unit cell discovery	[29]
Multi-source: as-designed model (3D CAD), material, and process (tabular)	Encoding, 3D transformation, AM KB engineering	CAD to voxelization, fusion of features for learning	Hybrid and explicit design, material, process features	Manufacturability assessment	[30]
As-designed models (triangular mesh)	AM KB engineering	Engineering informed knowledge to extract predictor which capture complex shape	Position, material, surface	Primitive part errors	[31]
Biological design (graphic)	AM KB engineering	Graphic transformation from biological design to image features	Graphic features	Bioinspired design	[32]
As-designed models (CAD)	AM KB engineering, feature subset selection, and tabular transformation	Knowledge-based extraction, subset selection, normalization	AM knowledge	Improvement of design inventories	[33]
As-designed models (3D CAD) and as-measured	AM KB engineering	Built-designed registration → Geometric feature extraction	Distance and curvature features	Geometric defect prediction	[34]

(continued)

Table 3.1 (continued)

Feature source (rep.)	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Design features	Application of design features	References
As-designed (3D CAD) and simulated model (AM knowledge)	AM KB engineering, synthetic feature generation, feature subset selection	Embedded feature selection by evaluating binary feature interactions, relative feature importance from model output	Design and defect feature	Influence of defects on ME properties	[35]
Multi-source: design, process, and material (3D and tabular)	AM KB engineering, one-hot encoding	Geometric feature extraction, encoding of feature types	Critical geometric features	Geometry prediction, part, and feature classification	[36]
Spatial structure (3D) and temporal response (sequence)	Feature learning, 3D transformation	Geometric structure to spatial coordinates, convolutional bidirectional LSTM for feature learning	Learned spatiotemporal features	Stress–strain prediction	[37]
Auxetic material design (3D CAD)	AM KB engineering	Manual extraction of unit cell geometry	Geometric feature of unit cell	Design optimization of modified auxetic structures	[38]
As-designed models (3D CAD)	3D transformation	CAD → Voxelization → Sparsity	Sparse voxels, material, and process parameters	Manufacturability assessment and printability map	[39]
Multi-source: as-designed models (3D CAD) and process parameters (tabular)	AM KB engineering	Manual extraction of CAD dimensions	CAD dimensions and process parameters	Dimensional accuracy	[40]

(continued)

Table 3.1 (continued)

Feature source (rep.)	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Design features	Application of design features	References
Geometric parameters (tabular)	Tabular transformation	NA	Geometric parameters	Prediction of geometric deviations	[41]
As-designed models (3D CAD)	3D transformation	CAD → Surfel construction → Surfel extraction as a multi-channel image	Surface elements	AM support prediction	[42]
As-designed models (3D mesh data)	Feature learning	Design of special convolution operation to capture invariant shape features from mesh data	Learned representations	Learning of invariant shape features for shape correspondence in mass customization	[43]
Composite design (graphic)	Graphic transformation	2D matrices to 1D arrays	1D arrays	Composite design classification	[44]
As-designed 2D structure (graphic)	Feature learning	2D binary image of desired structure → Custom architecture for feature learning	Learned representations of structure	Optimal digital mask	[45]
As-designed models (3D CAD)	3D transformation	Voxelization	Voxels, learned maps	Manufacturability prediction	[46]
As-designed lattice models (graphic)	KB feature extraction	Preprocessing (polygon preparation) and feature extraction (hand-engineered)	Graphic feature	Path planning	[47]
As-designed CFRP (graphic)	AM KB engineering	Evaluation of feature importance → Feature extraction	Microstructural features	Flexural strength	[48]

computational efficiency and performance of the model improved when predicting additive-based manufacturability of as-designed models [39]. 3D models are also indirectly used in the design feature generation process by serving as a reference to printed parts. Examples include registration against in-process (e.g., 3D models to sensor signals) or post-process (e.g., 3D models to printed structures) data to generate defect or deviation features for ML pipelines [27]. These features can serve as inputs or outputs of ML models for AM design-related predictions. Polygonal mesh and point clouds are other 3D representations that have been used for feature generation via miscellaneous transformations. For instance, Zhang et al. used normal vectors of polygons as features to cluster similar facets for build orientation prediction of a part [8].

AM design data in graphic form (e.g., 2D) is relatively less frequent in the feature engineering pipelines of ML as presented in Table 3.1. The approaches to prepare this type of data for ML are quite diverse and hence lack an overall leading trend. Encodings of graphic data are, however, quite common. Encodings are seen as a distinct trend at the design phase and almost always concern geometric information.

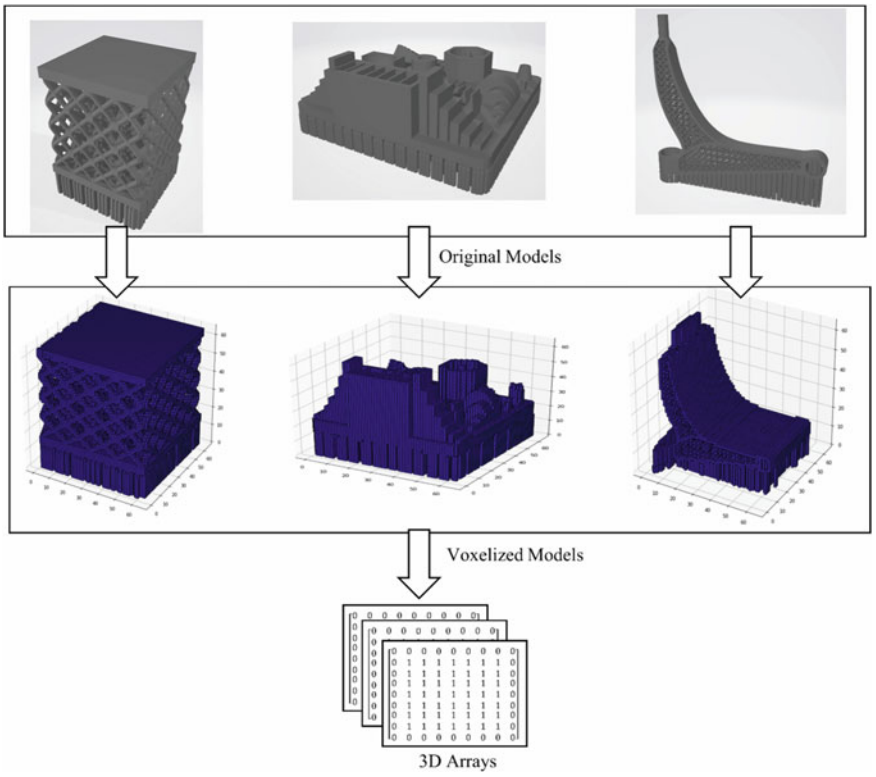


Fig. 3.1 Voxelization of CAD geometries resulting in 3D binary arrays for the classification model of manufacturability. Figure used with permission from Elsevier from [30]

Examples include numeric [4], categorical [25], or graph-based encodings [24]. One example of numeric encoding is shown in Fig. 3.2. These can make otherwise abstract design information ready for ML in a simple and explicit way. The example also illustrates the combinatory power (e.g., substituting features with encoded digits) of encodings that can be used to explore AM design space. Graphic data with design information is discretized too, with binarization as the main technique [14]. Another example of graphic transformation at design phase is to extract bioinspired design features from images, as in [32] where geometry information from the image of a bird is extracted. As can be seen in Table 3.1, tabular sources are least common at the design phase. One example of tabular transformation is PCA applied to geometric features of a CAD model [5]. AM knowledge has different levels of applications at the design phase. The knowledge to drive engineering of design features can be geometric, social, economic, user-based, application-oriented, or generated by physical models. The most common application of this knowledge has been to extract the geometric parameters. Additionally, it can be used to either transform certain representations (e.g., graphic data) or to select their features [32]. For example, a curvature feature was defined to capture the critical geometry information more effectively when predicting shape deviations in ML models [34].

In addition to transformations and KB engineering of design data, learning of latent design features is a useful engineering technique. In ML, the term "latent" refers to a new compressed feature space where similar features are expected to be closer to each other. Feature learning with deep architectures has been used to

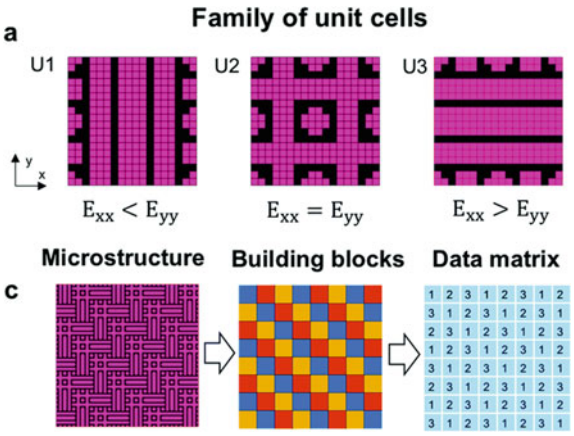


Fig. 3.2 Encoding of design microstructure into a data matrix using multiple digits each corresponding to one family of unit cells. U1, U2, and U3 each represent a unit cell from a distinct family of microstructure. E_{xx} and Eyy represent stiffness in the x- and y-directions, respectively. Once combined, each unit cell is represented with a different color (e.g., blue, red, and orange) and later with a different digit as shown by the building blocks and data matrix, respectively. The data matrix is used in a CNN to phase out inferior designs by predicting mechanical properties. Figure used under Creative Commons Attribution 3.0 unported License from Royal Society of Chemistry [4]

explore and simplify vast AM design space for certain applications [24]. Murphy et al. used transformed version of as-designed models in an AE model to learn features for design automation application [13]. Similarly, latent features of 2D structures in graph representations were learned to design and analyze microlattices [4]. Selecting a subset of features for data-driven models is often a downstream task in feature engineering pipelines. The leading trend for the selection of AM design features is based on knowledge. Yang et al. selected a geometry vector from CAD model while predicting power consumption of 3D printing process using design knowledge [20]. Similarly, a subset of AM feasibility features was selected to recommend designs that are printable with AM [19]. In an example of embedded feature selection, Hu et al. used LASSO regression to select important features of architected materials while investigating the influence of defects on their mechanical properties [35]. Overall, engineering pipelines of design features are customized and knowledge-driven, and this trend is analyzed and discussed in relevant sections on feature spaces.

3.2 Feature Engineering at AM Process Phase

AM process features are by far the most diverse class of features for ML applications. This is fueled by the process complexity (e.g., hardware, software, and material aspects), range of physical scale (e.g., microscopic to macroscopic), domain diversity (e.g., spatial and/or temporal), information acquisition rate (e.g., high frequency), and information magnitude (e.g., big data). The "unknowns" resulting from complicated process nature where both material and part are fabricated simultaneously add to the inspiration behind rich AM process features. The existing AM reviews on empirical modeling are inspired by the applications at process phase, and hence, the categorization of data-driven models is always based on the output being modeled [49]. However, this approach omits the sources of data (or features) driving AM applications. To the best of our knowledge, there is no existing text that classifies in detail all possible sources of AM process feature for data-driven techniques. The work by Liu et al. is an exception where "data items" relating to design, process parameters, process signatures, post-processing, and product quality were comprehensively identified for the sake of collaborative data management in MAM [50]. They identified a total of 19 subcategories belonging to the AM phases mentioned above. Following the DPSP paradigm, we rearrange these feature sources in the subsequent sections for a typical AM process. Another example exists in the form of identifying major sources of monitored signals in MAM processes by Lin et al. [51]. They summarized the information in a tabular form which is specific to metallic AM. In addition to metallic AM, measurement science sources of plastic AM and scientific modeling techniques for all AM processes could cover the whole spectrum of these sources but is difficult to compile.

As far as the process phase is concerned, a source-based approach is used to categorize major feature types. These process features are based on all physical objects (e.g., melt pool, layers, under-built part) and process activities (e.g., planning) which

carry with them distinct and representative information suitable for featurization techniques of data-driven approaches. The resulting features are arranged in the order of a typical AM process. These features have significant influence on the structure, property, and performance of printed parts. AM process planning (path/scan strategy, build orientation, and support structures) activities are identified as the source for the first category of AM process features and are grouped under "Planning Features (Sect. 3.4)". AM process parameters (variables relating to software, hardware, or material) are the second dominant category of process features at this phase. These features are conventionally deployed in-process optimization scenarios and are placed in "Parametric Features (Sect. 3.5)" within the scope of this text. Since AM processes are completed in a layer-upon-layer fashion, we identify layer features as the next category in an AM process flow. "Layer Features (Sect. 3.6)" are related to either part or material features in a spatial 2D sense. When the focus is shifted to capturing or modeling in-situ part (beads or multiple layers alongside substrate) in 3D, resulting features are classified as "In-Situ Geometry Features (Sect. 3.8)". "Melt Pool Features (Sect. 3.7)" constitute the next group of AM process features. We will pay special attention to the featurization of melt pool in metallic AM. This is largely inspired by the efforts being made to mature metallic AM for high-volume production scenarios. A separate category of AM process features is defined under "Generic Process Features (Sect. 3.3)" where identification of the actual source of information is a challenge. Almost all feature sources at AM process phase are covered in this way. The by-products from a particular feature source (e.g., spatters from melt pool) are covered under the respective feature category. Physical nature (thermal) and type of AM feature sources (synthetic) are sometimes made the basis to identify a feature group in AM. We will highlight these aspects of the features under each feature category but stick with the physical source or activity-based approach mentioned above.

3.3 Engineering of Generic Process Features

AM processes can be roughly seen as either metallic or non-metallic in nature. This review covers both classes alongside their representative processes (FDM for plastic AM, and PBF/DED for metallic AM). Owing to the complexity of an AM process, it is occasionally difficult to explicitly identify the source of a feature. For instance, a MAM process can involve several complex phenomena including heat transfer, laser powder interaction, melt pool flow, mass transfer, and phase transformation leading to numerous signals being emitted throughout the process. A few of the process signals can either have multiple contributors (e.g., airborne sound), unknown origin (e.g., vibration) or both (e.g., 1D optical signal) [52]. Furthermore, to diversify the input features of a learning model, multiple sensors each capturing a different aspect of an AM process are usually deployed. These efforts are sometimes aimed at identifying abstract process states or conditions because actual source of signal generation is never the focus. Feature resulting from these sources are processed representative

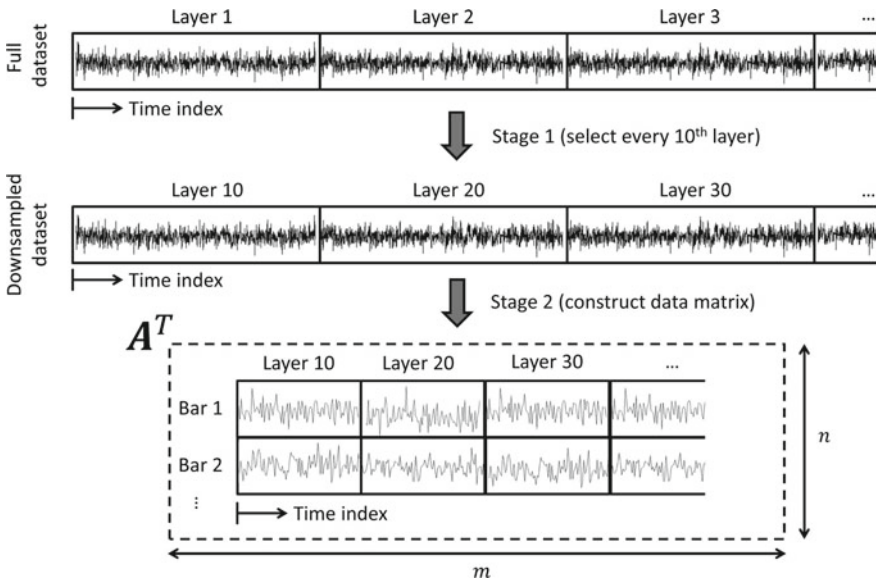


Fig. 3.3 Downsampling of photodiode signals from build chamber to consider data from the tenth consecutive layer. Layer-wise signals are arranged in row (for each part being built) and column (for each layer considered) fashion. This results in a data matrix which is featurized using randomized SVD transformation. The result is a two-dimensional feature vector for each part (or bar) being printed. Figure used under Creative Commons CC-BY license from Elsevier [53]

due to their aggregate nature. For cases such as these, the resulting features are placed in the category of "Generic Process Features" and are covered within this section. Figure 3.3 shows one example of data preparation at the process phase from generic sources.

Table 3.2 provides a summary of generic process features in the context of data preparation for ML. Light, sound, and vibration represent major generic process signatures in AM. The advent of multi-sensor monitoring has led to miscellaneous sensors being used in-situ. This hybrid setting results in diverse features being captured during an AM process. As a result, heterogeneous in-process monitoring represents another major source of generic process features. As far as the techniques to process these features are concerned, timeseries signal transformations are found to be the most common to prepare ML features. In this regard, time domain, frequency domain, and time–frequency domain features count for most resulting features from raw signals. The feature engineering pipelines for generic category are quite sophisticated and are driven by the domain expertise of the relevant signals being captured. An interesting trend in the application of these features is to predict generic process states such as anomalies, quality deviations, and abnormal or undesired process states.

A few examples of engineering AM process features from generic sources are discussed. Wu et al. deployed acoustic sensor in FDM for process state prediction

Table 3.2 Feature engineering techniques, pipelines, resulting features, and their applications at the process phase of AM for generic data sources

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features	Application of process features	References
AE sensor (timeseries)	Signal transformation	Time and frequency feature extraction from segments of AE waveform	Amplitude, counts, absolute energy, peak frequency, and frequency centroid	Process state (normal vs. failed) prediction	[54]
Accelerometer for vibration signal (timeseries)	Signal transformation	Signal windowing → Signal transformation (statistical and FFT) → feature subset selection (multiple) iterative	Selected features of frequency domain	Interference detection	[55]
Multi-sensor: IR camera (graphic), accelerometer (timeseries), thermocouple (timeseries), borescope (3D video)	Sensor data modeling based on dirichlet process (DP) mixture model and evidence theory	Univariate signal representation → Integration of heterogeneous signals	Miscellaneous signal (source dependent) representations	Process state prediction (normal, abnormal, failure)	[56]
Acoustic signal (timeseries) and G-Code (AM representation)	Data preprocessing, signal transformation	Low- and high-pass filters → Feature extraction in time and frequency domain	Features corresponding to STFT, ZCR, MFCC, and frame energies	G-Code reconstruction	[57]

(continued)

Table 3.2 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features	Application of process features	References
Acoustic signal (timeseries)	Signal transformation	Frequency domain transformation → Adaptive WT → Feature input strategies	Signal spectrogram	Porosity level (high, medium, and low) prediction	[58]
Multi-sensor: thermocouples (timeseries), infrared temperature sensors (graphic), and accelerometer (timeseries)	Signal transformation, feature subset selection	Time and frequency feature extraction → Feature subset selection (tree-based)	Time (8 statistical) and frequency (4 FFT) domain features	Surface roughness prediction	[59]
Multi-source: IR camera (graphic) and simulations (temperature, displacement, and stress field maps)	Signal transformation, graphic transformation	Alpert tree-based wavelet compression	Compressed spectral representation	AM data compression	[60]
Multi-source: process (tabular), property (tabular), and microstructure (graphic) from simulations and experiments	Multiple AM KB feature engineering techniques	KB FE → Feature fusion	Process (laser power, mass flow rate, energy density, and cooling rate), Microstructure (dilution and dendrite arm spacing), and property (microhardness) features	Production, visualization, and design of PSP linkages	[61]

(continued)

Table 3.2 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features	Application of process features	References
Three photodiode signals (timeseries)	Signal preprocessing, feature subset selection	KB signal removal and truncating → Feature subset selection (SVD)	Constants from the first basis vectors of each sensor data matrix	Build quality (w.r.t. density) prediction	[62]
Process AE signal	Signal transformation	Signal to graphic transformation	Spectrogram	Poor, medium, and high	[63]
Acoustic signal	Signal transformation	Wavelet transform	Spectrogram	Levels of porosity concentration	[64]
Heterogeneous in-process sensing	Signal transformation	Feature extraction from two different signals (camera + spectrometer) → network graph → product of graphs	Kronecker product of graphs	In-situ lack of fusion detection	[65]
Simulation of the manufacturing process	KB transformation of 3D voxel data (FEM) to structured data with neighboring temperature and tool path	Multi-step featurization (involving voxel spatiotemporal features + laser information)	Historical, spatiotemporal, spatial, temporal features of voxels	Temperature profile prediction	[66]
High-speed imaging of incoming powder and as-built bead	Graphic transformation	(A) Image with powder stream → Find ROI → Extract ROI → Black pixels → Powder counts → Powder features (*2) (B) Image with bead → Line profile tool → Height feature	Powder stream features, bead height feature	Process parameter-build quality dependency	[67]

(continued)

Table 3.2 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features	Application of process features	References
Multi-sources: thermal images of the surface, process and design parameters	Feature learning, feature fusion	Thermal images → Feature extraction in CNN and Parameters → Feature extraction in NN	Extracted features from CNN and ANN	Point-wise distortion prediction → compensation plan	[68]
Photodiodes (*2) mounted outside of the build chamber	Hand engineering of signal	Timeseries photodiode signal → Manual analysis	Single and multi-layer melt pool event features	Lack of fusion	[52, 69]
Photodiode sensor	Signal processing	Timeseries signal → Part level aggregation → Feature extraction	Part-wise average intensity and its standard deviation, standard deviation of layer-wise intensity	Process deviations, severity of material defects	[70]
Sets of photodiode data	Signal preprocessing, signal transformation	Downsampling → Mapping with laser position → Signal data matrix construction → Randomized SVD	Two-dimensional feature vector for each specimen	UTS-based certification (faulty or acceptable)	[53]
Simulated melt pool, laser path, and XCT scans	AM KB feature engineering, feature learning	Data merging → Data registration → CVAE	Spatially resolved pore features, learned latent pore features	Subsurface pore prediction	[71]

(continued)

Table 3.2 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features	Application of process features	References
Multi-source: process parameters (tabular) and descriptors from SEM (graphic)	AM KB hand engineering	Hybrid feature extraction	Process parameters and SEM descriptors	Qualification based on design primitives	[72]
Process, material, and sensing (temperature and vibration) inputs	Feature subset selection	Iterative feature selection and feature relevance analysis	Selected and highlighted (important) features from process and sensors	Tensile strength prediction	[73]
Thermal radiation from laser area (photodiode 1) and layer area (photodiode 2) and melt pool intensity distribution	Feature fusion	Data fusion: Indicator determination → Filters → Filter calibration → Sensor level data fusion	Fused data (or features)	Process monitoring	[74]
Photodiode signal from melt pool region	Signal transformation	Extraction of different statistical feature types	Central tendency, dispersion tendency, distribution tendency, impulsive metric	Anomaly detection	[75]

[54]. Before captured signal could be used for prediction, it was transformed to frequency domain for feature extraction in both time and frequency domains. They captured only segments of acoustic emission waveform to cope with high-speed data stream. For each segment, multiple process features were extracted and used. Time-series vibration signals were used to identify manually induced interferences in an AM process [55]. An integrated pipeline was developed where multiple windowing, signal transformation, and feature selection operators were tested for the incoming signal. The best-performing features of frequency domain were finally used. A multi-sensor setup with IR camera, accelerometer, thermocouple, and borescope was used for process state prediction (normal, abnormal, and failure) [56]. Their work provides insights into sensor data modeling to understand the underlying data generation process in contrast to its direct usage in downstream applications. There are other instances of using multi-sensor setups to capture different aspects of an AM process as in [59, 61]. To understand the potential of G-Code reconstruction and subsequent risk of intellectual property theft, Mativo et al. used timeseries acoustic signals [57]. Raw signal was preprocessed with low- and high-pass filters, and features in time and frequency domain were extracted.

3.4 Engineering of Process Features: Planning

Unlike subtractive manufacturing, AM provides greater freedom on how a part can be printed. At the process phase, this freedom is usually represented by several deposition strategies, build orientations, and support configurations which could be adapted to print a specific geometry. However, the effects of these choices are unique and may not lead to an optimum structure being printed [76]. As a result, there is a need to narrow down and simplify the complex planning space before AM is used as a large-scale manufacturing technique for certain applications. Data-driven tools can be used to relate available options at the planning stage with quantities and characteristics of concern at process, structure, and property phases. The results of such predictive models can help choose a deposition strategy, a build orientation, or a support structure which minimize the micro- and macromechanical defects for a given geometry. Table 3.3 provides a summary for featurization of planning-related inputs.

Eulerian tool paths for rib-web structures were realized by extracting features of their junction geometries [77]. These geometric features included number of turning points at each junction and resulting angles. The relationship between these features and a tool path metric (e.g., path length value to avoid material deficit) was modeled in a data-driven fashion. Supportless fused deposition modeling, a type of MEX, was realized with the help of a trajectory compensation scheme [78]. A rich set of deposition features corresponding to paths were used in this regard. These include coordinates, quaternions, and time features to generate the compensated trajectory. Process and deposition parameters were used in order to predict connection status between different paths [79]. The features included filament extrusion speed, print

Table 3.3 Feature engineering techniques, pipelines, resulting features, and their applications at the process phase of AM for planning sources

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Planning features (forms)	Application of planning features	References
As-designed models of junction geometries	AM KB engineering	Geometric feature extraction	Junction geometries (explicit)	Path planning	[77]
Deposition strategy	AM KB engineering	N/A	Point coordinates and their differences with the previous point, quaternions, time stamps (explicit and hybrid)	Generation of compensated trajectory	[78]
Process and deposition parameters	AM KB engineering	N/A	Filament extrusion speed, print speed, layer height, line distance (explicit)	Connection status between deposition paths	[79]
Process parameters and build orientation	AM KB engineering	N/A	Layer height, extrusion temperature, print speed, print acceleration, and flow (explicit)	Surface roughness	[80]
Simulation of varied process parameters, tool path strategies, and shapes	Data preprocessing (norm. and noise addition), AM KB engineering	Preprocessing → Signal feature extraction	Tool path, temporal, laser, and layer features (explicit, synthetic, and hybrid)	Thermal history at a point	[81]
Deposition paths (random, hatch, spiral, fractal, and spline)	AM KB engineering	Trajectory transformation by customized decomposition	Implicit trajectory descriptors (short- and long-term deposition points)	Temperature and density evolution	[82]
G-Code	Customized 3D transformation based on HIZ	Physical significance-based featurization of deposition process → KB feature selection	Sets of distances and deposition times (explicit)	Thermal history	[83]

(continued)

Table 3.3 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Planning features (forms)	Application of planning features	References
Laser trajectories	Graphic transformation	AM knowledge-based extraction of laser motion	Parameterized trajectory vector (explicit)	Fabrication quality	[84]
Laser power, scan velocity, hatch distance, scan strategy	N/A	N/A	Raw variables (explicit)	Microstructure formation (grain growth size and grain growth direction angle)	[85]
Scan strategy and process parameters	KB featurization of scan strategy	NBEM for scan strategy representation	Laser power, scan velocity, neighboring points matrix (explicit)	Melt pool size prediction	[86]
Thermal history for scan path and component geometries	Timeseries signal transformation	Thermal history of a point from discrete source model → Signal Transformation (SAX)→	String representation of thermal history (synthetic and implicit)	Zoned thermal histories	[87]
As-simulated scan strategies	KB featurization of AM simulations	Multi-level discretization for synthetic data preparation	Discretized component, laser pattern, laser status, and temperature filed (synthetic and explicit)	Thermal history distributions	[76]
Process parameters, tool path	3D transformation	3D voxel representation of substrate to 2D height map	Another example of fusions, move to build or substrate section, synthetic (MP simulations)	As-manufactured shape (capturing machine-specific uncertainty)	[88]
Build orientation	Feature subset selection	Feature importance analysis	Orientation features (explicit)	Mechanical properties	[89]
Layer height, seam style, build orientation, linkage length	Randomization	N/A	Randomized explicit features	Dimensional deviation and joint clearance	[90]

speed, layer height, and line distance. In metallic processes, it is not practical to carry out a comprehensive DoE for the purpose of planning data generation (e.g., varying tool paths, depositions, and build orientations). As a result, simulations are sometimes used as sources of AM planning features. Mozaffer et al. used FEA simulations to generate data for point-wise prediction of high-dimensional thermal history in DED process [81]. The input features used were a fusion of build, deposition, and process parameters. These synthetic features were normalized before being used in a sequence model to predict the temperature. Another example of simulating deposition process to generate data from ML models is shown in Fig. 3.4.

A representative example of deposition featurization has been reported where long- and short-term path features were extracted to be used in a sequence model of temperature prediction [82]. The work is another example of parent simulation replacement by data-driven model using representative deposition features. Roy and Wodo while acknowledging the importance of feature selection, adopted a featurization approach based on the physical characteristics of the deposition process in contrast to off-the-shelf feature engineering techniques [83]. They defined a Heat Influence Zone (HIZ) and extracted three distinct feature types from it. HIZ features included distances from the cooling surfaces, distances from the heat source, and a set of deposition times. These features resulting from the geometric transformation of

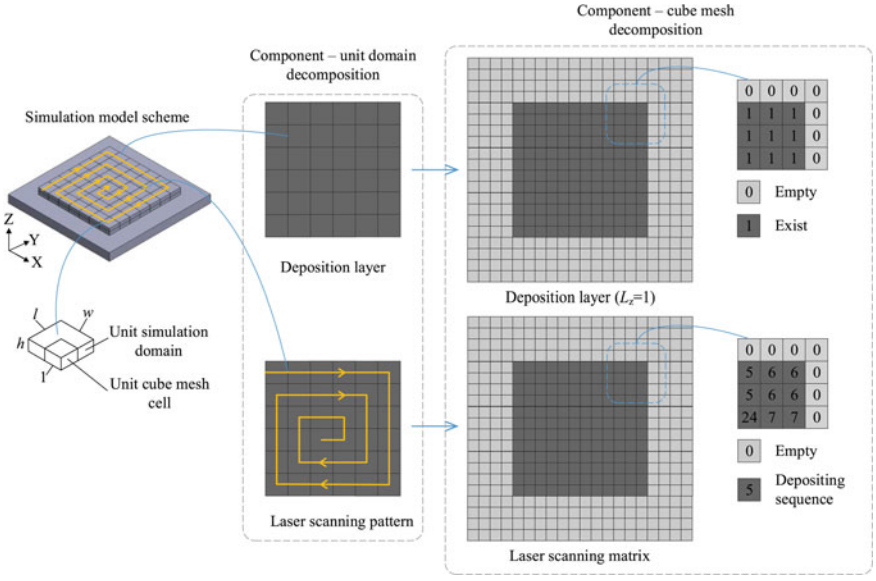


Fig. 3.4 Component-cube mesh architecture for decomposing a unit simulation domain. The architecture is used to uniformly discretize layers and laser scanning pattern. Later, laser scanning status matrix (50, 50) is used to predict temperature field (41, 41) in an optimized RNN-DNN architecture. Figure is used from [76] with permission from Elsevier. Readers are referred to the "dataset preparation" section for detailed description of the steps followed to generate the thermal history database

G-Code were used to predict thermal history. In another example of using deposition information for empirical modeling, Bauhofer and Daraio featurized laser trajectories into numeric feature vectors of size four with each representing laser motions in x- and y-directions [84]. The resulting features of trajectories were used to predict quality of parts in terms of material shrinkage.

3.5 Engineering of Process Features: Parametric

Parametric features with tabular representations are easily the most frequent process features as their use in data-driven models precedes big data-based graphic or sequence representations. The main motivation behind the applications based on parametric features is their ability to unearth complex nonlinear relations with characteristics of concern (CoC) which may not be possible with low-capacity function approximators (e.g., shallow models as opposed to deep models capable of learning complex functions). The parametric features are specific to a given process and are self-explanatory for common AM processes. In MAM-based processes, these are usually power (laser, electron), deposition (speed, spacing), or material (feed rate, layer thickness) parameters. In FDM-based processes, these can include feeding parameters (material and nozzle). These can also include pure material and print environment parameters. Table 3.4 summarizes the engineering of AM process parametric features alongside their sources.

Wang et al. used Min–Max normalization for mechanical and thermodynamic properties of material particles in cold spraying [107]. They justified this selection over Z-Score for its applicability to non-normal distributions. For feature subset selection, tree-based learners are widely used. Kamath et al. used regression trees for process feature selection to predict melt pool geometry [117]. There are also examples of KB selection of process features, e.g., using only significant or proven features in data-driven models [109]. Finally, Pearson correlation coefficient has been used to evaluate the strength of linear relationships among process features and eliminate those with a strong positive or negative relation [92]. Some exceptions to tabular representations of process features also exist. For instance, Yi et al. extracted contour features from laser temperature field (e.g., power features) on the sample and used them to predict laser temperature [103].

3.6 Engineering of Process Features: Layer

Layers are inherent to all seven categories of AM, and it makes sense to relate their features with CoCs. A "layer" in this text refers to 2D surface information (part or material area) information (captured or modeled) for use in data-driven models. We separate these sources from 3D capturing or modeling of in-situ geometry (with or without substrate), and these are covered under "In-Situ Geometry Features". Layers

Table 3.4 Feature engineering techniques, resulting features, and their applications at the process phase of AM for parametric sources

Feature source	Feature engineering technique (FET)	Parametric features	Application of planning features	References
Laser power, scan speed, hatch spacing, laser pattern increment angle, heat treatment condition	None	Explicit and hybrid features from process, planning, and preprocessing	Surface roughness	[91]
Material and process parameters (total 23)	Tabular transformation: maximal information coefficient and pearson coefficient → Feature selection	Selected material and process features in explicit form	Melt pool geometry	[92]
Hatch space and scan length (variations in FEA)	None	Explicit and synthetic features	Melt pool temperature, melt pool depth, and overlap rate between adjacent tracks	[93]
Extrusion temperature, layer height, and material density	None	Raw inputs in explicit form	Tensile strength	[94]
Printing speed(s)	Tabular transformation: multivariate ANOVA, group method of data handling	Explicit printing speeds for each layer	Maximum residual stress	[95]
Wire feed rate, travel speed, inter pass time	None	Raw inputs in explicit form	Deposition quality	[96]
Laser scan speed and laser power	None	Explicit and hybrid (exp. and simu.) features	Remelted depth of single track	[97]
Laser power, scan speed, layer thickness, beam diameter, and hatch spacing	Tabular transformation: global sensitivity analysis (S/N ratio) and uncertainty analysis (ANOVA)	Explicit influential features	Defects and properties	[98]
Laser power, velocity, hatch spacing	None	Raw inputs in explicit form	Strain recovery rates and transformation temperatures	[99]

(continued)

Table 3.4 (continued)

Feature source	Feature engineering technique (FET)	Parametric features	Application of planning features	References
Power, speed, feed rate	None	Raw inputs in explicit form	Deposition width and height	[100]
Plastic viscosity, yield stress of the cementitious material, and nozzle travel speed	None	Raw inputs in explicit form	Deformation of the printed filament	[101]
Process parameters, melt pool images	None	Fusion of process parameters with other features	Width and height of the clad bead	[102]
Laser temperature field (graphic parameter)	Graphic transformation: grayscale binarization → Denoising → Contour-based feature extraction	Implicit extracted image features	Laser temperature	[103]
Build job height, number of layers, packing density, number of parts and room temperature	None	Raw inputs in explicit form	Cooling time for different build jobs	[104]
Filament speed and nozzle temperatures	Tabular transformation: Z-score-based standardization	Transformed tabular features	Force within the nozzle	[105]
Spreader speeds	None	Synthetic feature from FEA	Spread layer quality	[106]
Material particle properties (mechanical and thermodynamic)	Min–max normalization (also talks about its edge over *Z*-score) → Tree-based feature subset selection (tree-based)	Selected particle features in normalized form	Cold spray critical velocity	[107]
Layer-wise process parameters in SLM	Tree-based feature subset selection	Layer clusters	Identification of defective pattern	[108]
Linear energy density and hatch spacing	KB critical feature selection	Selected features	Printability	[109]

(continued)

Table 3.4 (continued)

Feature source	Feature engineering technique (FET)	Parametric features	Application of planning features	References
Oxygen concentration of the printing chamber, laser power, and scanning speed	None	Raw inputs in explicit form	Magnetic characteristics of printed composites	[110]
Angle of incline, overlapping length, and number of shells	ANOVA	Selected features	Tensile strength	[111]
Material parameters (three)	Tabular transformation: 0–1 normalization	Normalized material parameters	Force displacement curve (FDC) error difference	[112]
Laser power, laser scanning speed, layer thickness, and hatch distance	Tabular transformation: *Z*-score standardization	Standardized features	Print properties (density ratio and surface roughness)	[113]
Laser power and scan speed	Tabular transformation: normalization (non-dimensional and zero)	Normalized input parameters	Relative density	[114]
Laser power, layer thickness, scanning speed	ANOVA for parameter ranking	Ranked features	Energy consumption, tensile strength, surface roughness	[115]
Laser power, scan speed, and laser beam size combination	KB filtering of collected data	Filtered features	Melt pool depth	[116]
Speed, power, beam size, absorptivity, noise	Feature subset selection: pearson correlation, regression tree	Selected explicit features	Melt pool width, length, and depth	[117]
Laser power, scanning speed, and uncertainty source (simulations and experiments)	None	Synthetic and hybrid raw features	Melt pool geometry	[118]
Power, speed, powder feed rate	None	Raw inputs in explicit form	Bead height	[119]

(continued)

Table 3.4 (continued)

Feature source	Feature engineering technique (FET)	Parametric features	Application of planning features	References
Layer thickness, hatch spacing, laser power, scanning speed, temperature of working environment, interval time, and scanning mode	None	Raw inputs in explicit form	Part density	[120]
Laser power, scanning speed, and hatch distance	Tabular transformation: min–max normalization	Normalized features	Relative density	[121]
Layer thickness, laser power, and laser scan speed	None	Raw inputs in explicit form	Open porosity	[122]
Part bed temperature, laser power, scan speed, scan spacing, and scan length	None	Raw inputs in explicit form	Shrinkage	[123]
Laser power, speed, and hatch spacing	None	Raw inputs in explicit form	Process maps for part surface roughness and density	[124]
Laser power, scan speed, and powder feed rate	Min–max normalization of inputs and outputs	Normalized dataset early example	Deposition height control	[119]
Layer thickness, laser power, scanning speed	None	Raw inputs in explicit form	Bead width	[125]
Sheath gas flow rate, Carrier gas flow rate, stage speed	None	Raw inputs in explicit form	Line width and line roughness in aerosol jet printing	[126]
Spray angle, traverse speed, and standoff distance	Scaling input and outputs to [−1 1]	Scaled dataset	Track profiles in cold spray AM	[127]

(continued)

Table 3.4 (continued)

Feature source	Feature engineering technique (FET)	Parametric features	Application of planning features	References
Laser power, laser speed, spot size, layer thickness	KB transformation	Normalized machine independent parameters (laser energy density, laser radiation pressure)	Porosity prediction (pass, flag, fail)	[128]
Layer thickness, printing speed, nozzle temperature, layer index, printing pattern direction, and neighborhood time difference	Featurization of as-simulated process, spatiotemporal registration	Synthetic printing process variables	Layer-wise thermal field prediction	[129]
Process parameters	Feature fusion	Experimental and synthetic features	Bond quality and porosity	[130]
Process parameters	None	Explicit parameters and measurements	Deviation prediction	[131]

are indicative of defects which may be specific to one type of AM. For instance, several examples exist where powder layer features were used in data-driven models to detect anomalies of PBF-based AM. In addition to material (powder) layers, a part's layer can be captured or modeled to extract different feature types (visual, thermal, etc.). The physical significance of these layer features is dependent on the appearance or emission (optical, thermal, or acoustic) being captured or property being modeled. For example, visual features usually highlight the anomalies which cannot be scene with human eye due to their scale of existence. Thermal layer features, on the other hand, provide overall temperature distributions that can help identify certain defects (e.g., porosity) and change the course of an AM process. Layer features are usually explicit or simple in nature where inception and evolution of visual defects can be identified, and closed-loop systems for single variable (e.g., power, material) could be developed. Table 3.5 is a summary of engineering layer features for data-driven AM.

A multi-sensor system was deployed to capture layer-wise temperature (IR and TC) and vibration (accelerometer) signals [132]. These were fused with non-sensory material and direction parameters to predict part's tensile strength. Before predictions were possible, non-sensory features were encoded using one-hot encoding and statistical sensor features were extracted and normalized. A recursive selection of features was performed at the end to gauge the performance on prediction task. The featurization pipelines for graphic sources of AM layers are more sophisticated and customized. Next Manufacturing Center at Carnegie Mellon University has made representative efforts to engineer powder layer features from graphic sources in visual range. In one such work, multiple patches at different scales, each capturing a different aspect, were extracted from visual images of powder layers [133]. The patches were manually labeled and used to predict anomalies caused by interaction between powder spreader and powder bed. In another refined pipeline, layer features were extracted using several image filters and clustered together based on their similarity. Each of the clusters was later averaged to get mean feature vectors for downstream classification tasks. Their work mostly focused on capturing different powder bed anomalies in an unsupervised way. Figure 3.5 is an example of multi-step processing pipeline embedded with ML to detect defects of AM layers.

In addition to these sophisticated pipelines with one type of transformation (e.g., graphic-based), integrated feature engineering is prevalent for this category of features where different types of transformations are used to generate better representation of features. Yazdi and colleagues developed a feature fusion pipeline where learned spatial (CNN-based) and extracted spatiotemporal (wavelet transform-based) features were concatenated to detect powder layer defects [137]. Researchers at Argonne National Laboratory (ANL) in the USA also made representative efforts by developing complex engineering pipelines of layer features to detect porosity. In one such work, a sparse dictionary of thermal layer features was learned using K-means SVD. A validation of learned dictionary validated noise reduction and sharpness of learned features [145]. In a companion work, they developed parallel FEPs to evaluate and compare learned features [149]. Both pipelines followed graphic transformation (wavelet transform for graphic denoising), feature extraction (PCA

Table 3.5 Feature engineering techniques, pipelines, resulting features, and their applications at the process phase of AM for layer-based sources

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features (forms)	Application of layer Features	References
Multi-input: layer thermal history (IR and TC for point temperatures, accelerometer for vibration), print direction, and material	Encoding, signal transformation	Encoding of non-sensory features → Extraction of sensor features → Normalization of extracted sensor features (zero mean and unit variance)	Encoded directions and material features, extracted and normalized sensor (mean, standard deviation, skewness, kurtosis) features (explicit)	Part tensile strength	[132]
Visual images of powder bed layer	Graphic transformation	Manual extraction of multi-scale image patches	Extracted image patches (implicit)	Six powder bed anomalies	[133]
Visual images of part and powder surface layer	Feature learning, feature fusion	Separate feature learning → Fusion of learned features	Learned feature maps (implicit)	Defects related to different process conditions	[134]
Visual images of FDM process	Graphic transformation	Image size augmentation	Transformed images	Part CoC: under extrusion, over extrusion, normal	[135]
Visual images from post-scan and post-recoat	AM data registration, feature fusion	Fusion of raw images (different lightening conditions), fusion of learned features	Raw image with different lightening conditions (explicit), learned features (implicit)	Microstructure defect prediction	[136]

(continued)

Table 3.5 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features (forms)	Application of layer Features	References
Layer-wise optical images	Graphic transformation, sequence transformation	Wavelet transform → Texture analysis (on wavelet coefficients) and CNN for feature extraction	Transformed graphic and sequence features (implicit)	Powder layer defects	[137]
Layer-wise optical images (FDM)	AM data registration, graphic transformation	Registration → Pixel-based segmentation (feature extraction) → Normalizing	Extracted and normalized image features (implicit)	FDM layer-wise quality	[138]
Layer-wise images (FDM)	Image preprocessing	NA	Processed images	Geometric defects in the layers	[139]
Spread powder images	Graphic analysis	3D digital image correlation to identify powder bed anomalies	Identified powder anomalies	Microstructural defect prediction	[140]
Layer-wise visual images of powder bed	Graphic transformation	Raw images → Padding → Bank of feature extractors (filters) → Clustering of resulting feature vectors → Mean feature vector for each cluster	Extracted feature vectors of images (implicit)	Six types of powder bed anomalies	[141]
Layer-wise images (EBM)	None	N/A	Raw images	Layer defects	[50]
Video of powder bed surface	Sequence transformation, graphic transformations	Video → Frames → Noise removal and resizing and cropping → Image labeling → Data balancing (RUS and ROS)	Transformed image dataset	Quality (OK or defective)	[142]

(continued)

Table 3.5 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features (forms)	Application of layer Features	References
Thermal images of layer and part surface	None	N/A	Raw images	Defect detection	[143]
Layer image from different printers	Graphic transformation, transfer learning	Real-time pixel-wise semantic segmentation	Extracted and learned image features (implicit)	Anomaly detection and classification	[144]
Infrared camera for layer thermal monitoring	Graphic transformation	Image denoising with sparse coding and *K*-means SVD	Denoised and reconstructed thermal image	Subsurface defects (e.g., porosity)	[145]
Layer-wise optical images of the build	Graphic transformation	Spectral graph theoretic and multifractal[1] analysis	Statistical image features, spectral graph theoretic Laplacian eigenvalues, multifractal, and lacunarity features	Lack of fusion porosity	[146]
Paraxial optical images of layer and extruder	Graphic transformation	ROI detection → Cropping → Image resizing	Segmented image	Under and over extrusion	[147]

(continued)

[1] A fractal is defined as a shape that embodies geometric similarity across multiple scales.

Table 3.5 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features (forms)	Application of layer Features	References
Pyrometric images of layer	Graphic transformation	Raw image → Size reduction → Rotation → Contrast → Binarization-based segmentation	Segmented image	Porosity prediction	[148]
Thermographic images of layer	Graphic transformation, feature learning	Method I: wavelet transform for denoising → PCA → ICA Method II: wavelet transform for denoising → SVD → SDL	Denoised, decomposed and learned features	Porosity defects	[149]
Thermal image of the layer	Graphic transformation	High-resolution layer with registered melt pool images → Grayscale conversion → Binarization of Islands (or hotspots) → Spatial SIZER → Pixel-wise Flagging → Feature extraction	Ratio of flagged pixels, number of flagger Island, and maximum Island size on the layer	Evaluation of process control (in or out of control)	[150]
Powder bed images	Data augmentations graphic transformation	Image thresholding (Otsu's binarization) → Rescaling → Normalization	Improved images of powder bed	Porosity identification	[151]
Powder bed images	Graphic transformation	Raw images → Normalization → Contrast enhancement → Object identification → Morphological filtering	Processed images of powder bed	Five powder bed anomalies	[152]

(continued)

Table 3.5 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features (forms)	Application of layer Features	References
High-resolution layer images	Graphic transformation	Layer images → 3D convolution filters → Feature matrix	Feature matrix corresponding to different 3D filters and sensor modalities	Layer-wise defect detection	[153]
Layer-wise images of PBF	AM KB feature engineering	Raw images → Adaptive local thresholding → Intuition-based feature selection → Extraction	Mathematically formulated features from observations	Feature-based build defect classification	[154]
Highly resolving layer-wise images in PBF	AM KB feature engineering	Layer images of surface → 9 image patches → Feature-based analysis → Feature extraction	Mean and median gray values, variance, dynamic range, surface reflectivity, periodicity (*2), periodicity ratio	Process monitoring	[155]
Visible layer-wise images	Graphic transformation	3D array of images → Vectorization → Matrix → VPCA → PCs (with weights for each pixel) → Hoteling T square distance	Resulting statistical descriptor of each frame	Spatial identification of layer defects	[156, 157]

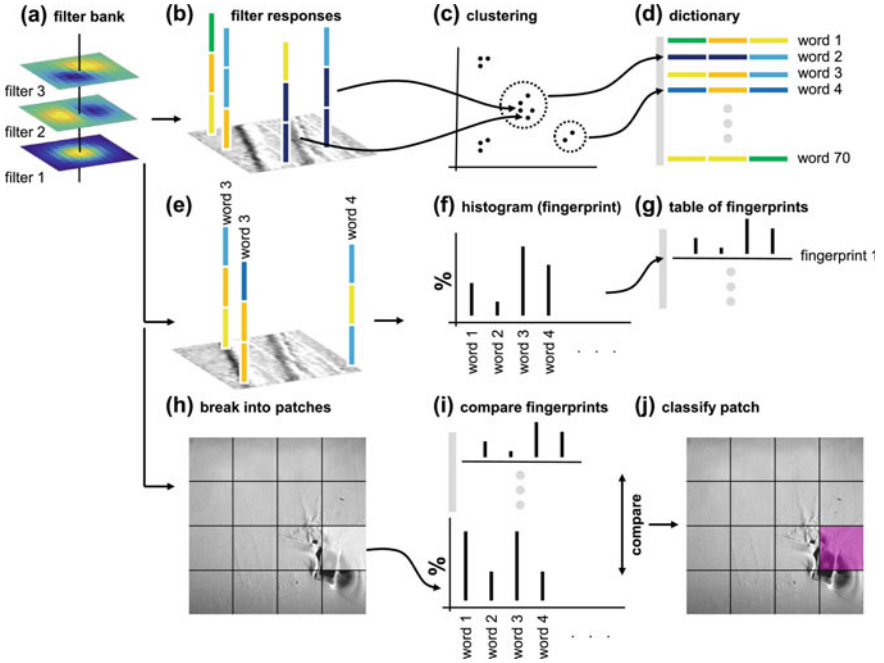


Fig. 3.5 Figure shows multi-step processing framework to classify the defective patches of powder layer during the powder spreading process of a PBF process. Figure used with permission from Elsevier [141]

or SVD), and feature learning (NN or SDL). They found that the performance of both pipelines was similar when detecting subsurface defects. Interested readers are referred to their work for details. Lastly, examples of simple transformations can be found which include mere cropping [142], resizing [135], or denoising by preprocessing [139]. In some cases, raw images were directly employed in the deep models for implicit feature learning [50, 143].

Feature operations such as fusion [4] and transfer [15] are common for layer features. Layer features are often learned implicitly. Caggiano et al. separately learned features for powder and part from visual images and later fused the learned feature maps to identify defective process conditions in a relatively simple learner [4]. Features were also fused in another work to predict structural defects at microscale but their sources were purely based on powder layers with different layer lightening conditions (e.g., brightness) [6]. Mahmoudi et al. captured high-resolution thermal images of the layer with registered melt pool images. These were converted into grayscale representation and binarized for the identification of hotspots (aggregate of melt pools at high temperature). Notable methods that relied on features from layer-based sources for applications outside of data-driven models also exist. For instance, Bartlett and colleagues used full-field infrared thermography of layers to identify subsurface defects in SLM [158]. Features for this identification were based

on layer-wise surface temperatures. Digital image correlation, a technique that allows to track and measure changes in images over time has been applied to powder layer images. 3D DIC was employed to detect anomalies of powder bed which served as inputs to a Bayesian classifier for identifying microstructural defects [140].

3.7 Engineering of Process Features: Melt Pool

Melt pool features are representative of process parameters, scan strategy, substrate geometry, material properties, and more. There has been a special interest among the AM community on featurizing melt pool for downstream data-driven applications. The underlying idea is to discard the previous sources (process, material, deposition, etc.) and solely work with melt pool representations to engineer features that can be modeled for diverse applications such as control, security, defect detection, and process stability. The list could certainly continue based on the specific applications. Since melt pool is a very small (e.g., micrometers in PBF and millimeters in DED) and fast-changing 3D object, its capturing or modeling is not without bottlenecks; this results in limitations (w.r.t. data fidelity) regarding the accuracy of the resulting representations. Vision-based areal capturing of melt pool is common (off-axial or coaxial) resulting in a 2D graphic representations. Depending on the working range (band of electromagnetic spectrum) and the characteristics (ROI, resolution, arrangement) of the sensor, different types of raw image streams can be generated. Interested readers are referred to the excellent review in AM which could provide insights on monitoring techniques [51]. We should also acknowledge that there is a wealth of image processing techniques (in-process monitoring) for AM data that precedes the feature engineering pipelines of data-driven models. Due to the high frequency of melt pool featurization, works with typical transformations (e.g., standardization), analysis, object detection, and single CoC-based control loops (e.g., vision-based width control) are excluded from comparison and subsequent analysis. Similarly, this review is not focused on monitoring. Monitoring is a source of vast and diverse AM data but only the works processing this raw data to generate features for data-driven applications are considered. As a result, two categories of melt pool features are considered: feature obtained (via selection, extraction, or learning) for downstream learners or special melt pool features (in contrast to conventional descriptors) engineered with significant AM knowledge involvement to drive data-based applications. Featurization of melt pool for data-driven AM has been summarized in Table 3.6.

The work by Gawade et al. is a highlight of the significance of feature engineering when shallow learners are involved for downstream prediction tasks [186]. They improved the performance of porosity predictor by fusing features extracted from virtual and experimental sources. The two data streams were first mapped for feature extraction and extracted features were placed in three categories (pure empirical, pure synthetic, and hybrid) for systematic selection. In their results, model trained with a specific hybrid feature set was found to capture the strengths of both data streams

Table 3.6 Feature engineering techniques, pipelines, resulting features, and their applications at the process phase of AM for melt pool data sources

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features	Application of process features	References
IR melt pool images	Data preprocessing	Standardization of LSTM inputs	Standardized data	Highest melt pool temperature	[159]
Melt pool images	KB graphic transformation (4-step feature extraction)	Transformation of initial coordinate information and interpolation of temperature response → MPCA on tensor data structure of melt pool image sequence → Low-dimensional tensor subspace	Tensor subspace of PCs (volume of the bounding convex hull and maximum norm of residual)	Defect distribution in fabricated layer (layer-wise quality)	[160]
FE-based thermal simulation data	3D transformation	Featurization of as-simulated melt pool: approximation of original simulation data using SVD	SVD-based latent space from LHS-designed FEM simulation, implicit	Robust manufacturing conditions	[161]
Melt pool visual images	Data registration, feature learning	Melt pool spatial registration with the measured porosity, learning of melt pool porosity features in CNN	Learned activations in CNN	Automated porosity predictions	[162]
Melt pool videos	Data preprocessing	Having videos of the fixed length frame and resolution, omission of first/last frame and high-speed videos resulting in only 10 middle frames of the video were selected (10 ms) also frames were center cropped	Preprocessed images	Monitoring of laser track welds w.r.t. height	[163]

(continued)

Table 3.6 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features	Application of process features	References
Melt pool digital images (from different scan strategy while keeping parameters the same)	Graphic transformation	Raw image in grayscale → Thresholding for boundary detection → Melt pool shape approximation using least squares → Image resizing	Transformed MP image	In-situ melt pool characterization (based on area), no melt pool observed (Z), smaller melt pool (S), normal melt pool (N), and larger melt pool (L) show an example of melt pool labels	[164]
MP digital images	AM KB graphic transformation	Melt pool image stream → Kalman filter-based tracking → Melt pool centroid → Detection of three ROIs (MP, plume and spatter) → MP features → Feature fusion	MP features (keyhole area + Grayscale bins), plume shape features, segmented spatter features	Process state classification	[165]
FEM generated MP image from 9 simulations (under variations of laser power)	Feature transfer, feature learning	Feature transfer (learned) from AlexNet (aka pretraining of its backbone), feature learning in dense layer (high-level features for the prediction task)	AlexNet feature maps	Disturbance detection	[166]
Melt pool thermal images	Graphic transformation	Thermal image → Polar transformation → FPCA	Principle components	Porosity prediction	[167]

(continued)

Table 3.6 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features	Application of process features	References
High-speed digital camera	Data registration or mapping	Neighboring effect modeling method (NBEM)	Build time, laser power, scan speed, and neighboring effect	Melt pool size prediction	[168]
MP optical images	Graphic transformations	Raw images → Noise removal → Boundary detection → Area calculations	Melt pool areas	Process defect classification	[169]
MP thermal image and FEM	Feature learning, feature fusion	Raw images → CNN for Feature learning → Feature fusion	Physical features from FEM (real and synthetic features)	Porosity prediction	[170]
Melt pool digital images	Graphic transformation	Raw image → SIFT → HoG → Word dictionary → Multi-modality feature histogram	Encoded melt pool nose, melt pool tail, and spatter features	In-situ detection of keyhole porosity	[171]
Thermal history videos (IR images)	AM KB graphic transformation	IR images → Extraction of temperature histories (at hotspots) → LOGO-CV-based recursive feature selection using GA	Feature set of thermal histories	Porosity	[172]
Melt pool visual images (PBF)	Feature learning	Melt pool images → Cropping → 2D sharpening kernel matrix (*2) → Convolutional autoencoder	Learned low-dimensional deep representations of MP	Data annotation and process monitoring	[173]
X-ray DED melt pool videos	Graphic transformation	Raw images → Edge images → Templates → Scoring using chamfer distance → Bayesian decision	Melt pool shape and dimensions	Automated feature extraction	[174]

(continued)

Table 3.6 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features	Application of process features	References
Process optical images	Graphic transformation	Raw images → Enhanced images → Customized computer vision algorithm	Quantity and trajectory of spatters	Powder bed thickness prediction	[175]
Optical images with spatter features	Graphic transformation	Captured images → NN architecture for image segmentation → Extracted spatter features	Contours of extracted spatter	Image segmentation	[176]
Captured melt pool video	KB graphic transformation	Raw image → Extraction of spatter → Melt Pool → Extraction of melt pool motion feature	Melt pool motion features	Characterizing the dynamic variation of melt pool	[177]
Melt pool thermal images	KB graphic transformation	Raw thermal image → Gray image → Dataset augmentation (mirroring and flipping) image enhancement GAN	Melt pool images with better CII	Melt pool image segmentation	[178]
Melt pool optical images	Feature learning, feature visualization	Images → Direct CNN → Activation maps, deconvolution, grad-CAM	Learned melt pool implicit features	MP classification	[179]
Melt pool optical images	Feature visualization	Grad-CAM-based visualization of features	Learned melt pool features	MP classification	[180]
Melt pool optical images	Graphic transformation, feature selection	Melt pool frame → ROI detection → Grayscale processing → Gaussian filtering → Binarization → Contour extraction → Features → Feature smoothing → EEMD → Z-score normalization → PCA → PC selection	Selected principal components	Porosity classification	[181]

(continued)

Table 3.6 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features	Application of process features	References
Melt pool data from coaxial video monitoring	Graphic transformation, signal transformation	Video → Frames → MP characteristic extraction → Time dependencies of melt pool characteristics → STC	Local features of joint behavior of melt pool characteristics	Analyzing melt pool dynamic (behavior) features	[182]
Melt pool visual images in DED	Graphic transformation	Images → Min–max scaling → (a) Magnitude of intensities → Histogram of intensities (b) Magnitude of gradients → Histogram of gradients → Stacking and normalization	Intensity and gradient features	Melt pool classification	[183]
FE simulation	Graphic transformation	Power field → Artificial image → Segmentation (Otsu thresholding) and fitting (Suzuki's contour) algorithms → Ellipse fitting by least squares Solution → Feature Extraction	Length and width of the melt pool	Build geometry prediction	[184]
Melt pool infrared temperature signal in FDM	KB FE of signal	Temperature signal → Point-by-point average and standard deviations of good signals → Selection of multiple for standard deviation	Upper and lower temperature control limits	Real-time defect detection	[185]
Melt pool images from pyrometer camera and simulations	Graphic transformation	Empirical and simulated melt pool mapping based on spatial and temporal similarity → FPCA of melt pool images → Feature extraction → Systematic feature selection	Selected hybrid feature set from empirical and simulated sources	Porosity prediction	[186]

(continued)

Table 3.6 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features	Application of process features	References
Melt pool video stream	Graphic transformation, feature learning	Collected frames → Process zone extraction (w.r.t. MP centroid) → CNN-I for spatial feature learning from a single frame → Sequential rearrangement of learned feature → CNN-2 for prediction using temporally arranged spatial features	Learned spatial features in temporal sequence	Process monitoring	[187]
Spectrometer and plume intensity camera	Graphic transformation, signal transformation	Fusing data from sensors as network graph (single layer) → Kronecker product of graphs (multiple layers)	Dictionary of signal patterns	Layer-wise lack of fusion defect prediction	[65]
Melt pool thermal images	Graphic transformation	Raw thermal images → Data rescaling → Spherical transformations → Interpolation via bi-harmonic surface interpolation method	Continuous temperature models with identical function support	Porosity prediction	[188]
High-speed visual images	Graphic transformation	Raw image → Otsu-based thresholding for segmentation → Connected component analysis → Spatter and laser heated zone components → Extraction of statistical descriptors	Spatter (number, average area, and area of enclosing Convex Hull) and laser heat zone (area) features	In-situ monitoring in PBF	[189]
Thermal monitoring data	Graphic transformations	Reference image → Extraction and matching of melt pool boundary → Similarity feature and judgment threshold	Melt pool similarity feature	Online defect detection	[190]

(continued)

Table 3.6 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features	Application of process features	References
Visual image melt pool and plasma arc in PAM	Feature learning, feature fusion	Raw image → (a) Learning of deep feature map in encoder (b) Learning of shallow feature map in CNN → Fusion of feature maps → Decoding of fused maps for segmentation	Shallow and deep feature maps	Pool and plasma arc segmentation	[191]
Melt pool thermal image series	Graphic and signal transformation	Video → Image series analysis formulation → Image registration between Two Consecutive frames → Error modeling → Feature extraction	Registration (distance and rotation shift) and error modeling-based (absolute value of intercept, variation, and covariance from GPM) features	Anomaly detection	[192]
Timeseries melt pool temperature signal	Signal transformation (timeseries analysis)	Raw signal → Dynamic time warping → Timeseries distances	Timeseries distance feature	Anomaly detection	[193]
Thermal images having melt pool and boundary	Graphic transformation, image registration	Raw images → Image filtering → Customized feature extraction → 3D spatial registration of features for visualization	Thermal gradient at the solidus-to-liquidus region, the maximum temperature, the area, and the length-to-width ratio of the melt pool	Thermally driven material characteristics and part quality	[194]
High-speed videos of powder-melt pool interactions	AM KB signal analysis	Videos → Particle behavior (pre- and post-impingement) analysis	Particle behavior features	Observation of particle melt pool impact events	[195]

(continued)

Table 3.6 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features	Application of process features	References
Optical images containing melt pool, spatter, and plume	Graphic transformation, feature learning	Images → Identification of plume and spatter signals, images → DBN for feature learning	Implicitly learned spatter and plume feature (areas, lengths, widths, orientations, perimeters, pixels, situations, mean intensities)	In-situ monitoring	[196]
Melt pool thermal images	Data augmentation, feature learning	Unbalanced image data → SVM-SMOTE → VAE	Melt pool sparse representation	Anomaly detection	[197]
Melt pool visual images	Graphic transformation	Image → Thresholding → Masking connected components → Energy consumption	Melt pool region and by-product features	Classification of process zones	[198]
Thermal MP data	Graphic transformation	Images of MP → PCA → Transformation of PCs (zero mean)	Transformed principal components analysis	Cost of the spatial distribution of defects	[199]
Melt pool optical images	AM KB graphic transformation	Photodiode signal or camera images → Melt pool feature extraction	Melt pool intensity, area, length, and width	Quality control or feedback	[200]
Melt pool images from vision sensor	Graphic transformation	Adaptive image thresholding → Connected component analysis → Iterative energy minimization → Region selection	Features of lack of fusion, conduction, and keyhole melting	Adaptive feature detection	[201]
MWIR camera	Graphic transformation	Raw image stream → Grouping of pixels into super-pixels → Segmentation → Melt pool optical flow → Feature extraction	Geometric and thermal features of melt pool motion	Melt pool monitoring	[202]

(continued)

Table 3.6 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Process features	Application of process features	References
X-ray imaging	Graphic analysis	KB analysis of X-ray images	Features relating to the formation of spatters	Spatter formation mechanism	[203]
Melt pool optical images	Graphic transformation	Raw image → Thresholding and binarization → Image moments → Melt pool dimensions → Mapping with layer	Reconstructed layer with melt pool images	Surface defect evaluation	[204]
High-speed camera capturing melt pool surface	Graphic analysis	Customized method for the analysis of powder particles	Melt pool flow features	Melt pool dynamics	[205]
Melt pool visual images	Graphic transformation	Melt pool video frame → Spatial feature extraction (area, width-length ratio, amount of spatter, spatter quantity, melt pool intensity, spatter direction, texture feature (HOG)) → Temporal feature extraction	Melt pool temporal features (mean and variance of time-varying spatial features)	Quantification of printing stability	[206]
Optical emission spectra signals	Signal transformation, feature learning	Optical spectrum → AE → Learned features → Semi-supervised clustering	Implicitly learned spectra features	Prediction of quality labels	[207]
Visual images of melt pool region	Graphic transformation	Image → Feature extraction	Feature descriptor of spattering pattern	Spatter-based classification	[208]

and hence outperformed other data streams. Hossain et al. hand-engineered infrared signals and defined an upper and lower control limit on the signal magnitude based on the data from good prints in FDM [185]. First, the temperature signals from good quality parts were averaged in a point-by-point fashion. The standard deviations of these signals were similarly determined, and a multiple of three standard deviations from the average was found to be an ideal choice to define upper and lower limits with minimal false positives. Recognizing the information loss when only spatial features of melt pool are learned in deep architectures, Zhang et al. engineered a hybrid deep setup for utilizing spatiotemporal features of a melt pool video stream [187]. A CNN was first used to learn spatial feature of each image frame. The learned features were then sequentially arranged to account for temporal features. The resulting 2D array was employed in a second CNN for prediction task. They used a sliding window to group melt pool frames and justified its size and sliding step based on AM knowledge (solidification time) and hardware setup (capture rate). Their companion works have focused on similar techniques to extract features from melt pool data streams [209]. Snyers et al. used simulation to generate monitoring data of the melt pool [184]. This is depicted in Fig. 3.6. Their example may be unique in the sense that an image (graphic representation) was extracted instead of usual tabular representations. The resulting images were subjected to segmentation, fitting, and elliptical approximation before geometric features could be extracted.

There are other methods to take into account the spatiotemporal effect like the one used by Guo et al. for thermal images [210]. Decision-level fusion, once a prediction has been made, also exists in addition to data- and feature-level fusion. Tian et al. fused the outputs of two separate CNN which they named as PyroNet and IRNet in a weighted manner to reach final prediction (good or bad) [211]. Decision-level fusions are out of the scope of current text and usually fall under ensemble-based (e.g., aggregation) data-driven modeling. Seifi et al. developed a knowledge-based

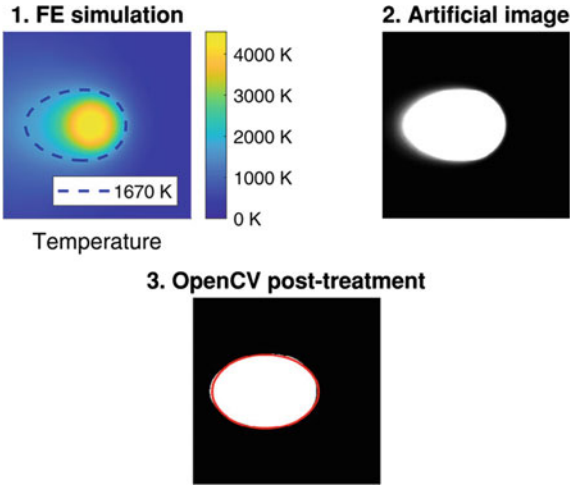


Fig. 3.6 Generation of melt pool data in graphic representations from simulations. An artificial image is extracted from the surface of FE simulation based on melt pool boundary temperature. The extracted image is processed (e.g., thresholding and binarization) to get a melt pool contour. Figure used with permission from Laser Institute of America from [184]

transformation pipeline for thermal stream of melt pool images [160]. Multilinear PCA was used to obtain low-dimensional representation of the input tensor. Features of subspace defined by the principal components were used to identify defect distribution in the printed layer. Wang et al., used SVD to reduce high-dimensional thermal response of as-simulated melt pool. The resulting latent space was used to directly link simulation control variables with their defined manufacturing conditions (and noise factors) in a surrogate modeling, and hence, the computational expenses by running the actual simulation were avoided [161].

Due to the high dimensionality of melt pool data, feature learning to generate low-dimensional representations deserves special mention. Fathizaden et al. learned a deep representation of melt pool visual images in a conditional AE [173]. A unique application of these representations was automatic annotation of melt pool images into normal and abnormal categories. The learned low-dimensional feature vector was later employed for anomaly detection. The work by Guo et al. is an example of both learning melt pool features and testing different fusion scenarios [170]. Features of thermal images were learned in a convolutional architecture and later combined with layer and melt pool features (synthetic) for porosity classification. They found that a full-fusion scenario (learned, layer, and synthetic melt pool features) provided a better performance. Zhang et al. employed feature learning of visual images for segmenting melt pool from plasma arc [191]. Two types of implicit features, namely shallow (CNN-based) and deep (encoder-based), were learned and fused. The fused feature vector was used in a simple decoder to segment the input images. Zhao et al. stressed on learning a sparse representation from thermal monitoring of melt pool by using a variational AE [197]. Synthetic minority over-sampling technique (SMOTE) was used to augment the dataset by balancing the minority (low number of defective images) class before features could be learned. Bypassing the design of a specialized feature learning architecture, Zhang et al. used conventional CNN to learn features of melt pool visual images and used these for automated porosity predictions [161].

There exist some unique melt pool features or descriptors that are engineered by fusing AM knowledge with melt pool data streams. To overcome the limitations of conventional melt pool descriptors when characterizing its dynamic response, Li et al. introduced a melt pool motion feature as highlighted in Fig. 3.7 [177]. Melt pool was first processed to identify its moving direction and centroid. It was later separated from its by-products (e.g., spatter) which resulted in an approximate shape-based ROI. Finally, melt pool motion feature (distance between centroid and boundary) describing changes in its size and vibration direction was extracted. The extracted features were employed in a simple clustering algorithm for process monitoring. Feng et al. introduced temperature distribution similarity detection method for online defect detection [190]. They refer to their technique for melt pool images as "external boundary alignment, internal correlation detection". As such, a similarity feature of detected melt pool relative to the reference melt pool was designed using standardized covariance correlation function.

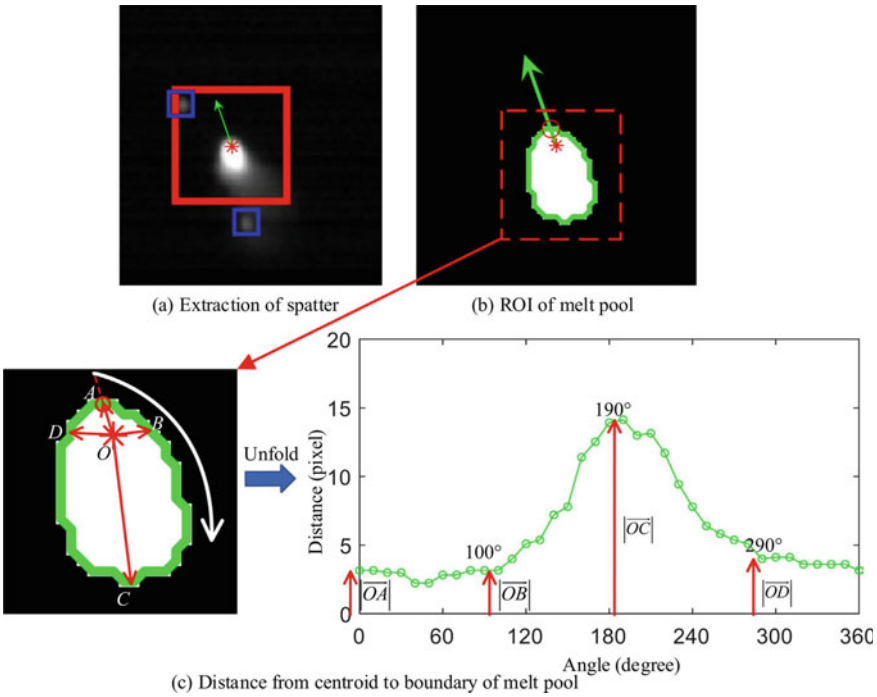


Fig. 3.7 Melt pool motion feature to capture dynamic behavior of MAM processes. Melt pool was first processed to identify its moving direction and centroid. It was later separated from its by-products (e.g., spatter) which resulted in an approximate shape-based ROI. Finally, melt pool motion feature (distance between centroid and boundary) describing changes in its size and vibration direction was extracted. Used with permission from Elsevier [177]

3.8 Engineering of Process Features: In-Situ Geometry

2D-based capturing of a layer brings limited information on a part being printed. To enhance information fidelity or to perform complex prediction tasks (e.g., in-situ 3D surface defect detection), it is appropriate to capture or model the in-situ part geometry. This approach also introduces a third category of AM models in addition to as-designed or as-printed models. We will categorize these models using as-simulated or as-captured in this text, depending on their source or origin. Table 3.7 summarizes the featurization of geometries captured or modeled during the process.

Korneev et al. simulated the build process using a sliding subdomain approach [88]. The resulting 3D voxel representation was transformed to 2D images. A height map of images was constructed and used to predict the 3D shape of solidified droplets. As-simulated model images (layer-wise) were used in conjunction with as-designed and as-built model images to detect cyber-attacks based on the unexpected geometry variations [215]. This was accomplished by extracting statistical features from different patches of these images. The work also sheds a light on the importance of

Table 3.7 Feature engineering techniques, pipelines, resulting features, and their applications at the process phase of AM for geometry-based sources

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	In-situ geometry features	Applications	References
Simulation of the build process using sliding subdomain approach	3D transformation	Substrate shape (a 3D voxel representation) to 2D image	Height map images	3D shape of the solidified droplet	[88]
In-situ imaging of the part (corner images of the part being printed)	Graphic transformation	Images → Corners of the component → ROI as input to CNN	Part corners in graphic representation	Probability of a corner being warped	[212]
Images of FDM part being built	Graphic transformation	Cropping, shift, flip, brightness, saturation, channel shift → CNN-based feature extraction	Implicitly extracted features of CNN	Spaghetti error prediction in FDM	[213]
Sideway images of the part being built	Data preprocessing	Layer-wise images → Resizing → Normalization	Transformed images	Five class classification of AM process quality	[214]
Multi-sources: as-designed, as-simulated, and as-captured images	Graphic transformation	Raw images → Patches (8) → Statistical feature extraction (3)	Statistical features of image patches	Cyber-attack detection	[215]

(continued)

Table 3.7 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	In-situ geometry features	Applications	References
Optical images of the layer from off-axis camera	Graphic transformation	16-bit TIFF images → Cropping → Background removal using intensity difference → Feature extraction	Mean and median of image pixels	Drift layer or non-drift layer	[216]
In-situ point cloud	3D transformation	Point cloud processing: filtering, segmentation, surface-to-point distance calculation, point clustering, feature extraction by ML	Extracted surface features	Detection and classification of surface defects	[217]
EOSPRINT slice (.SLI) files	3D transformations	Layer screenshots → 3D point cloud extraction → Overlaying of CAD information on powder bed	Shape orientation and position of parts being built	Complementing powder bed monitoring algorithm	[144]
Optical microscopy of printed beads	Graphic transformation	Image → Processing → Extraction of solidified bead parameters	Solidified bead features	Process parameter optimization	[218]
Real-time scanning of deposited track	3D transformation	Scanned profile → Local to global coordinate transform → Matching of as-designed and as-built geometry → Geometric feature extraction	Track width, deposition height, discrepancy, and discrepancy area	Online geometry monitoring	[219]

(continued)

Table 3.7 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	In-situ geometry features	Applications	References
EPMP-based monitoring of 3D surface topography	AM KB feature engineering	Image → Layer-wise 3D contour detection → Stacking of contours → Comparison of digital 3D volume with CAD	Layer-wise geometric accuracy of contour	In-situ 3D monitoring tool	[220]
Layer-wise imaging of geometry	Graphic transformation	Layer-wise image → Smoothing → Intensity correction → Segmentation → Masking with nominal shape → Extraction of statistical descriptors	Deviation maps and statistical descriptors of edges and regions	Layer-wise detection of geometrical distortions	[221]
Layer-wise videos	Graphic transformation, sequence transformation	Video → Adaptive thresholding → Rasterization → MPCA → Hoteling's T square distribution	Feature of altered layers	In-situ process authentication	[222]
Geometric features (width, depth, height) of deposited metal trace	Tabular transformation	Construction of feature pattern vector	Explicit geometry features	Power, speed, and material feed rate	[223]
Paraxial thermal images of the DED process	Graphic transformation	Center cropping	Cropped images	Process conditions	[224]

feature extraction and relates it to expertise in data processing. There are examples of generic featurization of as-simulated models which could be employed for data-driven task as in [225]. In contrast to simulating in-situ geometry, Chen and colleagues directly captured a 3D point-cloud representation of a part for rapid surface defect detection [217]. Their main contribution is a sophisticated point-cloud processing pipeline for under-build AM parts. This work is also an example of how intricate the feature engineering pipelines in AM could be. The steps are listed in Table 3.7 (see line for Ref. [217]), and interested readers are referred to their work for detailed description of each step.

In-situ graphic data is often generated in a fashion that can capture part information (e.g., geometry) paving the way for its featurization and subsequent prediction of macrolevel defects. For example, images of a part being built in FDM were captured and subjected to graphic transformation and feature extraction in Ref. [213]. In a related example, single melt track was recognized and later classified using deep learning [226]. The featurization pipeline was the simplest which included manual annotation of the melt track images. Instead of manual or knowledge-driven engineering of features, these were implicitly extracted in a CNN for ensuing task. Saluja et al. processed sideway images of parts being built and used them to predict in-situ warping in FDM [212]. In a related example, off-axial optical images of layers were transformed for feature extraction [216]. The extraction was based on statistical image analysis which led to mean and median of image pixels being identified as the key features. These features were used to identify drifts in the layers. Simple geometry control systems, such as height and/or control, must also rely on some extraction process resulting in features. Since the focus is limited to data-driven pipelines, such works are excluded from this review. Figure 3.8 highlights another example of in-situ featurization of AM geometry for data-driven solutions.

3.9 Engineering of Macrostructural Features

Evaluation is fundamental to qualification and certification of AM-printed parts. Macrolevel CoC is the first ones to come under radar once a part is printed. In the context of data-driven applications, these features are either employed in reverse engineering pipelines (as inputs to model design or process-based CoC) or serve as ground truths to design and process features. Same goes for microstructural features of the next section. Table 3.8 contains featurization of as-built AM models for macrolevel sources.

Registering data of as-built models against process (laser positions) or design (CAD models)-based references is common. Features registered with these references are usually deviations or more sophisticated macrolevel anomalies such as surface defects. The registering process sets the stage for systematic extraction of macro- and microfeatures. The point cloud of AM printed part was registered against laser positions (x, y, and z) for mapping with thermal images to predict point-wise distortions in a DED process [68]. Figure 3.9 shows the recorded deviations of

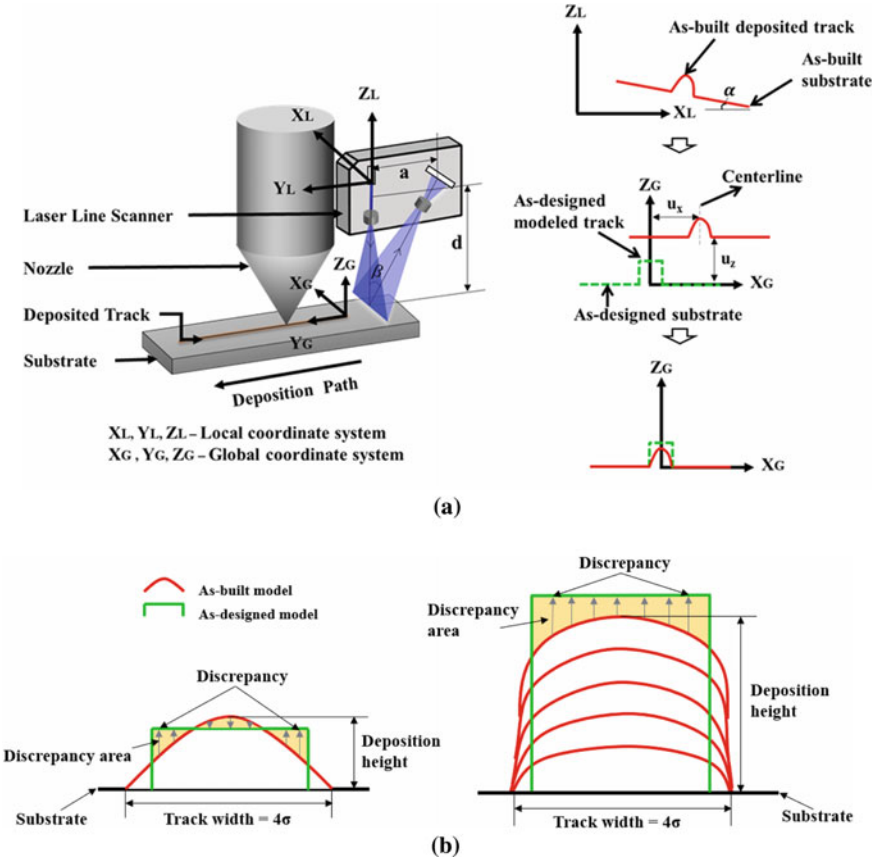


Fig. 3.8 Use of laser line scanner to monitor in-situ geometry. **a** Comparison of as-designed and as-built geometry. **b** Comparison of geometries for different deposition scenarios. Figures used with permission from Elsevier [219]

printed part. Similarly, selected points from as-designed CAD models were used to align point clouds of as-built models for dimensional variation-based classification [228].

Featurization of point cloud-based representations of as-built parts usually follows application-driven customized pipelines. Tootooni et al. proposed a novel technique based on spectral graph theory to extract features from 3D point cloud data obtained with a laser scanner [228]. Their approach enables dimension reduction by using a sparse representation of the point cloud in ML models. Graphic data capturing macrolevel information has been transformed into features as well. Images of printed cross sections were used in a computer vision algorithm to extract geometric features for mechanical response prediction [232]. In another example, images of spanning prints were subjected to PCA to extract principal components for the identification and correction of self-supporting structures [234]. Defects can be identified visually

Table 3.8 Feature engineering techniques, pipelines, resulting features, and their applications at the post-process phase of AM for macrolevel sources

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Post- process features (forms)	Application of post-process features	References
XCT of lattice dome structure with circular cross sections (mm scale)	3D transformation	X-ray images → 3D volumetric models (STL surface) → Point cloud	Point cloud in cartesian coordinates of as-built structure cross sections, as-built polar radii and angle	Polar radii of as-designed point-cloud having the same polar angle as	[227]
3D point cloud of as-built parts	3D transformation	3D point cloud data → Extraction of the Laplacian eigenspectrum → Graph topology quantification → Laplacian eigenvalues	Spectral graph Laplacian eigenvalues as extracted features	Dimensional variation-based classification	[228]
As-printed parts (deviations)	AM KB feature engineering	NA	Deviations (explicit and experimental)	Prediction of unexplained geometric deviation	[229]
Medical imaging (XCT)	Graphic transformation, feature learning	Customized fe pipeline: image alignment (masking) → Patch extraction (cropping) → Feature learning (CNN)	Learned features from CT images (Implicit and experimental)	Bone and background separation	[230]
As-built models (point-cloud)	KB 3D transformation	Customized transformation on point cloud data	Angles and radii of points for in-plane and out-of-plane deviation (explicit)	Geometric shape deviation modeling	[231]
Images of design cross section	Graphic transformation	Computer vision-based geometric feature extraction from image	Geometric features (explicit)	Mechanical response prediction	[232]
As-built models (point cloud)	3D transformation	3D representation comparison (CAD points with scanned points) for deviation feature generation	Geometric deviations (explicit)	Deviation prone process state identification	[233]

(continued)

Table 3.8 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Post- process features (forms)	Application of post-process features	References
Images of spanning prints	KB graphic transformation	Image feature extraction with principal component analysis	PCs of images (implicit)	Identification and correction of self-supporting structures in AM	[234]
Point cloud of point-wise distortion	3D transformation (point cloud registration with thermal images)	Parsing of build code for laser location extraction → Thermal image timestamp → x, y, z location of images	Image locations (x, y, and z) corresponding to distortions	Distortion prediction	[68]
3D scanning of as-printed parts	3D transformation, feature learning	CAD → STL → Transformations (translation, scale up, scale down, rotate) → Voxelization	AE-based voxel representations	As-designed to as-built and as-designed to as-compensated prediction	[235]
Synthetic images from physics simulator	Graphic transformation	Automated annotation generator with gray-thresholding	Mask annotations of printed objects (polygon)	Automated processing of as-built parts	[236]
Electro-optical images of the weld tracks	Automated graphic transformation	Ex-situ height map analysis algorithm	Track width average and standard deviation, continuity	Automated track geometry prediction	[163]

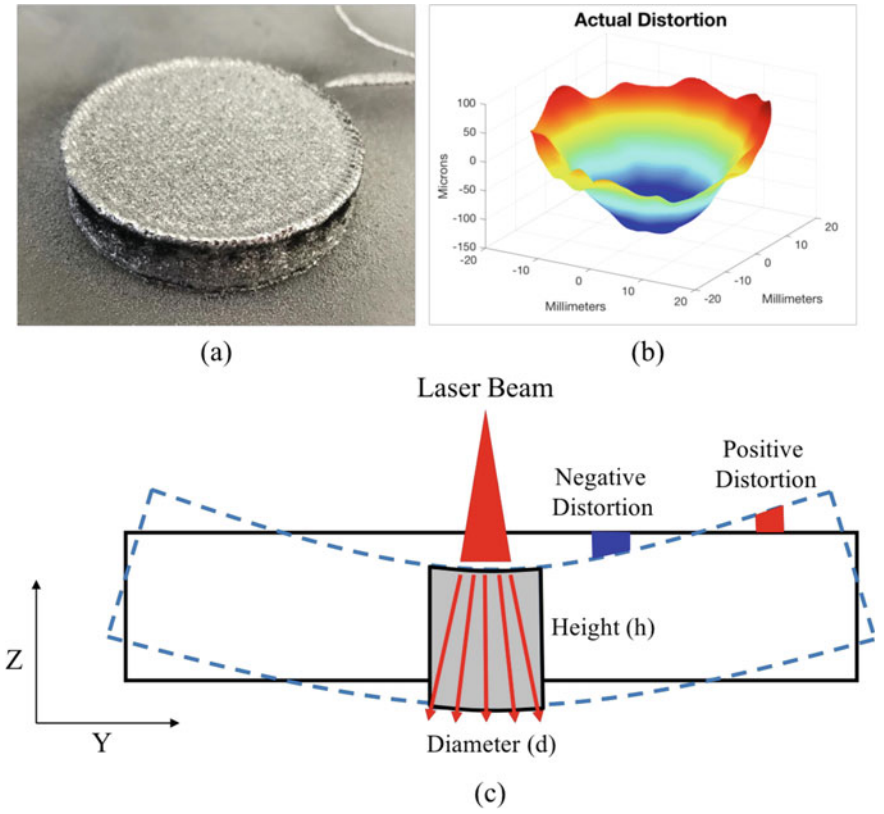


Fig. 3.9 Measurement of deviation features and their subsequent registration against laser positions. The registered features serve as outputs (or labels) against the melt pool images in a deep learner. **a** Fabricated disk. **b** Distortion of the disk. **c** Side view during fabrication. Used with permission from Elsevier [68]

for labeling input data streams. As a result, some macrofeatures correspond to raw measurements or explicit labels. Yadav et al. visually inspected the layer images to assign quality labels [216].

Due to the way XCT data is generated, its featurization can follow graphic and/or 3D transformations. For graphic transformations, a layer-wise (e.g., slices of 3D models) approach is usually adopted leading to transformed images or 2D extracted features. At the 3D level, voxelized XCT representations can help extract features of concern to drive data-driven models. Data from XCT of lattice structures was converted into point cloud representation to get as-built polar radii and angles [227]. These were used to predict as-designed polar radii in a NN. 3D point cloud dataset was converted into a graph using spectral graph theory for feature extraction [228]. The extracted features, graph eigenvalues, were used in a data-driven pipeline for dimensional variation prediction.

3.10 Engineering of Microstructural Features

Microstructural features concern physical characteristics of additively printed parts on a microscale (between macro and nano). For the bulk of literature published in data-driven AM, these represent anomalies in metal-based microstructures such as pores and cracks. These features could also relate to microstructural descriptors, such as grains and phase boundaries or their physical aspects including angles, sizes, or distributions. In the context of empirical modeling, these features usually serve as ground truths to different CoCs but also support inverse modeling of structure-to-process or structure-to-design when treated as direct inputs. The engineering of microstructural features follows their extraction from printed samples which is usually done through metallographic procedures. The resulting microstructural representations are mostly graphic. XCT can also help extract microstructural features, especially anomalies, depending on its resolution. This is done by retaining microscale pixels (if treated layer-wise) or microscale voxels (if treated volumetrically) from obtained CT data. Accordingly, the resulting representations are either 2D or 3D. It's interesting to highlight that an aggregate of microscale voxels (or pixels) could fall into macroscale depending on the flaw size, in which case it will be treated as a macroscale anomaly feature (e.g., macrocrack or severe lack of fusion). In addition to metallography and XCT, KB-extraction of microstructural features is quite common, where resulting features could be morphological descriptors of metallographs with tabular representations. Like for the process phase, virtual or simulated sources (e.g., microlevel simulations of structure) also exist to feed data-driven pipelines and come in 2D or 3D representations.

Table 3.9 provides a summary of featurization efforts aimed at the microscale sources of AM-printed parts.

Engineering of microstructural features using domain knowledge where sophisticated pipelines are involved is quite common. For instance, engineering of graphic representations widely employs regular computer vision transformations in conjunction with KB feature extraction. Popova et al. engineered simulated microstructures for ground truth generation [240]. First, a KB transformation was applied where simulated microstructures were quantified using a known descriptor (Chord Length Distribution). Later, the high-dimensional feature space was reduced using PCA and resulting PCs were used as inputs to data-driven models. Liu et al. applied graphic transformations to SEM images and extracted optimal features to support process optimization pipeline [114]. Their approach is a composite one where generic morphological features were extracted first and later transformed with PCA to a compact representation of the microstructure. Simple graphic transformations were applied to microstructural images of part surfaces for weld quality prediction [242]. Cross-sectional images from standard metallography procedure were processed with a series of graphic transformations listed in Table 3.9 (see line for Ref. [242]). The

Table 3.9 Feature engineering techniques, pipelines, resulting features, and their applications at the post-process phase of AM for microlevel sources

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Post-process features	Application of post-process features	References
Microstructure simulation	AM KB feature engineering	Transformation of thermal latent space to original temperature field for feature extraction	Simulation control variables, melt pool latent response, random variables	Robust conditions for equiaxed material microstructure	[161]
Microstructural images of different material systems	Feature learning	Convolutional deep belief network	Learned representations (low-dimensional latent space) of microstructures	Microstructure reconstruction	[237]
Microstructure (SEM) images, cooling, and latent embeddings	Feature learning	Auxiliary classifier wasserstein gan with gradient penalty	Synthetic SEM image (1024 dim feature vector) again, great, modeling of processing structure relations	Predict cooling methods, assign image score (real or synthetic microstructure)	[238]
Microstructure image	Feature transfer, feature learning	Original images → Encoder → VGG-layers → Encoded reconstruction → Decoder → Images with labeled material phases	Learned microstructural features	Microstructure reconstruction and structure–property predictions	[239]
Simulated microstructure	AM KB feature engineering	Microstructure quantification via chord length distributions → PCA → Feature extraction	Principle components of microstructure	Labels to process–structure predictor	[240]
Microstructure, processing conditions	Feature learning	GAN: generator (condition to synthetic SEM), discriminator (real vs. synthetic SEM)	Learned microstructural features	Process–structure predictions	[241]

(continued)

Table 3.9 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Post-process features	Application of post-process features	References
Microstructure images at different process conditions	AM KB feature engineering	Image of microstructure → Binarization → Noise removal → Extraction of morphological features → PCA → Feature extraction	Dimensional-scale index (Id) and the shape index (Is)	Labels to process optimization	[114]
Microscopic images of part surfaces (metal AM)	Graphic transformation	Graphic transformations: resizing by cropping and segmentation, random noise, and blur to make the model robust	Transformed images	Weld quality prediction (three types)	[242]
Traverse section images of the samples	Graphic transformation	Sequential transformations: random rotation from − 180° to 180°, horizontal flipping, random crop, Gaussian noise, and blurring	Transformed images	Quality prediction (good quality, crack, gas porosity, and lack of fusion)	[243]
Microstructure (SEM) images	Feature learning	SEM images → CNN (adopted from VGG16) → Image feature vector → Tree-based classifier VGG16 (excluding fully connected layers) for image featurization	Learned feature vectors from CNN	Featurization and microstructure classification	[244]
Microscale geometry parameters of porous sound absorbers	Tabular transformation	Raw data → *Z*-score-based transformation	Normalized bar width, bar spacing, and bar height	Biot parameters (acoustic material properties)	[245]
Microscopic images of polished surface	Graphic transformation	Raw images → Segmentation → Patch extraction	Extracted patches	Optimal laser polishing condition for AM part surfaces	[246]
Microstructure simulation	Feature learning, feature importance comparison	3D simulation → KB feature extraction for shallow models and iterations of 3D CNN with different voxel-based features	KB microstructural descriptors, 3D image of microstructure	Mechanical properties	[247]

(continued)

Table 3.9 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Post-process features	Application of post-process features	References
Tabular microstructural features, along with oxygen weight percent, and porosity	Tabular transformation	Microstructural data → Min–Max normalization	Normalized microstructural features	Mechanical properties	[248]
Microstructure (SEM) images	Graphic transformation	Micrographs → Smoothing and unblurring → Hessian response → Pore segmentation → Feature extraction	Extracted porosity features	Ex-situ porosity classification	[249]
Microstructure (SEM) of as-built part (graphic)	AM KB graphic transformation	SEM image analysis in MATLAB for grain growth size and grain growth direction angles	Grain growth size and angles	Process parameters and scan strategy	[85]
Microstructural (SEM) images of as-deposited track (graphic)	Image preprocessing, feature learning, graphic transformation (PCA)	Customized FEP: Segment the images → Feature learning from segments in AE → Graphic transformation	PCs of reconstructed images	Process parameters optimization	[250]
Micro-XCT of as-built parts	3D transformation, graphic transformation	Dimension reduction (3D to image) → Cropping and denoising	Orientation, high-level graphic features	Fiber orientation in microstructure	[251]

(continued)

Table 3.9 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Post-process features	Application of post-process features	References
Micro-XCT of as-built FFF parts	Graphic transformation, 3D transformation	Project images from tomographic scans → full view reconstruction by fusion of feature maps in a CNN (also HSCNN is capable of feature recognition) → registration and parameterization of raw voxel-based uCT 3D data → PCA on parametrized morphology → Pore Clustering	Principle components of parameterized pore	Pore classification (four morphological groups based on pore types)	[252]
Microstructural images of as-manufactured part surfaces	KB graphic transformation	Images → Denoising → Grayscale conversion → Brightness threshold and radial filter → KB feature extraction (feature vectors) → Normalization of feature vectors → Feature selection (PCA)	Transformed pixelic features	Detection of boundaries for fused tracks and layers	[253]
Microscopic images of ajp-based lines	Graphic transformation	Raw images → Overspray removal → Binarization → Line detection → Feature calculation	Mean line width and edge roughness	Process modeling	[126]
Micro-CT of as-built parts	Graphic transformation	TIFF images → Processing (8-bit grayscale images, rotation with bilinear algorithm, brightness, and contrast variation) → Binary segmentation → Porosity identification and quantification registration with pyrometry results	Tomograms representing layers in 3D volume	Characterization of as-built structure (external) and porosity (internal)	[148]
NiCrBSi-WC optical microscopy images	Graphic transformation, feature learning, feature transfer, and freezing	Image → Preprocessing (non-local means denoising, pixel mean shift) → CNN-based (U-Net) feature learning → Pixel-wise segmentation	Implicitly learned low-dimensional carbide and non-carbide features	Automated semantic segmentation	[254]

(continued)

Table 3.9 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Post-process features	Application of post-process features	References
High-speed micro-X-ray imaging	Signal transformation, AM KB graphic transformation	Video → Frame-wise analysis → Manual identification of porosity origin	Porosity pixels	Ground truths for porosity prediction	[172]
XCT of printed parts (Voxel = 3.5 μm)	3D transformation	3D XCT → Slicing → 2D image → Cropping and denoising → 2D image	Graphic features	Build orientation classification	[255]
As-built thin-walled models (XCT with 14 μm^3)	AM data registration, KB graphic transformation	XCT scan image (of a layer) → Contoured image → Binarized image → Thin wall feature extraction	Thin wall thickness, thin wall roughness, thin wall density	Design defect characterization	[27]
XCT scans of cylindrical coupons (voxel = 60 μm)	KB graphic transformation	Selected coupons → Scanning → Defect localization with automated defect recognition (ADR) algorithm	Normal and abnormal voxels as labels	Prediction of lack of fusion defects	[136]
XCT of as-printed parts (16 μm)	Graphic and 3D transformation	CT scan image → Binarized CT scan image → Complemented CT scan image → Noise reduced image → Spatial distribution analysis	Pore spatial distribution count, size, and location	Quantifying the effects of LPBF process parameters	[146]
XCT of as-built parts (15 μm)	3D and graphic transformation	3D XCT data → Slice extraction → Cropping and rotation (to eliminate edge defects and align CT with machine data) → Binarization → Slice to layer conversion → Identification of defect feature	Length of lack of fusion defect (major axis encompassing defect area)	In-situ prediction lack of fusion	[65]
XCT images (voxel size = unclear)	Customized graphic transformation, feature learning	3D U-Net	Learned features for probability maps of background and defect	Automated volume segmentation	[256]

(continued)

Table 3.9 (continued)

Feature source	Feature engineering technique (FET)	Feature engineering pipeline (FEP)	Post-process features	Application of post-process features	References
XCT (with SME and XRD), voxel = 10 μm	Graphic transformation and analysis	AM KB manual extraction of porosity	Porosity in closed- and open-loop situations	Closed-loop control of melt pool temperature	[257]
XCT (with SEM), voxel = 50 μm	AM KB graphic analysis	AM KB manual extraction of porosity	Size of the defects (effective diameter)	Classification of internal defects	[258]
XCT scans, voxel = 15 μm	Graphic transformation, 3D transformation, feature learning	OTSU-based normalization on complete 3D voxel representation → CNN for pixel-based segmentation (here images are obtained from 3D XCT and are processed in CNN in a piece-wise manner) → Segmented pores	Implicitly learned features in CNN for segmentation U-Net and Res-NET-based architecture for feature learning	Automatic porosity segmentation	[153, 259]
XCT images	Graphic transformation, AM KB feature engineering, feature learning	XCT data → Otsu thresholding → Normalization (based on mean and STD of foreground (part) pixels → PNG file → Customized CNN (feature learning inspired by U-Net)	Implicitly learned foreground and background features	Porosity segmentation in XCT	[260]

transformed images were employed in data-driven weld quality prediction tasks. García-Moreno et al. developed a multi-step computer vision pipeline for processing micrographs. Porosity features were extracted and used in a tree-based classifier for ex-situ prediction of porosity [249]. Wang et al. extracted microstructure simulation features (both inputs and outputs) to train a computationally inexpensive surrogate model [161]. The input features were simulation design variables, thermal response, and noise factors while the outputs were microstructural statistical moments.

Microstructural feature learning for dimension reduction or subsequent applications (e.g., classification in shallow learners) warrant special mention as their motivation rests in the rich feature spaces that can't be captured by the simple descriptors or transformers discussed earlier. Cang et al. learned a low-dimensional (with 1000-fold reductions) feature representation of microstructures (e.g., metallography images) from complex material systems and used it to reconstruct material representations close to the original samples [237]. This work is a great highlight of feature engineering in ICME and can be extended to additively manufactured materials by developing process–structure or structure–property models. Figure 3.10 shows learned activation map from a SEM image. Iyer et al. used a modified version of GAN to generate a latent representation of microstructure [238]. The learned representations were employed in the discriminator to predict the cooling method of a microstructure in addition to its typical function of detecting whether the input belongs to learned or original feature space.

Li et al. developed a sophisticated pipeline for microstructure reconstruction and structure–property predictions. They first transferred VGG-19 features trained on ImageNet [239]. Original microstructural images were encoded in appropriate representation and used to tune transferred layers for the reconstruction task. This

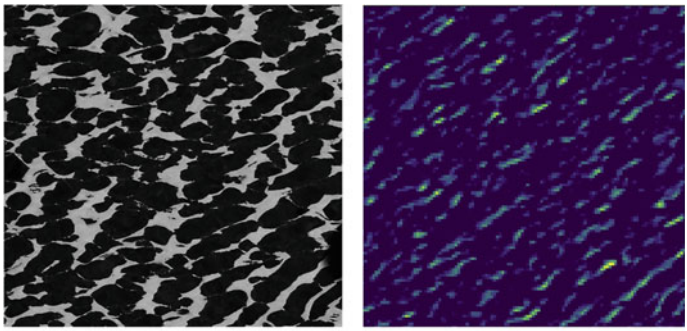


Fig. 3.10 Feature learning of microstructural images in support of material design in ICME. The figure (left) shows a SEM image, and the figure (right) shows the learned feature map from a convolutional layer of the ML model. Use with permission from Elsevier [244]

was accomplished using feature-matching-based optimization. Correlations between microstructural features and layer hierarchy were investigated and used to develop structure–property predictors. In another application of GAN at this phase, Tang et al. used processing conditions to learn representations of microstructures [241]. The learned microstructural features helped link process parameters with resulting structures and were found suitable for the new processing states as well. The work from Citrine Informatics had an interesting scope, e.g., generalization and interpretability, while classifying microstructural images in deep data-driven models. The generalization was aimed at well-realized predictions on different datasets while interpretability was focused on visualizing and understanding the impact of SEM featurization. In this regard, they were able to compare different featurization strategies and found that mean featurization of texture features (spatial aggregation of a convolutional layer's output) achieved best performance. This work is also a distinct example where CNN was employed for the sake of featurization instead of regular prediction task (e.g., classification) which was accomplished using a tree-based classifier.

Engineering of features from micro-XCT constitutes third major approach at this phase. Since featurization of as-built AM parts is expensive and labor-intensive, some research works have focused on its automation using data-driven techniques. An Automated Defect Recognition (ADR) algorithm was purposed to distinguish normal voxels from abnormal voxels for efficiently identifying lack of fusion defects from XCT scans [136]. Their featurization pipeline (XCT mask corresponding to the part via image segmentation → segmentation analysis to identify voxels with varying Intensities → grouping of identified voxels → location, size, morphologies of anomaly candidates → manual verification by operator and parallel extraction of normal voxels away from anomalous voxels) is quite sophisticated. A 3D Gaussian filter was used to extract anomaly features from XCT of printed parts [153]. The filter, as shown in Fig. 3.11, is capable of extracting voids and inclusions based on intensity variations in XCT data. These features were related with in-situ high-resolution images of PBF layers for supervised binary classification. Imani and Khanzadeh decomposed XCT into layers and applied several graphic transformations to extract macrolevel features of the part (thin wall sample) being printed and used these for geometric defect characterization [27]. Micro-CT (uCT) images were used to predict the layer-wise fiber orientation in 3D-printed composite materials [251]**.** Images with one clear fiber orientation (0°) were selected for dataset generation through augmentation. This was done by rotating the images with a step size of 1° which resulted in a larger dataset for the training. Images were later cropped to circular shapes for avoiding sharp corners in the learning process. The fiber orientation predictions enabled reverse engineering of the composite part by tool path reconstruction. A self-contained application of the structure data is its segmentation where results could be processed to generate features for ML applications [254].

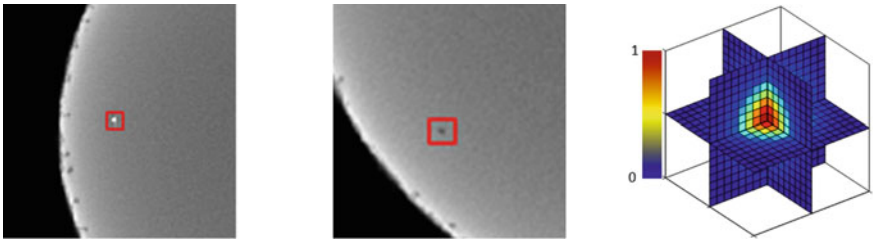


Fig. 3.11 Example of ML-enabled feature generation of labels for data-driven tasks. 3D Gaussian filter (on right) is able to identify the inclusions (on left) and pores (in middle) in XCT data from layers of a custom-built part. Figure used with permission from Elsevier [153]

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Chapter 4
Analyzing Additive Manufacturing Feature Spaces



4.1 Design Feature Space

The section deals with insights and trends discovered in the design space of AM features.

The major sources of AM design features found in data-driven applications include as-designed models (e.g., 3D), graphic design information (e.g., 2D), AM knowledge (ontologies and more), design theory (miscellaneous forms), and design space parameters (tabular). There are also instances of using material and process features in a hybrid setting to drive DfAM applications of specific AM processes [1]. The designed models are mostly used in their well-known 3D representations such as native CAD or STL files. In addition to volumetric models, a low-dimensional representation of as-designed models has been used as well, either in the form of graphic (e.g., image of cross section, image of sliced layer, or image of composite material structure [2]) or tabular (e.g., 3D primitive-based or AM knowledge-based features [3]) data. Featurization of as-built models, which is discussed in the post-process section, often works in conjunction with as-designed features to drive applications on geometric deviations or dimensional accuracy [4]. The remaining sources of AM design features follow miscellaneous representations such as graphic [5], tabular [6], ontologies [7], and more.

Design features are engineered with generic or AM-specific transformations. There are relatively few instances where tabular transformations are applied to AM design data. Feature generation through transformation is the most common engineering technique and follows data-specific approaches. Among these, 3D transformations are most prevalent. KB transformation of AM design data is common as well. The main technique in KB transformations is to extract design features directly from source representations using knowledge. Feature learning and subset selection usually work in conjunction with upstream transformations. These pipelines have

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been presented in Chap. 3 in the context of applications. Since numerous configurations of these pipelines exist and many more are possible, it is difficult to identify unifying themes to summarize them.

The majority of design features are explicit geometric or knowledge representations with a few generated by process, material, experimental, or surrogate sources. Implicit design features are either learned through deep models or generated with the help of physical models and customized transformations. Pure design-oriented features are sometimes embedded or fused with non-design features to generate hybrid feature vectors or matrices. As indicated by Yang et al., hybrid features can potentially improve the performance of design-oriented ML models, but it is not always the case [8]. There are also examples of new features (e.g., curvature feature of [4]) being introduced to capture critical design information and be used for specific applications. The primary domain concerned with AM design featurization is geometric ML while scale of featurization can be macro or micro depending on the source. The feature engineering pipelines used to process data are mostly customized with several steps between raw data and engineered features.

The applications of design features closely relate to efforts focused on manufacturability enhancement. The manufacturability can be judged based on several criteria. In this regard, data-driven models support design modification or selection with the potential to minimize unwanted variations of quality. In addition to supporting manufacturability, other aspects of design such as cost, mechanical properties, printing time, design-specific process recommendations, and design space exploration are also possible. Figure 4.1 provides a summary of the AM design featurization landscape. The design feature space is divided into four quadrants using AI level and AM frequency as two dimensions. AI level refers to improved quality of features in their ability to support data-driven tasks. AM frequency represents the popularity of a given feature type and is simply a count of its occurrence in the literature. The same AI and AM dimensions will also be used when discussing the process and post-process feature spaces.

We chose four criteria to visualize the trends in design feature space. These include feature sources, engineering techniques, breakdown of transformations, and feature applications. Figure 4.2 highlights the trends of design feature space. Among feature sources, as-designed models and graphic data with design information are the most common. While knowledge-based engineering leads to design FETs, data-specific transformations are a close second. Among transformations, encodings and 3D-based transformations are most common. As expected, the majority of the design features are used in DfAM-based targets with some applications at the process (e.g., path planning), structure (e.g., bead geometry), and property (e.g., flexural strength) phases.

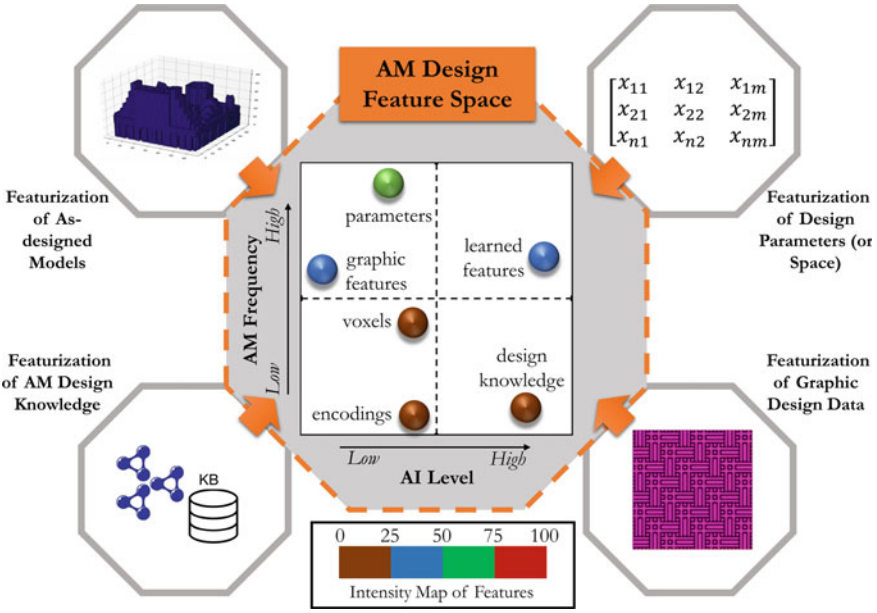


Fig. 4.1 Four major sources of AM design features contributing to feature space. The resulting space is presented in terms of AM and AI context. Some of the features are highlighted in the 2D feature space with their respective frequencies in the literature. Figures used with permission from Elsevier and Royal Society of Chemistry [1, 2, 9]

4.2 Process Feature Space

The section deals with insights and trends discovered in the process space of AM features.

The feature sources available at the **planning phase** of AM process include scan strategies (tool path and scan pattern), build orientations, and specific geometries to be printed. Fusion of parametric features with deposition features is common as well. The material and system parameters are kept separate from planning parameters. Similarly, process parameter selection and optimization are seldomly referred to as process planning but these are discussed separately in this text due to their diverse sources and applications which do not align with the planning definition considered. Knowledge-based transformations were found to be most prevalent. 3D transformations become relevant when the whole deposition zone is considered [10]. Graphic transformations have an equal representation at this phase. Thermal history prediction corresponding to scan strategies or build orientations is the main application of planning features.

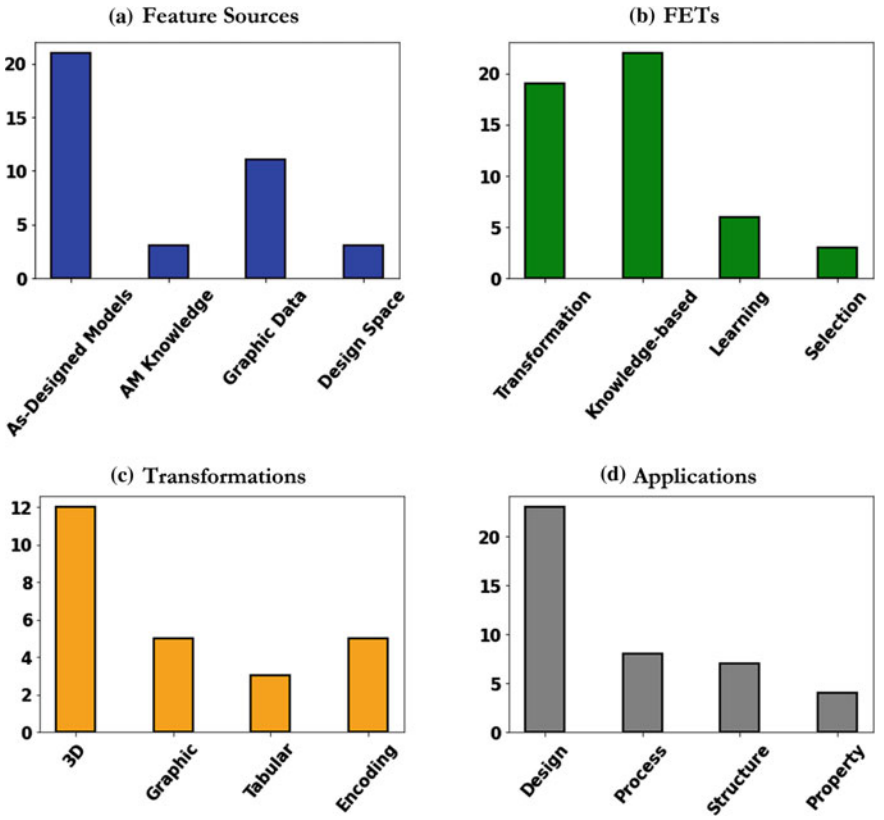


Fig. 4.2 Trends of design feature space w.r.t. feature sources, feature engineering techniques, feature transformations, and feature applications

Vision-based monitoring (visible-light cameras, IR cameras, pyrometry, X-ray imaging), signals (spectrometers, thermal videos), and simulations (melt pool thermal profile) are common sources of **melt pool data**. As a result, their corresponding representations are based on graphic, sequence, and 3D data. Point-based features of melt pool, such as those captured with a single photodiode, are also common. However, as with acoustic sensor's nature of integrating sources of airborne sound, point-based photodiode could also potentially combine different optical emissions and is therefore moved to the generic process feature section. Melt pool feature engineering techniques are most diverse among their counterparts. Graphic transformations lead the trend followed by sequence and 3D transformations. Feature learning is greatest for high-dimensional melt pool data and several deep architectures have been customized to learn a low-dimensional useful melt pool feature vector for downstream applications. KB engineering of graphic sources (e.g., dynamics, solidification) is a quite common in melt pool featurization pipelines. Feature operations such as transfer,

fusion, and visualization have been reported as well. Similarly, generic and AM-specific preprocessing is common to prepare melt pool streams for data analytics. Finally, a few analysis-oriented studies focused on melt pool graphic and sequence representations have been included in the respective table as well based on their potential for data-driven AM.

Understandably, the most common transformations for **parametric feature** are of tabular nature. These include Min–Max, *Z*-Score, and PCA. The engineering of parametric features is relatively straightforward as compared to other feature sources. Their numeric and explicit representations are directly employed in feature selectors or tabular transformers. Hence, sophisticated and customized feature engineering pipelines are not involved. Min–Max normalization or *Z*-Score-based standardization is the main transformation tool for parametric features in tabular forms. Both techniques bring features on a common scale and preserve the mutual differences in values. A rare example of graphic transformation was found when laser profile was featurized to predict its temperature [11]. Selection of tabular features is also common for parametric sources. Methods of selection include ANOVA, Pearson Correlation, and tree-based techniques. There are two dozen or so variables that can influence the print quality and are sometimes selected using feature selection techniques. As a result, KB transformation, selection, and filtering of parameters is usually performed. An interesting observation at this phase was the lack of featurization for most of the literature where raw tabular data was directly used in the learners. The applications of parametric features are diverse with an equal usage for process and property-based targets while structure leads the overall trend.

Given the consideration of 2D part or layer surface, it is not surprising that the majority of **layer feature** sources have graphic representations. Graphic representations can be broken down into thermal or visual categories based on the working principle of the sensor or camera used to capture them. Among these, visual layer images are the most common. Thermal images of layers are present as well which capture infrared radiations from the layer surface. Equivalently, the graphic representations can be arranged in terms of their physical sources, namely material or part layers. In one example with non-graphic source representation, the point-wise thermal history of a layer was captured and fused with inputs from material, path, and other sensory measurements [12]. Graphic transformations are the most common feature engineering technique followed by sequence transformations. Feature learning and KB engineering are also involved in the featurization spectrum of AM layers. A noticeable trend is to register layer data before its subsequent featurization. Application-wise, layer features are most frequently related with structure-based CoCs, followed by process and property. While no applications for design-based targets were found, the registration of layer data with as-designed models is usually performed to evaluate their quality.

The top sources for the information for parts being built are graphic representations containing **geometry information**. If a specific work does not highlight the fact that geometry information was handled, it was placed under generic or layer features depending on the source. In this section, we consider works that mostly deal with geometry-based inputs. The second most common representation of these sources is understandably 3D-based where in-situ scanning or modeling of a part is involved. Other examples include geometric features of printed beads or layers in tabular representations and multi-input sources with as-captured or as-simulated geometry. Graphic, 3D, sequence, and tabular transformations all exist in the same order of usage. While feature selection and learning is absent, applications of KB engineering and data preprocessing exist. Finally, the applications of in-situ geometry features are limited to process and structure-based CoCs with a roughly equal share.

Generic process feature sources usually combine simulations, sensors, and parameters. Among singular sources, acoustic and optical sources are most prevalent. FETs are dominated by transformations and among them signal processing is the leading method. Knowledge-based engineering of process features also exists with a few examples of learning and subset selection. Generic features are of importance for process and design-oriented applications, although their main application is for structure.

Figure 4.3 presents an overview of the process feature space of AM. Six major sources of features namely process parameters, layer, melt pool, planning, in-situ geometry, and of generic type have been identified. Process parameters engineered features of melt pool and layers, in-situ geometry, and process knowledge represent the majority of resulting features. The frequency of process parameters, layer, and melt pool features is found to be high relative to other feature types. Learned features and process knowledge has higher potential of aiding the prediction task at hand.

Figure 4.4 highlights the trends of process feature space. Melt pool features are most common in the reviewed literature which points to the fact that significant research efforts are focused on metallic AM to realize high-volume production. Parametric features remain popular at the process phase. Featurization of layers, due to their broad applicability, is also common. Data-specific transformations are clear choice to engineer AM process features. Among these, graphic and sequence transformations are most common. The resulting features are mostly used for process and structure-based applications.

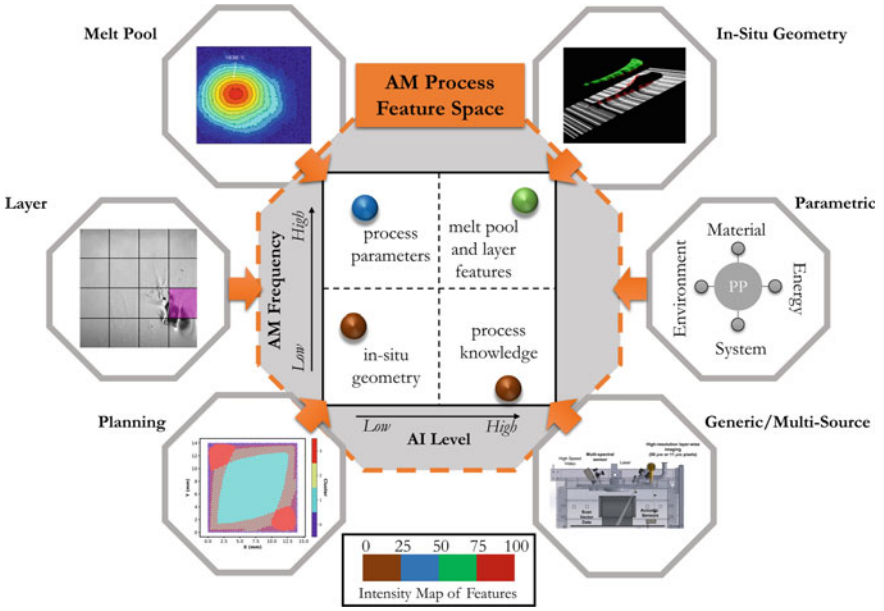


Fig. 4.3 Six major sources of AM process features contributing to feature space. The resulting space is presented in terms of AM and AI context. Some of the features are highlighted in the 2D feature space with their respective intensities in the literature. Figures from [13–17] used with Elsevier permission

4.3 Post-process Feature Space

The section deals with insights and trends discovered in the post-process space of AM features. Since most of these are based on structure information from as-built parts, the word "structure" is used interchangeably with "post-process". Figure 4.5 presents an overview of the post-process feature space of AM.

The common sources of macrostructural features include XCT (medical imaging and macroscale sources), laser-based scanning, graphic data, visual inspection, or manual measurements. XCT-based macrostructural features come from low-resolution scans for the purpose of 3D geometry reconstructions. XCT and laser-based scanning both result in 3D representations with the former being primarily volumetric (e.g., voxels) and the latter having point cloud as its typical representation. A few of the references have relied on graphic sources to extract macrolevel part features. These images are highly context dependent but usually capture geometric aspects of as-built parts (cross sections or surfaces). Finally, manual measurements

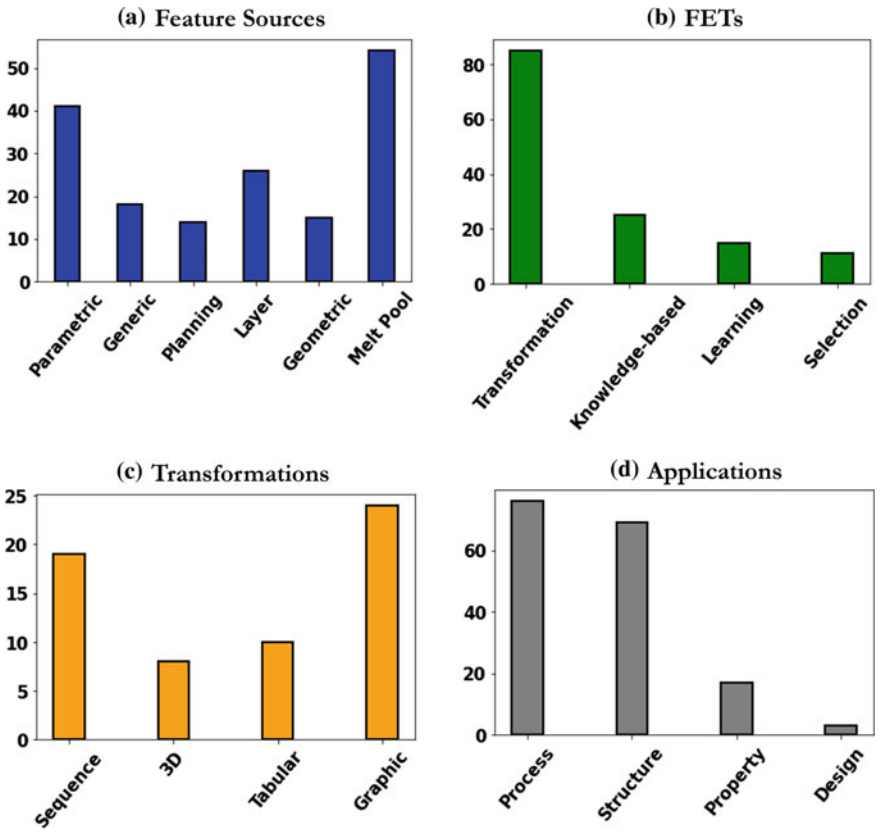


Fig. 4.4 Trends in process feature space w.r.t. feature sources, feature engineering techniques, feature transformations, and feature applications

(visual defects) or simpler instruments (macroscales) can be employed to collect data (particularly ground truth values) from as-printed parts [18]. SEM and micro-XCT are found to be the leading sources of microstructural features in data-driven applications and mostly concern cross sections and volumes of printed samples, respectively. Microscale surface images are sometime used to make predictions based on surface quality of the parts. The high fidelity (e.g., 3D) of micro-XCT is balanced by its ability to scan small-sized parts. Researchers at the Argonne National Laboratories have pioneered the use of high-speed X-ray imaging for sideway visualization and analysis of microscale AM features but its usage in data-driven pipelines is limited [19]. Numerical simulations resulting in microstructures have been used as a source

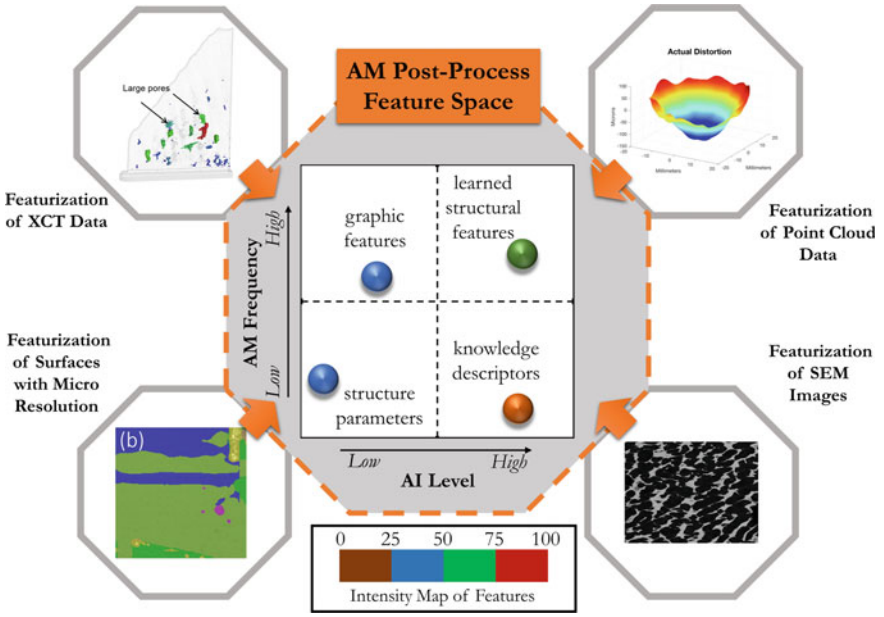


Fig. 4.5 Four major sources of AM post-process features contributing to feature space. The resulting space is presented in terms of AM and AI context. Some of the features are highlighted in the 2D feature space with their respective intensities in the literature. Used with permission from [20–23]

of AM features as well. Lastly, tabular representations of features extracted from microstructures are common too.

Transformations, 3D or graphic, are most common feature engineering techniques for macroscale features. Outside of transformations, feature learning, and KB engineering is sometimes preferred. Feature engineering pipelines are highly customized and vary based on geometries and applications. The resulting features are usually explicit geometric variables but can include learned features and transformed images of printed part. The applications of macro-AM features are mainly concerned with the identification of anomalies at the macrolevel of printed parts. Feature learning and transformations (mostly graphic) represent almost all the FETs deployed to engineer microscale features. One distinct trend at this phase is the diversity of feature engineering pipelines that span representation learning, customized and automatic feature extraction, KB graphic transformation, and graphic analysis. This is inspired by the research in materials informatics that far precedes materials development in AM. The resulting features are latent representations of microstructural spaces, transformed images, explicit microstructural descriptors, transformed tabular representations or

identified defects such as pores. The leading application of these features is to serve as labels for learners. Reduction of high-dimensional feature space, segmentation, and miscellaneous prediction applications are also common.

Figure 4.6 highlights trends of feature engineering in the post-process space of AM. Although several sources are available to characterize materials printed with AM, SEM and XCT are found to be the most common methods when generating data for ML applications. Transformations, especially graphic, are the most common FETs followed by knowledge engineering and learning. These features are used to support self-contained applications at the structure phase. Additionally, microstructural features from XCT, SEM, and other instruments serve as labels or targets to miscellaneous data-driven applications at the process phase.

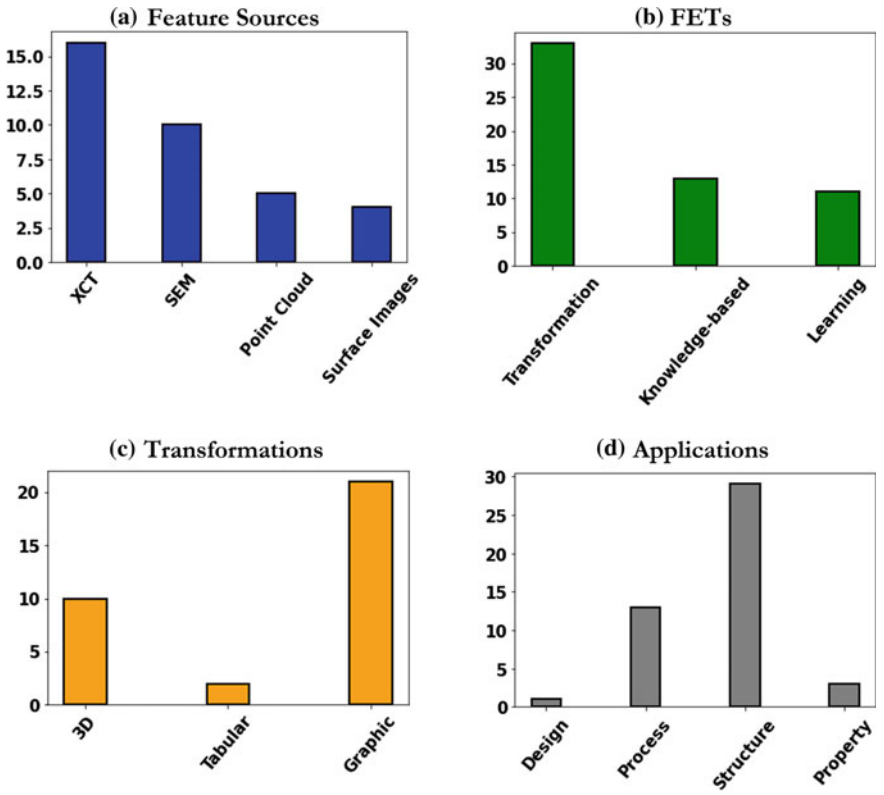


Fig. 4.6 Trends in post-process feature space w.r.t. feature sources, feature engineering techniques, feature transformations, and feature applications

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Chapter 5
Challenges and Opportunities in Additive Manufacturing Data Preparation



5.1 Challenges

Lack of FAIRness: The great majority of challenges faced while preparing AM data are caused by a severe lack of its fairness [1]. The fair principles (e.g., findable, accessible, interoperable, and reusable) can be used to evaluate raw data, processing techniques, and resulting features in AM. In this regard, openness is a challenge in AM community. The data of R&D activities in AM is usually proprietary. As a result, most of the works reviewed do not openly provide data. Open-source AM data is scattered across WWW limiting its access and usage. Even if the data was somehow organized and made available through a single platform, it is not always readily usable. The reusability of AM data concerns issues related to its diversity (multi-mode, multi-scale, multi-phase, multi-spectrum, multi-physics) and machine representations (noise, low quality, missing data). In the first case, AM data sources have not been benchmarked for their reusability in specific scenarios. This requires efforts on the user's end to iterate over all available data sources which could solve a given problem (e.g., porosity prediction from different types of images). The second issue hinders efficient reuse of a specific dataset. For instance, noisy representations (graphic, 3D, spectrum, tabular) with missing or added features will require significant processing effort before these can be reused. AM data processing pipelines are also heavily embedded with knowledge. These steps may not be easily transferable into algorithms. Sometimes, portions of processing techniques are left unexplained limiting their reproducibility. Sophisticated engineering pipelines are usually data-specific limiting their generalizability. Knowledge-driven processing brings interpretability but lacks automation. Finally, learned features are neither benchmarked (learners and representations) nor made available as part of architectures (like trained computer vision architectures such as AlexNet with ImageNet weights).

Knowledge Barrier: Data-driven AM lies at the intersection of different domains. There exists a knowledge gap between AM and AI experts. AM experts are mostly

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not well trained in data-driven techniques and might not assess its true potential. They are the main players in data-driven paradigm as they hold the key to datafication. By knowing the inner working of data processing and data-driven tools, effective polices can be adopted at the organization level (digitization, high-quality datasets, etc.). AI experts are mostly not knowledgeable about the intricacies in AM and might often not be aware of the complexity of AM problems. This can significantly reduce the generalizability of solutions being developed.

Experimental and Computational Costs: Implementing data-driven solutions is expensive, especially regarding the initial cost of hardware (e.g., sensors, GPUs). Similarly, generating sufficient and representative data with high quality can be costly for small and medium-sized enterprises (SMEs). This can limit the diversity and quality of data needed to drive empirical solutions. More importantly, the significance and potential offered by different data types and data sources cannot be evaluated. Some feature learning pipelines require access to high compute resources. While research activities can benefit from national or institutional resources, SMEs usually have reduced access to these sources.

Insufficient Standardization: AM is still maturing into a large volume production technology. The standardization landscape in AM is in its infancy. Therefore, standardization of ML-driven AM has a long way to go before the technology is mature. This exposes researchers and practitioners to a range of bottlenecks when implementing data-driven solutions. For instance, there exist no methods and metrics to evaluate the potential of feature sources, feature engineering techniques, and resulting features. This has led to highly customized solutions being developed, solutions which lack generalization.

5.2 Opportunities

Open Repositories: Systematic open repositories can make data findable, accessible, interoperable, and reusable. Encouraging AM researchers to share data on such repositories can expedite the adoption of data-driven AM in industry, particularly in SMEs. So far, the efforts on this end in AM are limited. McComb of Carnegie Mellon University has evaluated the readiness of design repositories (e.g., 3D models) for ML applications and presented useful insights in support of AM design data readiness [2]. Similarly, process and structure repositories can be developed. In this regard, collaborations are needed to manage highly diverse process and structure datasets of AM. A practical solution, as pointed out by [3], is to bring AM datasets for ML applications on a singular platform and publish it openly. NIST's Additive Manufacturing and Materials Database (AMMD) is an example. One current challenge though is the diversity of the available datasets which are mostly limited to the PBF process.

Support for Benchmarking: Benchmarking test beds, datasets, engineering techniques, and resulting features presents an important research direction in data-driven AM. Benchmark test beds can help evaluate the potential of different measurement techniques for data-driven pipelines. In this regard, NIST's Additive Manufacturing Metrology Testbed (AMMT) is an important first step. Governments need to invest in the development of AM test beds in support of measurement science needs for SMEs looking to embrace data-driven AM. High-quality datasets optimized to contain the most important features for downstream tasks need to be developed to tackle a range of issues. The measurement science spectrum in AM is broad. Each of its domains (e.g., signal processing) offers multiple processing techniques for a given data type (e.g., molten streams of MAM). Research on benchmarking these techniques can help to identify the most relevant methods for a given problem. Similarly, benchmark datasets are expected to lead to reusable features for future tasks (analogous to learned representations of computer vision such as ImageNet).

Automation of Data Processing: An algorithmic equivalent of open repositories is to bring openness and automation in AM data processing. Existing pipelines of data processing are heavily dependent on AM know-how with significant manual involvement. Bringing automation to these procedures can be a game changer for supporting data-driven AM in the industry. Interestingly, ML can come to the rescue at this phase as well. One key example is to automate label (or ground truth) extraction or generation processes at the design, process, and structure phases of AM. Some examples of such implementations already exist and were highlighted in respective sections such as in [4] for structure information extraction. In addition to labeling, AM data processing libraries can be developed to automate feature engineering pipelines covered in this review. So far, well-known libraries in signal processing, computer vision, geometric ML, and multi-dimensional arrays are being used to process data. AM-oriented libraries (e.g., design feature extraction, melt pool feature extraction, microstructural feature extraction) can be implemented and made open source to support data-driven AM.

Improving Feature Fidelities: Major sources of AM features either belong to measurement science (e.g., sensors) or scientific modeling (e.g., simulations). Several researchers have pointed out the possibility of enriching these features through hybrid approaches. Multi-sensor fusion is one way to capture features with unique physical significance. Another approach could be real-virtual fusion where synthetic features are fused with their real-world counterparts. The scope of improving feature fidelities applies both to measurements as well as the models. Sensors capable of capturing high-fidelity data can be developed (e.g., 3D melt pool data as opposed to 2D images). Similarly, the realness of computer simulations can be improved by developing novel approaches (e.g., multi-physics and multi-phase scripts) specific to an AM process. Pursing research in the direction of high-fidelity features will eventually lead to better predictions in data-driven models.

Enabling Knowledge Transfer: Raw and processed data (e.g., features) are important components of the informatics chain that supports AM knowledge development

and discovery. There are inherent similarities among DIKW components (e.g., data-level, feature-level) of AM semantics. This opens exciting opportunities to transfer knowledge across regimes and scenarios. As a result, data-based (e.g., melt pool sequences, microstructural images), feature-based (e.g., learned representation of deep models), and knowledge-based (e.g., optimized model architectures) exchanges among different processes, printers, and material systems are possible. In the pursuit of knowledge transfer, transfer learning (TL) and domain adaptation (DA) are two important techniques and are expected to significantly grow enabling knowledge exchanges from developed areas to relatively less explored scenarios.

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Chapter 6
Summary



Recent research and development activities in AM have shown great potential to resolve challenges related to design, process, and post-process phases. The motivation behind the text rests in the rich spectrum of additive manufacturing data in terms of its sources, representations, and handing techniques. Data-driven AM is a hot topic and offers one way to approach key issues of process repeatability and reliability. AM data in numerous computer representations and at multiple scales drives these applications. The necessary handling techniques to process data for these applications depend on its physical and virtual context. The physical context is primarily concerned with the physical source and process phase at which data is generated. The virtual context of the data is representative of the modeling technique employed to generate data and the specific computer format used to represent it. AM engineers, practitioners, and researchers have a range of techniques from measurement science and scientific modeling at their disposal to acquire data. This makes AM data landscape extremely rich. Similarly, there exist a range of techniques and tools to process AM big data, each specific to its context. The growing interaction of AM big data with processing techniques from several domains has not been covered before. To address this issue, this book is an effort to group AM data processing under the umbrella of feature engineering by linking specific handling techniques with upstream sources and downstream applications.

We started by briefly introducing AM in terms of its capacity, existing challenges, and potential solutions. Data-driven AM is considered as the primary solution of interest based on its merits and popularity. Featurization is defined as the process to improve AM data quality in support of data-driven solutions, and all of its handling techniques were grouped under this root term. We comprehensively evaluated and compared existing reviews and surveys on data-driven AM for their featurization levels. This led to the identification of existing gaps and the need to provide an overview of AM featurization landscape. The review methodology to collect literature for this text was introduced and linked to techniques utilized in peer-reviewed AM literature.

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Feature engineering in AM was connected to existing domains and paradigms to help visualize its complete spectrum. Major data sources across the lifecycle of a typical AM process were identified and arranged according to the AM digital thread. The text then introduced a rough taxonomy of the feature engineering technique linked to AM data. Generic data preparation techniques from data mining were identified as the basic and frequent tools to process raw AM data. AM research has inspired certain handling techniques that were grouped under AM-specific data preparation. Selection of key AM features from raw data based on either AM knowledge or statistics was grouped under feature subset selection. Generation of new AM features from raw data through mathematical transformations was introduced. These transformations are mostly related to specific representations of AM data (e.g., tabular, sequence, graphics) and can be seen as conventional or advanced in their scope. ML techniques to generate new features from raw data were grouped under feature learning. The main motivation behind feature learning is to obtain a compressed representation of raw AM data to support subsequent learners. AM knowledge plays a key role in processing raw data. In line with the existing literature, we defined generic and mechanistic feature engineering as the two main types of engineering techniques based on AM knowledge. The feature engineering techniques can be integrated in different ways. These integration techniques were grouped under integrated feature engineering techniques. The most frequent feature operations are identified, and AM data processing libraries are introduced.

The major sources of AM features were identified by decomposing process lifecycle phases (e.g., design, process, post-process). Once these sources have been identified, the associated processing techniques and specific applications were summarized in tabular representations. This starts with AM design phase which contains the first source of AM features. Virtual AM designs (both 2D and 3D), design knowledge, and parametric design spaces were found to be the main sources of design data. The second type of AM feature sources belongs to process phase. The specific feature categories at this phase include generic, planning, parametric, layer, melt pool, and in-situ geometry. Generic process features represent data sources where exact physical source of data is unclear (e.g., vibration or photodiode signal) or when multiple in-situ data sources are integrated (multi-sensor fusion). Data from AM planning activities (e.g., path planning, scan strategies) forms another important source of AM process features. Parametric features represent numeric values associated with energy, environment, material, and system variables in AM. In-situ layer-wise data collection (material or specimen) is quite common in AM and represents another source of features at this phase. There has been a special focus to capture in-situ melt pool (molten material) data as this is representative of subsequent structure and properties. As a result, we identified melt pool features as an important feature category at the process phase. Finally, in-situ geometry features representing the under-build or incomplete part come from 3D process data and can be used to driven solutions on geometry control. The third type of feature sources belongs to the structure of AM printed parts from post-processing phase and is usually concerned with macro or microscales. As a result, macrostructural and microstructural features are the last two major sources of features in MA lifecycle. AM data is then processed based on

the specific data type or form and is used to support downstream tasks in data-driven solutions. The specifics were detailed in respective tabular summaries. The property space in AM remains sparse and without big data.

After identifying the featurization spectrum and their existing applications in support of data-driven AM, the design, process, and post-process phases of AM were analyzed in terms of data sources, feature engineering techniques, resulting feature and key applications. The results of this analysis were illustrated based on AM frequency and AI importance of design, process, and structure features. Additionally, data sources, feature engineering techniques, transformations-based featurization, and feature applications were plotted to show the popularity trend at each AM lifecycle phase. Some key challenges and potential opportunities to support high-quality AM data preparation were identified.

The authors expect that this text will provide an overview to readers of the relevant feature engineering techniques to handle AM big data. The reviewed applications of AM data preparation can be adapted to similar scenarios in industry and academia. Solving the existing issues related to AM data can expedite industrial adoption of data-driven AM.